

The Triadic Measurement Substrate

$$\bar{B} = 3\bar{\alpha}$$

Volume 1

Paper 5: Reinterpretation, Field Emergence, and the Big Bang Correspondence: From Exhausted Substrate to a Universe

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Paper 5

Reinterpretation, Field Emergence, and the Big Bang Correspondence: From Exhausted Substrate to a Universe

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Status of Results

This paper distinguishes four categories of claim, each labeled where it appears in the text. The categories continue the labeling conventions established in Papers 1 through 4.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and the structures established in Papers 1 through 4 by explicit construction or proof. The reinterpretation operator $\hat{R}$ as the unique PGE-selection at $\tau > \tau^*(R)$ (§ 2.3); the recursion-preservation theorem stating that $\hat{R}$ produces a substrate structurally identical to the original under the same five axioms applied at the new scale (§ 2.5); the maximum novelty output theorem as bifurcation-forced strict maximization, with the doubling factor empirically verified by Stage 5 (§ 2.6, Appendix E); the wave equation unification theorem under the PGE-forced wave-form argument of Paper 2 § VII (§ 3.2); quantization as trough-finding in the recursive wave hierarchy (§ 3.5); the sublattice bifurcation at every hierarchical level $k \geq 1$, with the two-sublattice structure verified through Stage 6 (§ 5.3 Claim 1, § 6.1, Appendix E); the rigorous derivation $L^{p(1)} = 3\sqrt{2} \ell_p$ from the stella-octangula tile plus cardinal boundary, with the underlying bifurcated structure empirically confirmed (§ 6.1.3, Appendix E); the $c^{(k)} = (1/\sqrt{2})_k c$ sequence at saturation as a structural law of the FCC face-diagonal-to-edge ratio recurring at every level by scale invariance (§ 7.2 sharpened, § 6.2); gravity as cross-level dispersion, with § 7.2–§ 7.4 unified as one structural picture (§ 7.5); the closed form of $f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}}, k)$ under additive Kolmogorov bounds (§ 6.4, Appendix C); the substrate-level $Z_2 \times Z_2$ stabilizer structure at every hierarchical level (§ 8.1); the cross-level commutator mechanism from the sublattice bifurcation as the substrate-level origin of non-Abelian gauge closure (§ 8.2); the complete five-irrep field decomposition under O_h on the FCC nearest-neighbor space, exhausting all 12 dimensions (§ 4.2). The colour gauge group as $\text{su}(3)$ closing in the push-forward transport with no overshoot, and the global form $\text{SU}(3)$ (with unbroken $\mathbb{Z}_3$ centre realizing confinement) fixed by the fundamental triality of the agents, Derived under the § 4.1 continuum lift (§ 8.3). The level-1 $\text{SU}(2)$ factor and its $\text{U}(1)$ trace-mode partner on the same footing: $\text{su}(2)$ closing from the A/B doublet transport with no overshoot, global form $\text{SU}(2)$ (not $\text{SO}(3)$) fixed by the faithful fundamental doublet, so that the full gauge algebra $\text{su}(3) \oplus \text{su}(2) \oplus \text{u}(1)$ is Derived (§ 8.3); the electroweak global quotient, the chirality of $\text{SU}(2)_L$, and the fermion generations remain structural correspondences.

Structural correspondence. Results in which a TMS construction faithfully reproduces the skeleton of a known framework without claiming reproduction of every feature. The Big Bang Correspondence (§ V) is the central case: the structural features of the observed cosmological initial state — hot dense early phase, near-uniform background with small fluctuations, large-scale flatness, horizon-connected past — are reproduced as consequences of a reinterpretation event acting on a saturated biographical record $\mathcal{B}(R)$; numerical pinning of CMB temperature, scalar spectral index, and analogous quantitative observables depends on the specific $\mathcal{B}(R)$ and is reserved for simulation. The matter-antimatter structural correspondence (§ 5.3 Claim 2): the two-sublattice $A \leftrightarrow B$ exchange at every level $k \geq 1$ is the substrate-level analog of QFT’s matter / antimatter sector decomposition, with the parallel to Sakharov’s three baryogenesis conditions developed structurally in § 5.3 Claim 3. The field emergence (§ IV) is a structural correspondence to quantum field theory in the QCA-derived sense (Mlodinow & Brun 2021; Brun & Mlodinow 2025): the polarization landscape propagating on the FCC lattice with $[[4,2,2]]$ stabilizer structure satisfies the structural conditions for QCA-to-QFT correspondence in the continuum limit. The hierarchical $[[4,2,2]]$ carrier structure for the $SU(2) \times U(1)$ factors (§ 8.3) is a structural correspondence: the TMS framework provides the substrate-level $Z_2 \times Z_2$ stabilizer hierarchy with cross-level commutator structure, with the explicit Lie-group identification of those factors reserved for subsequent representation-theoretic work. The colour $SU(3)$ factor is the exception: as of § 8.3 it is Derived under the § 4.1 lift (see the Derived block) — the push-forward transport closes to $\mathfrak{su}(3)$ with no overshoot and the fundamental triality of the agents fixes the global group as $SU(3)$. The erosion picture as a substrate-level account of the apparent permanence of standard-model field structure at human-accessible timescales (§ 3.5).

Predictions / Conjectures. Falsifiable claims that follow from the framework but whose closed forms or numerical values are not derived within this paper. The matter-antimatter asymmetry of the observed universe as a generic consequence of reinterpretation from a non-perfectly-symmetric parent record (§ 5.3 Claim 3), with the three TMS conditions (sum-mod-4-changing transitions, inversion-asymmetric $\mathcal{B}(R)$, $\hat{R}$ phase-transition character) jointly producing the asymmetry. The hierarchical level identification of our universe — which level k we occupy — is a prediction whose empirical bridge requires comparing measured cosmological observables against the $c^{(k)}$ and $L^{p(k)}$ sequences (§ 6.3, § 9.5). The cosmological-natural-selection signature predicted by repeated reinterpretation across cycles, in the spirit of Smolin (1992) and Dowker–Zalel (2017), is a structural prediction (§ 5.8). The CMB-anisotropy correspondence to the spectral content of $\mathcal{B}(R)$ prior to reinterpretation (§ 9.2) is a falsifiable claim whose empirical content depends on simulation work. The three-generation fermion content from three-level hierarchical embedding (§ 8.4) is a structural prediction whose quantitative consequences (mass spectrum, mixing angles, CP-violating phases) require detailed simulation. The cross-level relativistic phenomena are derivable structurally (§ VII) but the explicit functional form recovering the Einstein equations in the continuum limit is a structural conjecture pending detailed computation (§ 7.6).

Deferred closures. Two classes of result remain deferred beyond Paper 5, by the structural-vs-numerical division of labor across the volume. First, exact numerical pinning of $L^{p(k)}$, $T^{p(k)}$, $c^{(k)}$ at $k \geq 2$ requires combinatorial enumeration of stabilizer-satisfying T-closed configurations at each level — a task whose structural framework Paper 5 specifies (§ VI and Appendix B) and whose exhaustive enumeration is reserved for simulation. Second, the explicit representation-theoretic mapping from the meta- $[[4,2,2]]$ stabilizer hierarchy with cross-level commutator structure (§ 8.2) to the $SU(2) \times U(1)$ gauge structure (§ 8.3) requires the apparatus of QCA-to-QFT correspondence developed by Mlodinow, Brun, and the D’Ariano–Perinotti program, applied to the FCC lattice with hierarchical $[[4,2,2]]$ symmetry; the colour $SU(3)$ factor is no longer in this deferred class, having been closed in § 8.3 via the push-forward transport.

This four-way labeling applies throughout the paper. Every claim in the body text falls into exactly one category, and every category's status is visible at the point of claim.

Preface to Paper 5

This paper is the fifth and concluding installment in Volume 1 of the Triadic Measurement Substrate program. Volume 1, Paper 1 established the geometric and ontological architecture of the substrate. Volume 1, Paper 2 developed the substrate's computational dynamics, the Born-rule structural correspondence, and the formal program definition. Volume 1, Paper 3 derived computation from informational substrate via surround-driven repair, polarization-coherence bistability, recursive coarse-graining, and the first computing subsystem. Volume 1, Paper 4 developed the macroscopic consequences — hierarchies, self-modeling, multi-subsystem stressor dynamics, region biography, and novelty exhaustion — and closed at the structural problem: at the exhaustion threshold of any bounded region, the substrate holds informationally rich confirmed bits with no remaining direct-read protocol capable of extracting novelty from them.

Paper 5 inherits that structural problem and closes it. The reinterpretation protocol that takes the substrate from exhausted $\mathcal{B}(R)$ to a new generative phase is derived here. The fundamental fields, the initial conditions, and the Big Bang correspondence follow from that protocol applied at the appropriate scale. As in Papers 1 through 4, no new axioms are introduced. No new free numerical parameters are introduced. The Principle of Generative Economy continues to govern every selection.

Paper 5 is also the numerical-closure paper of the volume. Several results stated structurally in Papers 1 through 4 are closed here in their numerical or representational form: the minimum program extent $L_p^{(1)}$ at the substrate level, the closed-form $f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}}, k)$ under additive Kolmogorov bounds, the hierarchical $c^{(k)}$ sequence at the saturation bound, and the structural correspondence to the gauge symmetries of the Standard Model. The remaining numerical pinning at $k \geq 2$ is the domain of subsequent simulation work.

The promise of the volume, stated at the close of Paper 1 — that the absolute length scale, the absolute time scale, the propagation speed, the binary alphabet, the fault-tolerant lattice, the computational substrate, and the universe inside it follow as geometric consequences of five axioms and one principle — is brought to its completion here. The arc from the sphere primitive of Paper 1 § 1.2 to the cosmological initial conditions of § V is one unbroken geometric chain. The substrate generates universes inside itself because the geometry of confirmation, applied recursively across the structural problem of exhaustion, has no other available move.

Abstract. Building on the exhaustion analysis of Paper 4 and the recursive coarse-graining of Paper 3, we close the structural loop of Volume 1 by deriving the reinterpretation protocol that takes the substrate from an exhausted bounded region to a new generative phase. Eight principal results are established. First, the reinterpretation operator $\hat{R}$ acting on the biographical record $\mathcal{B}(R)$ of an exhausted region is the unique Principle-of-Generative-Economy selection at $\tau > \tau^*(R)$, forced by Axioms 3 and 4 jointly closing every other continuation. Second, the recursion-preservation theorem establishes that $\hat{R}$ produces a substrate $\Xi(R)$ structurally identical to the original substrate under the same five axioms applied at the new scale, with confirmed bits of $\mathcal{B}(R)$ playing the structural role of agents in Ξ . Third, the two-waves problem — the apparent tension between Paper 1’s first-order geometric wave equation and the second-order wave structure introduced by reinterpretation — is resolved by the unification theorem: there is one wave equation, applied at successive recursion levels, with level- k providing boundary conditions for level- $(k+1)$. Fourth, the continuum limit of the polarization field on the FCC lattice with $[[4,2,2]]$ stabilizer structure satisfies the structural conditions for QCA-to-QFT correspondence (Mlodinow & Brun 2021; Brun & Mlodinow 2025), yielding fundamental fields with the same skeletal structure as those of standard QFT. Fifth, a reinterpretation event acting on a saturated biographical record reproduces the structural signatures of the observed cosmological initial state: hot dense early phase, near-uniform background with small fluctuations, large-scale flatness, and a horizon-connected past — the Big Bang Correspondence. Sixth, the closed forms of the structural debts inherited from Papers 1 through 4 are supplied: $L_p^{(1)} = 3\sqrt{2} \ell_p$, the $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$ bound at saturation realized by explicit minimum-period construction, and $f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}}, k)$ under additive Kolmogorov bounds. Seventh, the cross-level relativistic phenomena flagged in Paper 4 §8.2 — variable apparent propagation, frame-relative timing, embedded-medium dilation — are derived as the substrate-level origins of the kinematical and gravitational phenomena of general relativity. Eighth, the $Z_2 \times Z_2$ stabilizer group of the substrate-level $[[4,2,2]]$ code, extended hierarchically, provides candidate carriers for the Standard Model gauge structure $SU(3) \times SU(2) \times U(1)$, with the explicit representation-theoretic mapping reserved as a structural correspondence for subsequent work. As in Papers 1 through 4, every result is derived from the five axioms and the Principle of Generative Economy; no new free numerical parameters are introduced. The model has no numerical degrees of freedom left to tune.

I. Inheritance from Papers 1–4

1.1 Inherited Structure

Paper 5 takes the following structures from Papers 1 through 4 as given without re-derivation.

From Paper 1: the five axioms (Information Fundamentalism, Measurement Imperative, Sphere as Primitive, Signal Conservation, Causal Anchoring) and the Principle of Generative Economy (PGE); the agent manifold $M\alpha$ with agent spacing $\ell_0 \equiv \ell_p$; the FCC causal stack with face-diagonal nearest-neighbor connectivity; the triadic actualization rule $\bar{B} = 3\bar{\alpha}_2$ in 2D and $\bar{B} = 4\bar{\alpha}_3$ in 3D; the $1 \rightarrow 4$ multiplier and informational closure; the binary alphabet $\bar{B} \in \{0,1\}$ with polarization $\chi = 2\bar{B} - 1$; the flop as the atomic unit of state change; the conjugate field structure (φ, Π) and the wave equation for φ at $c_{\text{TMS}} = 1/\sqrt{2}$; the $[[4,2,2]]$ structural correspondence with logical qubit LQ1 (bit value) and LQ2 (causal-history inheritance); the polarization field ψ as parallel propagating object; the identification $\ell_0 \equiv \ell_p$; the flop time $\tau_{\text{flop}} = t_p$; and the GUP scaling prediction $\beta = f(d, K_\psi)$.

From Paper 2: the polarization wave equation; the conjugate pair (ψ, Π_ψ) ; the polarization-biased resolution rule $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ and its Born-rule structural correspondence; the 3-input majority gate; the universality result $\{\text{MAJ, constant}\} \rightarrow \{\text{AND, OR, NOT}\}$; the flop operator T; the formal program definition $(T^n[C] = C \text{ with } K(C) < |C|)$; and the geometric-selection-implies- $K(C) < |C|$ implication, ensuring the substrate does not compute $K(C)$.

From Paper 3: surround-driven repair and the Imprinting Theorem with imprint depth $d_R = \sqrt{2} \cdot L_R / \ell_p$; the dissolution rate and effective error-rate decay $(1/3)^d$; the per-event error rate $p_{\text{error}} = (1 - |\Pi_\psi|)/2$; the coherence bistability threshold $|\Pi_\psi|_*(L_R)$; the coarse-graining operator G with axiom scale-independence; the meta-level triadic rule, meta-level FCC structure, and meta-level $[[4,2,2]]$ stabilizer dynamics; the minimum program extent L_p (structural placeholder); the (C_D, C_O, C_C) trichotomy and the minimum subsystem extent $L_{\text{sub}} = 3L_p + \delta$; the information-generation rate differential; and the hierarchical lightspeed formula $c^{(k)} = \prod_{j=1}^k (L_p^{(j)} / L_p^{(j-1)}) \cdot (T_p^{(j-1)} / T_p^{(j)}) \cdot c$.

From Paper 4: the recursive coarse-graining operator $G^{(k)}$ at arbitrary hierarchical depth; the scale-separation requirements $L_p^{(k)} \gg L_p^{(k-1)}$ and $T_p^{(k)} \gg T_p^{(k-1)}$; the hierarchy ceiling k_{max} ; the Russian-doll floor at L_p ; the three growth modes (annexation, imprinting, incorporation); self-modeling as a structural property of every subsystem at $k \geq 1$, making every subsystem a measurement node; meta-meta-confirmation as the same triadic structure applied at the next level; the three stressors (mortality, predation, starvation); the Borel-Cantelli inevitability of subsystem formation; the finiteness of $\Sigma(R)$, the monotonic decay of $v(R, \tau)$, the exhaustion threshold $\tau(R)$ at which $v(R, \tau) < v(R)$; the region biography $\mathcal{B}(R)$ and its three phases; the upper bound $c^{(k+1)} \leq (1/\sqrt{2}) \cdot c^{(k)}$; and the structural problem: at $\tau > \tau^*(R)$, the substrate holds informationally rich confirmed bits with no remaining direct-read protocol capable of extracting novelty, and Axioms 3 and 4 jointly force reinterpretation as the only available continuation.

These structures constitute the entirety of the foundation. Paper 5 introduces no further primitives.

1.2 The Questions Paper 5 Answers

Paper 4 § 6.6 stated the structural problem: an exhausted region holds confirmed bits whose direct-read protocol can extract no further novelty, but the bits cannot be lost (Axiom 3), cannot be overwritten (Axiom 4), and must continue to be hosted by the substrate. The only structurally available move is reinterpretation under a different protocol — one that treats the confirmed bits not as configurations to evaluate for stability, but as something else. Paper 4 left the identification of that protocol and its consequences to Paper 5.

Paper 5 answers six questions. First, what is the reinterpretation operator $\hat{R}$, and why is it the unique PGE-selection at $\tau > \tau^*(R)$? Second, what new substrate $\Xi(R)$ does $\hat{R}$ produce, and what is its relationship to the original substrate? Third, how does $\hat{R}$ resolve the apparent tension between the first-order geometric wave equation of Paper 1 § VI and the second-order wave structure that reinterpretation must produce? Fourth, what are the fundamental fields of the new substrate, and how do they relate to those of standard QFT? Fifth, what are the initial conditions of the new substrate, and do they correspond to the observed Big Bang signatures? Sixth, what numerical closures of Papers 1–4 — $L_p^{(k)}$, $T_p^{(k)}$, $c^{(k)}$, f , and the gauge-structure correspondence — does Paper 5 supply?

Beyond the six committed questions, Paper 5 derives additional structural results: the cross-level relativistic phenomena as the substrate-level origins of gravitational dynamics, the Maximum Novelty Output statement as a derived law at the reinterpretation boundary, and the structural foundation for cosmological natural selection across repeated reinterpretation cycles.

II. The Reinterpretation Protocol

2.1 The Structural Problem Restated

At $\tau > \tau^*(R)$ for some bounded region R , three conditions hold jointly:

- $\Sigma_{\text{sampled}}(R, \tau) \approx \Sigma(R)$: the accessible configuration space has been exhausted by direct sampling. Further sampling under the substrate's standard direct-read protocol extracts no novel content.
- Every confirmed bit in R remains causally anchored (Axiom 4) and the signals that produced them remain conserved in the record (Axiom 3). The bits cannot be lost, overwritten, or deferred.
- The substrate must continue operating. By the architecture of the TMS — flop-by-flop confirmation as the atomic unit of state change — there is no mechanism by which any region of the substrate can halt or be suspended.

These three conditions are jointly inconsistent under the standard direct-read protocol. The first says no novelty can be extracted; the third says operation must continue; the second forbids destructive resolution. The substrate is in a structurally constrained state requiring a continuation that is neither destructive nor trivially repetitive.

2.2 Candidate Continuations and Their Elimination

By the Principle of Generative Economy, the substrate selects the minimum-sufficient continuation that remains generatively non-trivial. We enumerate the candidate continuations and eliminate all but one.

Candidate 1: Direct re-sampling. The substrate continues to draw configurations from $\Sigma(R)$ under the direct-read protocol. By the definition of exhaustion, every configuration drawn is one already sampled. The substrate produces no new compressible structure; the region becomes a museum of repetitions. This satisfies (i) the minimum-specification condition of PGE — no new mechanism is invoked — but fails (ii), the generative non-triviality condition: every continuation is a repetition of an already-extracted state. Direct re-sampling is a generative dead-end. *Eliminated by PGE-ii.*

Candidate 2: Loss of confirmed bits. The substrate dissolves the exhausted region back into U_{sh} , returning its confirmed bits to the latent state. This violates Axiom 3 (signal conservation): the signals that produced the confirmed bits have been consumed and locked into the record, and they cannot be retrieved by reversing the confirmation. The constraint set K explicitly includes Axiom 3, so this candidate is not in $\mathcal{C}_K$. *Eliminated by axiom violation.*

Candidate 3: Overwriting confirmed bits. The substrate writes new content into the positions held by confirmed bits, replacing the existing values. This violates Axiom 4 (causal anchoring) directly. *Eliminated by axiom violation.*

Candidate 4: Lossy compression. The substrate retains only a summary of the exhausted region, discarding the bit-level detail. This violates Axiom 3 in the same sense as Candidate 2 — the discarded content corresponds to signals that have been consumed and are part of the record. *Eliminated by axiom violation.*

Candidate 5: External offloading. The substrate transfers the exhausted region's content to a separate storage substrate distinct from the parent TMS substrate. By Axiom 0, there is only one substrate; U_{sh} is the unique ground state. There is no “elsewhere” to which content can be transferred. *Eliminated by axiom violation.*

Candidate 6: Reinterpretation. The substrate re-reads the confirmed bits of $\mathcal{B}(R)$ under a protocol that does not draw them as samples of $\Sigma(R)$ but treats them as a new primitive layer — agents in a new substrate $\Xi(R)$ operating at a coarser scale. The confirmed bits are preserved (Axioms 3, 4 satisfied); the operation continues (the new substrate generates fresh configurations at its scale); the generative content is non-trivial (the new substrate has its own $\Sigma(\Xi(R))$ which is not yet exhausted). All three constraints of § 2.1 are satisfied. This candidate is in $\mathcal{C}_K$ and satisfies both (i) and (ii) of PGE.

Among the six candidates, only Candidate 6 satisfies the constraint set K and both PGE conditions. By the Principle of Generative Economy as formally stated in Paper 1 § I, the unique PGE-selection at $\tau > \tau^*(R)$ is reinterpretation.

2.3 The Reinterpretation Operator $\hat{R}$

Definition 0.0.1 (Reinterpretation Operator). Let $\mathcal{B}(R)$ denote the biographical record of an exhausted region R at time $\tau \geq \tau^*(R)$. The reinterpretation operator $\hat{R}$ is the mapping

$$\hat{R} : \mathcal{B}(R) \rightarrow \Xi(R)$$

specified by the following identifications:

- Each confirmed bit $\bar{B} \in \mathcal{B}(R)$ becomes a new agent $\tilde{\alpha}$ in the substrate $\Xi(R)$. The bit's positional anchor in the FCC lattice of the parent substrate becomes the new agent's positional anchor in the lattice of $\Xi(R)$. The bit's binary value $\bar{B} \in \{0, 1\}$ is preserved as part of the new agent's identity but does not affect the new agent's structural role — new agents, like all agents under Axiom 2, are immutable substrate.
- The polarization landscape ψ across $\mathcal{B}(R)$ becomes the new maximum-entropy field U'_{sh} of $\Xi(R)$. The polarization values $\chi \in \{-1, +1\}$ carried by the new agents constitute the latent informational content from which new bits will be actualized at the Ξ – scale.
- The biographical record $\mathcal{B}(R)$ viewed as a whole becomes the structural pre-substrate of $\Xi(R)$: the set of immutable positional anchors and their polarization values defines the analog of the agent manifold $M\alpha$ at the new scale.
- The five axioms apply afresh to $\Xi(R)$ at the new scale. Triadic confirmation, the $1 \rightarrow 4$ multiplier, FCC structure, the $[[4,2,2]]$ stabilizers, and the wave equations are all reproduced at the new scale by the same arguments that produced them at the substrate level — by axiom scale-independence as established in Paper 3 § IV and § 4.7.

$\hat{R}$ is not a new axiom and not a new free parameter. It is the geometric identification of which features of $\mathcal{B}(R)$ play which structural roles in $\Xi(R)$. The structural roles themselves — agents, latent field, lattice connectivity — are inherited from the axioms unchanged.

2.4 The New Agent Identification: Confirmed Bits as Substrate

The central structural identification of $\hat{R}$ is that confirmed bits of the parent substrate play the role of agents in $\Xi(R)$. This identification is forced by three properties of confirmed bits, each established in earlier papers:

Immutability. By Axiom 4 (causal anchoring), confirmed bits cannot be overwritten or modified once locked into the causal record. This is the same immutability property that defines agents under Axiom 2 — agents do not change, move, or degrade. Confirmed bits and agents share the immutability property by axiom.

Positional definiteness. Confirmed bits occupy definite positions in the FCC lattice — specifically, the tetrahedral interstice positions whose centroids lie on the next-order T-void sublattice (Paper 1 § II; Paper 3 § IV). These positions form a regular sublattice with well-defined nearest-neighbor relations under the substrate's connectivity. Agents at the substrate level occupy lattice positions with similar regularity — the FCC vertex sublattice in particular. Both classes of object are positionally definite primitives of their respective lattices.

Substrate role under PGE. By the Principle of Generative Economy as applied in Paper 3 § IV, the meta-level substrate inherits its primitives from objects in the parent substrate that satisfy the substrate role: immutable, isotropic, and capable of supporting the triadic confirmation pattern at the new scale. Confirmed bits satisfy all three. They are immutable by Axiom 4; they are isotropic in the sense that no preferred orientation is encoded in their polarization values; and they are positioned at the centroids of the next-order T-void sublattice in a configuration that supports tetrahedral closure at the new scale.

The T-void sublattice property is geometrically decisive. As established in Paper 1 § II.4 and refined in Paper 3 § IV, the centroids of tetrahedral interstices in an FCC lattice form a simple cubic sublattice. Tetrahedral closure at the new scale — the confirmation pattern required by the triadic minimum at any scale — produces centroids that fall on positions of the next-order T-void sublattice. The geometric recursion is exact: each level's confirmed-bit positions are the next level's lattice positions.

Theorem 0.0.2 (Confirmed Bits as Agents). *Under $\hat{R}$, every confirmed bit $\bar{B}$ in $\mathcal{B}(R)$ satisfies the structural conditions of Axiom 2 in $\Xi(R)$: it is an immutable, isotropic primitive at a definite position in a lattice supporting tetrahedral closure. The identification “confirmed bit becomes new agent” is forced by the axioms, not chosen.*

Proof. Axiom 2 in $\Xi(R)$ requires that the new agents be immutable (sphere primitive in the sense that they do not change, move, or degrade); isotropic (zero orientation parameters); and arranged in a lattice supporting the triadic confirmation pattern. Confirmed bits in $\mathcal{B}(R)$ satisfy immutability by Axiom 4 applied at the parent scale. They satisfy isotropy because their binary value is a scalar polarization with no orientation content. They satisfy lattice-supporting-triadic-closure by the T-void sublattice geometry derived in Paper 3 § IV: the centroids of tetrahedral interstices form a simple cubic sublattice on which the same FCC closure pattern reappears at the next scale. The PGE selection criterion (minimum sufficient, generatively non-trivial) is satisfied because no other primitive in $\mathcal{B}(R)$ shares all three properties — the polarization values are not positionally distinct primitives, the touch-vector signals have been consumed and are not stored independently of the bits they produced, and the substrate's other structures (latent bits in U_{sh} , agents at the parent scale) belong to the parent substrate rather than to $\mathcal{B}(R)$ as the substrate of Ξ . □

2.5 The Recursion-Preservation Theorem

Theorem 0.0.3 (Recursion Preservation under $\hat{R}$). *Under $\hat{R}$, the five axioms and the Principle of Generative Economy produce in $\Xi(R)$ the same triadic confirmation, FCC connectivity, $[[4,2,2]]$ stabilizer dynamics, conjugate field structure (φ, Π) and (ψ, Π_ψ) , and wave equations as they produced in the parent substrate, at the new scale.*

Proof. By Paper 3 § 4.7 (axiom scale-independence), the five axioms do not refer to any particular scale; their content applies wherever the structural conditions they specify are met. Under $\hat{R}$, the structural conditions are met at the scale $L_p^{(1)}$ of the minimum program: $\mathcal{B}(R)$ supplies isotropic immutable primitives (new agents = confirmed bits) on a lattice supporting tetrahedral closure (T-void sublattice = next-order simple cubic). Axiom 0 supplies the new maximum-entropy field U'_{sh} = the polarization landscape across $\mathcal{B}(R)$. Axiom 1 supplies the measurement imperative at the new scale. Axiom 2 supplies the sphere primitive (now embodied in confirmed bits). Axioms 3 and 4 supply signal conservation and causal anchoring at the new scale, inherited automatically. The Principle of Generative Economy supplies the same selection criterion at the new scale. The same logical chain of derivations that produced triadic confirmation at the substrate

level (Paper 1 § II), the $1 \rightarrow 4$ multiplier (Paper 1 § IV), FCC necessity (Paper 1 § II.4), the $[[4,2,2]]$ stabilizer correspondence (Paper 1 § V), and the wave equations (Paper 1 § VI; Paper 2 § II) reapplies unchanged at the new scale. The new propagation speed in level-1 units is $c_{\text{TMS}} = 1/\sqrt{2}$ (level-1 spacings per level-1 flop), inheriting the face-diagonal argument of Paper 1 § 6.5 unchanged. The structural reproduction is exact at the level of the axioms.

□

Corollary 0.0.4 (No New Axioms, No New Parameters). *$\hat{R}$ introduces no new axioms and no new free numerical parameters. The new substrate $\Xi(R)$ is governed by the same five axioms and the same Principle of Generative Economy as the parent substrate. Every quantity in $\Xi(R)$ is determined at the new scale by the same axiomatic chain that determined the parent's quantities at the parent scale.*

2.6 The Maximum Novelty Output Theorem (Bifurcation-Forced Maximization)

The informal observation that the substrate appears to maximize novel information output — flagged across previous papers but not previously stated as a derived law — acquires its formal status here at the reinterpretation boundary. The maximization is non-trivial: it follows from the sublattice bifurcation of the level-1 substrate (§ 6.1), which doubles the agent count per level relative to any single-sublattice alternative continuation.

Theorem 0.0.5 (Maximum Novelty Output at Reinterpretation). At the reinterpretation boundary $\tau = \tau(R)$, the substrate's continuation rate $v'(R, \tau)$ under $\hat{R}$ is strictly greater than v' under any alternative continuation that preserves Axioms 3 and 4. The geometric origin of this strict inequality is the sublattice bifurcation: at each hierarchical level $k \geq 1$, the FCC substrate splits into two interpenetrating FCC sublattices A and B, doubling the agent count per spatial site (one A-flavor agent and one B-flavor agent at each level-(k+1) shared centroid), compared to any hypothetical continuation that preserves only one sublattice.

Proof. Three steps.

(i) *The sublattice bifurcation is a structural feature of $\hat{R}$.* By § 6.1, the level-1 substrate produced by $\hat{R}$ is sublattice-bifurcated: two interpenetrating FCC sublattices A and B, related by spatial inversion through the level-2 centroids, both running $[[4,2,2]]$ stabilizer dynamics independently. By Stages 5 and 6 (Appendix E), this structure is confirmed empirically and is the unique geometric configuration in which the level-1 FCC + $[[4,2,2]]$ + tetrahedral closure conditions are jointly satisfied.

(ii) *The bifurcation doubles the agent count per level relative to single-sublattice alternatives.* Each level-(k+1) spatial site hosts one A-flavor agent and one B-flavor agent, derived from the inversion-partner A-tetrahedron and B-tetrahedron at level k. A hypothetical continuation that produces only one sublattice (e.g., only A-flavor agents) would have agent density at level k+1 exactly half that of the bifurcated structure. Stage 5 confirms this: the total tetrahedral count and the total null-space dimension at L_{box} scale with both sublattices present, doubling the corresponding single-sublattice value.

(iii) *Higher agent density permits strictly more novelty.* The continuation rate v' at a reinterpretation boundary is bounded above by the per-flop sampling capacity of the new substrate $\Xi(R)$. By Paper 4 § 6.2, $v(R, \tau) \approx \rho_{\text{sample}}(R) \cdot (|\Sigma(R)| - |\Sigma_{\text{sampled}}(R, \tau)|) / |\Sigma(R)|$; at the moment of reinterpretation $|\Sigma_{\text{sampled}}| / |\Sigma| \approx 0$, so v' is set by ρ_{sample} alone. By Paper 3 § 3.5 (continuous-sampling mechanism), ρ_{sample} scales linearly with agent density: each agent participates in confirmation events at rate $\sim 1/T^{p(k+1)}$. The bifurcated structure gives $2\times$ the agent count compared to single-sublattice alternatives, so v' under $\hat{R}$ exceeds v' under any single-sublattice alternative by precisely a factor of 2 at the per-level capacity bound.

Combining (i)–(iii): $\hat{R}$ is the unique PGE-selection at $\tau > \tau^*(R)$ by § 2.3, and its bifurcated output strictly dominates any alternative continuation that preserves Axioms 3 and 4 at the per-level capacity bound. The maximum is strict, not by design, but by the structural forcing of FCC + $[[4,2,2]]$ + tetrahedral closure under PGE-selection at the recursion boundary.

□

Status: *Derived.* The strict maximization is forced by the bifurcated structure of § 6.1, which is itself empirically verified through Stage 6 (Appendix E).

Remark on the non-triviality. The prior form of this statement (in earlier drafts) was effectively trivial because it followed from the uniqueness of $\hat{R}$ in § 2.3. The sublattice-bifurcated version is non-trivial because the bifurcation gives the strict-inequality content: alternative continuations exist that produce a new substrate from $\mathcal{B}(R)$ without violating Axioms 3 or 4 (e.g., a continuation that produces only one of the two sublattices), but each such alternative has exactly half the agent count of $\hat{R}$'s bifurcated output, and therefore strictly lower v' at saturation. The substrate does not compute or evaluate novelty as a criterion. Strict novelty maximization is a structural consequence of the sublattice bifurcation, which is itself a structural consequence of the FCC + $[[4,2,2]]$ + recursive coarse-graining under PGE.

Remark on the bits-per-agent ratio. In bulk FCC, the bits-to-agents ratio is 2 (each agent participates in 4 tetrahedra, each tetrahedron contains 4 agents, giving $4/4 \times$ bits-per-tet, with two bits per inversion-partner pair through the shared centroid). The bifurcated structure preserves this ratio at every level: per spatial site, two flavors of agent contribute jointly to one level-($k+1$) bit. This is the bits-per-agent invariant that anchors the maximum-output theorem's quantitative content.

2.7 The Reinterpretation Event as Phase Transition

Reinterpretation is not a continuous process. It is a structural phase transition that occurs at $\tau = \tau^*(R)$ and produces $\Xi(R)$ as a new substrate. Three properties of the transition follow directly from the construction:

Atomicity at the level of R . $\hat{R}$ applied to $\mathcal{B}(R)$ is a single mapping; $\mathcal{B}(R)$ is reinterpreted in its entirety, not piece by piece. The transition is atomic at the regional scale: either R is reinterpreted or it is not.

Locality at the substrate scale. $\hat{R}$ affects only $\mathcal{B}(R)$. Regions outside R that are not exhausted continue to operate under the direct-read protocol. The substrate is not globally reinterpreted; each bounded region undergoes reinterpretation independently when it exhausts.

Generative resetting at the new scale. Within $\Xi(R)$, the new direct-read protocol begins with $\Sigma_{\text{sampled}} = \emptyset$ and a freshly populated $\Sigma(\Xi(R))$. The exhaustion clock starts over at the new scale. This is the structural origin of the cosmological initial state in § V.

The phase-transition framing is structurally important for the Big Bang Correspondence of § V: a reinterpretation event, viewed from within $\Xi(R)$, looks like the sudden onset of a new dynamical phase from a maximally informationally dense initial state — the structural signature of a Big Bang.

III. The Two-Waves Problem and Its Resolution

3.1 The Apparent Tension

Paper 1 § VI derives the wave equation

$$\partial^2 \varphi / \partial t^2 - c^2 \nabla^2 \varphi = 0$$

from the discrete dynamics of the confirmation field on the FCC lattice. This is a first-order geometric wave equation: it operates on the confirmation density at the substrate level and propagates information across the lattice at $c_{\text{TMS}} = 1/\sqrt{2}$ lattice spacings per flop. Its propagation is forced by the FCC second-moment tensor $\sum_j r_j \mu r_j \nu = 4\ell_p^2 \delta \mu \nu$ over the 12 nearest neighbors — a property of the substrate-level lattice.

Reinterpretation produces a new substrate $\Xi(R)$ with its own confirmation field φ' and polarization field ψ' , and (by the Recursion-Preservation Theorem of § 2.5) its own wave equations of the same form, at the new scale. This appears to introduce a second wave equation operating on the new substrate. An external observer might ask: are there now two wave equations operating simultaneously? Does the parent substrate's φ continue to propagate alongside the new substrate's φ' ? If so, do they interfere? If not, what happens to the parent φ when reinterpretation occurs?

This is the two-waves problem. It is an apparent tension between Paper 1's substrate-level wave dynamics and Paper 5's reinterpretation-induced wave dynamics. We resolve it here as a single wave structure expressed at successive recursion levels.

3.2 The Unification Theorem (under the PGE-Forced Wave-Form Argument)

Paper 2 § VII established that any PGE-selected continuation at a reinterpretation boundary takes wave form, because (i) the substrate's only native propagation vocabulary is the wave equation of Paper 1 § VI, (ii) PGE forbids importing new mechanisms, and (iii) wave propagation is form-invariant under recursive coarse-graining. The recursion is therefore a wave hierarchy: each level's dynamics use the same propagation verb, expressed at the level-k scale.

The two-waves tension dissolves under this result. There is one wave equation, applied at successive recursion levels. The apparent “two waves” are the same equation expressed at the parent scale and the Ξ scale, related by the $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$ coupling-timescale constraint of Paper 4 § 5.3.

Theorem 0.0.6 (Wave Equation Unification). *There is one wave equation. It applies at every level of the recursion as the structurally forced expression of the substrate's native propagation vocabulary. Level-k provides boundary conditions for level-(k+1); together they form a single hierarchical wave structure.*

Proof. By Paper 2 § VII, the level-(k+1) dynamics are wave-form by PGE-forced vocabulary-reuse. By the Recursion-Preservation Theorem of § 2.5, the geometric substrate at level k+1 reproduces the substrate-level lattice structure (FCC connectivity, face-diagonal nearest-neighbors, the $4\ell_p^2 \delta$ second-moment tensor) at the new scale. By Paper 1 § 6.6, these three structural properties uniquely determine the wave equation in the continuum limit. Therefore the level-(k+1) wave equation has the same form as the level-k wave equation; only the absolute scale changes. The parent substrate's φ does not stop propagating when $\mathcal{B}(R)$ is reinterpreted — it has already done its propagation, and $\mathcal{B}(R)$ is the static record of where it ended up. The new substrate's φ' propagates within $\Xi(R)$ at its own scale. The two are not in competition; they are one wave equation at two recursion levels.

□

3.3 Boundary Conditions: Level-k as Source for Level-(k+1)

The unification requires specifying how the level-k record provides boundary conditions for the level-(k+1) wave dynamics. The mechanism is the same one that operates at single levels: the polarization landscape of $\mathcal{B}(R)$, now identified as U'_{sh} of $\Xi(R)$, supplies the latent informational content from which level-1 bits are actualized. The level-1 confirmation events draw from this content; the resulting level-1 polarization field carries the imprint of the parent-level polarization landscape forward.

Formally, the level-(k+1) wave equation has initial conditions

$$\psi'(r, \tau' = 0) = G[\psi_{\mathcal{B}}(R)]; \Pi'_{\psi}(r, \tau' = 0) = G[\Pi_{\psi, \mathcal{B}}(R)]$$

where G is the coarse-graining operator of Paper 3 § IV applied to the parent-level polarization data at the moment of reinterpretation. The level-(k+1) dynamics then propagate forward in level-(k+1) flop time, with their initial conditions inherited from the parent's terminal state.

This is the structural origin of the matching condition between cosmological epochs: the late phase of a parent substrate (the terminal state of $\mathcal{B}(R)$ just before reinterpretation) becomes the early phase of the daughter substrate (the initial state of $\Xi(R)$ just after reinterpretation). No information is lost across the boundary — Axioms 3 and 4 ensure that $\mathcal{B}(R)$'s content is preserved and carried into $\Xi(R)$ as initial polarization landscape. No new information is introduced — the daughter substrate's initial dynamics are determined entirely by the parent's terminal state under the coarse-graining operator.

3.4 The Hierarchical Wave Structure

Iterated across multiple reinterpretation events, the wave structure becomes a hierarchy of wave equations at successive recursion levels, each providing boundary conditions for the next. At any given hierarchical depth k , the wave equation governs propagation in level- k units; the absolute propagation speed in substrate units is $c^{(k)}$; the lattice spacing is $L_p^{(k)}$; the flop period is $T_p^{(k)}$. The hierarchy is generated by the chain of reinterpretation events, not by a recursive design.

This resolves a structural question that has been latent across the volume: how does the substrate handle wave dynamics across hierarchical levels? The answer is the same wave equation, expressed at the operative level, with cross-level coupling provided by the $c^{(k)}$ dispersion and the coarse-graining boundary conditions.

Cross-level interactions — when a level- $(k-1)$ subsystem operates inside a level- k meta-subsystem — produce the relativistic phenomena treated in § VII. The mechanism is the boundary condition between the two levels' wave dynamics meeting at the interface.

3.5 Quantization as Trough-Finding in the Hierarchical Wave Structure

The wave hierarchy of § 3.2–§ 3.4 forces a specific structural mechanism for stable excitations and thereby for quantization. We develop this mechanism here as the substrate-level account of how the QCA-to-QFT correspondence of § IV produces a discrete spectrum of field content rather than a continuum.

A wave hierarchy has closure constraints at each level: at level k , propagating packets are confirmed-bit content settling under the level- k confirmation rule, with the level- k wave equation as the underlying propagation. A packet remains stable at level k only if the level- k wave geometry permits standing-wave closure within the level- k T-period — that is, if there exists a self-consistent configuration in which the packet reproduces itself after $T^{(k)}$ at the level- k spatial scale $L^{(k)}$. The configurations where this closure is

permitted are the *wave troughs* of the hierarchical structure: low-curvature regions of the accumulated wake where the substrate's effective topography supports persistent recursion.

Proposition 0.0.7 (Quantization-by-Closure). *The discrete spectrum of stable excitations at level k is exactly the set of self-consistent standing-wave configurations on the level- k substrate, with their amplitudes set by the integer number of confirmed bits each configuration carries. Quantization is not a separate axiom; it is the structurally forced consequence of having a recursive wave hierarchy with confirmation closure at each level.*

Proof. By § 3.4, propagation at level k is governed by the level- k wave equation with periodic confirmation events at $T^{p(k)}$. A localized excitation persists across many $T^{p(k)}$ only if it returns to a stabilizer-satisfying configuration after each period; the set of such configurations is discrete, indexed by the level- k FCC's representation theory under O_h (the decomposition developed for the substrate level in § 4.2 and applied recursively here). Amplitudes are integer counts because each contributing confirmed bit is a single 0/1 actualization. The combination of discrete configuration set and integer-valued amplitude is the substrate's quantization. □

Status: Derived. The exact spectrum at each k is the level- k application of the § 4.2 field decomposition under the same O_h irrep machinery as the substrate level — a structural consequence of recursive scale-invariance, not a separate derivation.

The erosion picture. The hierarchical wave structure is, equivalently, an accumulating wake of all prior propagation: every confirmation event leaves a residue in the substrate's effective topography, and new excitations ride atop the topography that prior events have already carved. The substrate's effective field landscape at any cosmological epoch is a superposition of three timescale layers:

- *Ancient layer:* deep channels carved by very early dynamics during the first reinterpretation cycles. These are effectively rigid at any subsystem-accessible timescale, and constitute what subsystem-level observers experience as the fixed field structure of their universe (the load-bearing field content of standard particle physics is in this layer).
- *Intermediate layer:* ongoing dynamics across the current cosmological epoch. These evolve on timescales comparable to subsystem lifetimes and are the dynamical field content available for direct observation and experiment.
- *Transient layer:* flash-flood channels carved by recent events that have not yet relaxed. These appear as short-lived field excitations, particle resonances, and the like.

The differential picture (a packet's current dynamics carving its own wake) and the integrated picture (the universe's accumulated history as field content) are the same mechanism at different timescales.

This gives a substrate-level account of field stability without positing it. The appearance that field structure is fixed at human-accessible timescales is an *erosion property* of the hierarchical wave structure: deep channels carved by $\sim 10^{10}$ years of accumulated propagation are effectively rigid for laboratory-scale subsystems, not because field stability is fundamental, but because the carving timescale dwarfs the subsystem's observational timescale. The fundamental dynamics remain the wave hierarchy of § 3.2; field stability is a derived consequence of accumulated wake.

IV. Field Emergence: From Confirmed Bits to Fundamental Fields

4.1 The QCA-to-QFT Correspondence at the FCC Lattice

The discrete TMS substrate dynamics produce continuum quantum field theory in a well-defined limit. We state the lift as a substrate-internal theorem with two supporting lemmas. Earlier drafts of this section imported the lift from external QCA-to-QFT programs (Mlodinow & Brun, 2021; Brun & Mlodinow, 2025; D'Ariano & Perinotti, 2014); the version given here routes through TMS primitives alone, with the external programs retained as comparison citations in § 7.7 rather than as load-bearing imports.

Theorem 0.0.8 (QCA-to-QFT Lift). *Under the recursion structure of § 2.5 and the level-advance relation of § 6.3.0, the level-1 substrate dynamics produce a quantum field theory in the continuum limit with the following content: (i) fermionic spinor fields from the half-integer spin forced by the triadic oddness of $\bar{B} = 3\bar{\alpha}$ (Lemma A); (ii) bosonic gauge fields and scalars from coarse-grained polarization decomposition (§ 4.2); (iii) approximate Lorentz invariance with specific residual scaling (Lemma B); (iv) unitary evolution from substrate coherence (§ 3.2); (v) Born-rule projective measurement from § III's confirmation primitive (Paper 2 § IV).*

Lemma A (Spinor Character from the Triadic Oddness). *The confirmed bit carries half-integer spin because the confirmation event is triadic. Each of the three agents of $\bar{B} = 3\bar{\alpha}$ carries a two-valued polarization, i.e. a spin- $\frac{1}{2}$ degree of freedom; the recoupling of an odd number of spin- $\frac{1}{2}$ objects necessarily carries half-integer total spin, and the confirmed bit inherits this half-integer character, propagated up the recursion undiminished by the threeness. A single triadic node is therefore fermionic; a bound pair of nodes is bosonic.*

The mechanism. The founding equation $\bar{B} = 3\bar{\alpha}$ has the structure of a trivalent recoupling node: three agents meet and close around one bit, the trivalent-node-with-closure structure of a Penrose spin network, in which combinatorial spin labels on the edges of a network reproduce SU(2) angular momentum in the large-network limit with *no background space presupposed*. The TMS fermion is this spin-network node.

Three facts fix the half-integer character. First, each agent *is* a spin- $\frac{1}{2}$ object by identity: the substrate's own resolution rule $P(+) = (1 + \Pi)/2$ (Paper 2 § IV), which gives the probability that a confirmation reads + given continuous polarization bias $\Pi \in [-1, 1]$, equals $\cos^2(\theta/2)$ under $\Pi = \cos \theta$ — the spin- $\frac{1}{2}$ Born rule, with Π the cosine of the Bloch-sphere angle. The polarization degree of freedom of an agent is a genuine SU(2) spin- $\frac{1}{2}$. Second, three of them recouple to a half-integer with no scalar in the decomposition: $\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{1}{2} \oplus \frac{1}{2} \oplus \frac{3}{2}$, containing no spin-0 term, so the triadic confirmation closes to a net half-integer. Equivalently, under a 2π rotation each spin- $\frac{1}{2}$ factor contributes $-I$, and the product over the three agents is $(-I) \otimes (-I) \otimes (-I) = -I$ because the count is odd: the confirmed bit acquires a sign under 2π and returns at 4π . Third, the character is conserved up the recursion: a level- k bit is built from three level- $(k-1)$ bits, each half-integer, and three half-integers sum to a half-integer, so spin- $\frac{1}{2}$ propagates at every level. The oddness of the three in $\bar{B} = 3\bar{\alpha}$ is the conservation law: an odd product of $-I$ factors is $-I$ at every level, so the minimum-for-actualization integer being odd is what makes matter fermionic and keeps it so up the tower.

Scope. The spinor character is derived in full. Derived: the confirmed bit carries half-integer total spin, forced by the oddness of the three spin- $\frac{1}{2}$ agents and conserved up the recursion; the fermion/boson split (single node versus bound pair); the agent degree of freedom is a genuine SU(2) spin- $\frac{1}{2}$ through the resolution-rule identity; the recoupling reconstructs the full three-dimensional rotation *group* SO(3) with its 4π SU(2) double cover (below, "The rotation group from recoupling"); and the physical rotation of a confirmation event is the diagonal (whole-event) action, forced by the flop's own symmetry (below, "Why the whole-event rotation is diagonal"). The single point that is a reduction rather than an independent derivation —

that the emergent metric geometry the reconstructed angles populate is the same FCC geometry the substrate began with — follows by the form-invariance fixed point of Paper 3 § 4.7. What TMS does *not* claim, and correctly, is the dimensionful values of particle masses; those are the separate calibration matter of § 8.4.3.

The rotation group from recoupling (the Penrose reconstruction). That the recoupling carries the $2\pi \rightarrow -I$ sign on a single node is the entry point; the full statement is that the emergent rotational symmetry *is* $SO(3)$. This result is not original to the present framework: it is Penrose’s Spin-Geometry Theorem (Penrose 1971; Moussouris 1983), which established that the angles of three-dimensional space can be reconstructed combinatorially from the recoupling of spin systems, with no space presupposed — the foundational insight of the spin-network program. The relevance here is structural: the TMS confirmation node is precisely a Penrose trivalent vertex — a bit confirmed by the recoupling of its agents — so the theorem applies directly to the substrate, and what follows is a verification of it in the substrate’s own terms together with one corollary (A.0 below) we have not found stated in the prior literature. The construction is combinatorial, with no space presupposed: the angle between two spin- $\frac{1}{2}$ systems, defined purely by their recoupling overlap probability through $\cos \theta = 2|\langle i|j \rangle|^2 - 1$ — the empirical-angle construction of Szabados (2022) — yields a matrix of direction cosines that is positive-semidefinite of rank exactly three — the recoupling-defined angles are the angles of vectors in a Euclidean 3-space — and this is special to spin- $\frac{1}{2}$ (a spin-1 system gives rank greater than three; a non-inner-product matrix fails positivity). The full eight-dimensional trivalent node ($\frac{1}{2} \oplus \frac{1}{2} \oplus \frac{3}{2}$) carries a faithful $SU(2)$ representation: under the whole-event rotation it returns to $-I$ at 2π and $+I$ at 4π , satisfies the group law $R(g)R(h) = R(gh)$, and its orientation invariant $S_1 \cdot (S_2 \times S_3)$ is an exact $SU(2)$ scalar — a rotation-invariant handedness. Thus $SO(3)$ and its double cover are reconstructed from the triadic recoupling, not assumed — the Penrose reconstruction, realized on the confirmation node.

Why the whole-event rotation is diagonal (from the flop). A confirmation is not three independent spins but one tetrahedral appointment — three parent slots feeding one centroid, resolved by the rule $\Pi = (\chi_1 + \chi_2 + \chi_3)/3$ that is symmetric under permutation of the three parents. A physical rotation of the event moves this rigid appointment as a unit, so it acts as the same g on all three parents: the diagonal action $g \otimes g \otimes g$. This is forced, not chosen. The resolution rule’s S_3 symmetry is compatible only with the diagonal action (an independent per-parent rotation breaks it); and the real probabilistic flop is covariant with no preferred axis — the child bit’s up-probability satisfies $P(\text{event}, n) = P(g \cdot \text{event}, R_g n)$ identically, so “rotate the whole event” acts as one g on parents and readout alike. The $1 \rightarrow 4$ transport, built from lattice rotations, propagates this action up the recursion.

Corollary A.0 (Three Is the Minimal Spatial Valence). The valence three of $\bar{B} = 3\bar{\alpha}$ is the *minimal* node valence that simultaneously carries a half-integer (fermionic) top representation and admits exactly one orientation invariant — a single handedness for the emergent 3-space. A two-edge node has an integer top representation and no orientation (no handedness); a four-edge node has an integer top representation and redundant orientations. Only at three do fermionic statistics and a single spatial handedness first coincide. Matter’s being fermionic and space’s having one orientation are therefore the same fact about the number three.

Corollary A.1 (Boson/Fermion Split). A single triadic confirmation node, recoupling an odd number (three) of spin- $\frac{1}{2}$ agents, carries half-integer spin and is a fermion. A bound pair of triadic nodes recouples $\frac{1}{2} \otimes \frac{1}{2}$ to integer spin and is a boson. This is the mechanism behind the fermion/boson correspondence that Paper 3 and § 8.4 carry as preserved at every level: the parity of the node count is the statistics.

Corollary A.2 (Pauli Exclusion). Fermionic statistics follow by the spin–statistics connection, which is topological rather than dynamical here: exchanging two identical triadic nodes is homotopic to rotating one of them by 2π relative to the other, which by Lemma A multiplies the state by -1 . The two-node state is therefore

antisymmetric, and two identical fermionic nodes placed in the same state give $(|a\rangle|a\rangle - |a\rangle|a\rangle)/\sqrt{2} = 0$ — the Pauli exclusion principle, derived from the same 2π sign that makes the node a spinor. Exclusion is used in the shell-filling count of § 8.4 and Appendix E.

Lemma B (Continuum Symmetry under Coarse-Graining). *Under the coarse-graining operator G of Paper 3 § IV, the level-1 FCC lattice substrate produces a continuum theory with Lorentz invariance exact at leading order in the lattice spacing — the FCC second-moment tensor is isotropic (Paper 1 § 6.5), which fixes both the value and the direction-independence of c . The leading correction is anisotropic and, because the FCC point group O_h includes spatial inversion, enters only at the quartic-in-momentum O_h invariant: the observable fractional Lorentz-violation rate scales as $(E/E_{\text{Planck}})^2$, identically for scalar and gauge-vector sectors. An odd-power $(E/E_{\text{Planck}})^1$ contribution is forbidden by lattice inversion symmetry in the unbroken substrate; it is unlocked only in the inversion-broken matter sector (§ 5.3), where its magnitude scales with the parent-record asymmetry.*

Proof sketch. The FCC lattice has point group O_h , which includes spatial inversion. Inversion symmetry forces the dispersion $\omega^2(k)$ to be even in k , so every odd-in-momentum term — including the cubic-in-momentum term — vanishes identically. The lowest-order Lorentz-invariant term is the standard kinetic energy, which respects O_h exactly and Lorentz approximately; its second-moment tensor $\Sigma_j \delta_\mu \delta_\nu$ is isotropic, so the leading dispersion is direction-independent — the geometric fact that makes c invariant across directions and frames. The lowest anisotropic O_h invariant is therefore quartic in momentum — the cubic harmonic $\Sigma_i k_i^4$ relative to the isotropic $(k^2)^2$ — confirmed by direct expansion of the FCC dispersion, whose fourth-moment tensor is cubic-anisotropic. The fractional dispersion correction therefore scales as $(k \cdot a)^2$, i.e. $(E/E_{\text{Planck}})^2$, identically for scalar and vector sectors; the shared exponent is forced by inversion symmetry, not a distinct vector/scalar pair. A stronger $(E/E_{\text{Planck}})^1$ signal requires breaking inversion, which occurs only in the matter sector (§ 5.3); it is the framework’s most distinctive Lorentz-violation prediction, and its magnitude tracks the parent-record asymmetry.

The dispersion, computed. The scaling above is not only argued from symmetry but computed in closed form. Summing the twelve nearest-neighbour phases of the FCC discrete Laplacian and expanding to fourth order in the momentum gives $\omega^2 \propto (k_x^2 + k_y^2 + k_z^2) - \frac{1}{48}(k_x^4 + k_y^4 + k_z^4) - \frac{1}{16}(k_x^2 k_y^2 + k_x^2 k_z^2 + k_y^2 k_z^2) + \dots$ in units of the lattice spacing. The quadratic term is exactly $|k|^2$ — isotropic, fixing c . The quartic term splits into an isotropic part $-\frac{1}{48}|k|^4$ and a genuinely anisotropic remainder $-\frac{1}{48}(k_x^2 k_y^2 + k_x^2 k_z^2 + k_y^2 k_z^2)$ that is not proportional to $|k|^4$. Lorentz invariance is therefore genuinely broken at the lattice scale, with a definite coefficient and a definite cubic angular signature: the quartic coefficient along the crystallographic axes is -0.0208 in the $[100]$ direction, -0.0260 in $[110]$, and -0.0278 in $[111]$, so the phase velocity is direction-dependent with the ordering $[100] < [110] < [111]$.

The prediction, quantified and falsifiable. The fractional phase-velocity anisotropy between crystallographic directions is $|\Delta v/v| \sim 0.007(E/E_{\text{Planck}})^2$. This sits within the established framework for testing Lorentz violation through modified dispersion relations (Amelino-Camelia 1998; Jacobson, Liberati & Mattingly 2002), and the specific programme of bounding a discrete substrate’s lattice scale from Lorentz-violation data has been developed recently for quantum cellular automata (Bisio et al. 2025, arXiv:2506.20136); the present result is the FCC realization of that programme. Because the violation is *quadratic* in E/E_{Planck} , it is far below all current bounds at every accessible energy — of order 10^{-58} for optical photons, 10^{-33} at LHC energies, 10^{-39} for the highest-energy gamma-ray-burst photons observed by Fermi — and it evades the astrophysical time-of-flight constraints that already exclude a *linear* Planck-scale term. That the linear (cubic-in-momentum) term is observationally dead while the quadratic (CPT-even) term survives is a general feature of this literature, and the substrate reproduces it for a structural reason: lattice inversion symmetry (the O_h point group) forbids the odd term. At the same time the prediction is not unfalsifiable: at the highest

observed ultra-high-energy cosmic-ray energies (the GZK cutoff, $\sim 5 \times 10^{19}$ eV) the predicted anisotropy reaches $\sim 10^{-19}$, within one to two orders of magnitude of current sensitivity to directional Lorentz violation ($\sim 10^{-18}$). The distinctive fingerprint is not merely a magnitude but the specific cubic O_h angular pattern above, which a generic Lorentz-violating theory would not predict. **Status: Prediction.** The exact expansion, the angular coefficients, and the magnitude comparison are reproducible (*dispersion₂₅*) in the companion materials. The coefficient $1/48$ is specific to the minimal nearest-neighbour model; the quadratic scaling and the cubic symmetry are forced by the lattice and are robust to the coupling details.

Steps (ii)–(v) of the main theorem follow from the polarization decomposition of § 4.2, the substrate coherence properties of § 3.2, and the Born-rule structural correspondence of Paper 2 § IV.

4.2 The Field Decomposition Theorem

Theorem 0.0.9 (Field Decomposition on FCC). *The continuum limit of polarization propagation on the FCC lattice admits a natural decomposition into five irreducible components under the cubic point group O_h , jointly spanning the 12-dimensional nearest-neighbor displacement space: a scalar (A_{1g}), a two-dimensional deformation tensor (E_g), a three-dimensional tangential tensor (T_{2g}), a polar vector (T_{1u}), and a three-dimensional odd-parity tensor (T_{2u}).*

Proof sketch. The FCC lattice has 12 nearest neighbors, each at face-diagonal distance ℓ_p in the standard FCC parametrization (Paper 1 § VI). The 12 nearest-neighbor displacement vectors $\{r_j\}$ are symmetric under the cubic point group O_h , and their irreducible decomposition under O_h is

$$12 = A_{1g} \oplus E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u} \text{ (with dimensions } 1 + 2 + 3 + 3 + 3 = 12\text{),}$$

with character vector $[E, 8C_3, 6C_2', 6C_4, 3C_2, i, 6S_4, 8S_6, 3\sigma_h, 6\sigma_d] = [12, 0, 2, 0, 0, 0, 0, 4, 2]$ on the twelve nearest-neighbor displacements, verified two independent ways (character contraction and explicit class representatives). The parity split is six even (A_{1g}, E_g, T_{2g}) and six odd (T_{1u}, T_{2u}),

producing five irreducible components:

- *Scalar* (A_{1g} , 1-dimensional, parity-even): the local mean polarization $\langle \psi \rangle$ across the 12 neighbors.
- *Polar vector* (T_{1u} , 3-dimensional, parity-odd): the polarization gradient $\nabla \psi$ across the 12 neighbors — three independent directional components transforming as a vector under O_h .
- *Deformation tensor* (E_g , 2-dimensional, parity-even): the cubic-symmetric traceless symmetric-tensor component of the polarization quadrupole, with diagonal basis $(\partial_{xx} - \partial_{yy}), (\partial_{xx} + \partial_{yy} - 2\partial_{zz})/\sqrt{3}$.
- *Tangential tensor* (T_{2g} , 3-dimensional, parity-even): the off-diagonal symmetric-tensor component, $(\partial_{xy}, \partial_{yz}, \partial_{zx})$.
- *Odd tensor* (T_{2u} , 3-dimensional, parity-odd): the second parity-odd sector — the odd-parity partner completing the twelve dimensions alongside the polar vector T_{1u} . Physically: the inversion-odd off-diagonal response of the directional polarization across the 12 nearest neighbors, the odd counterpart of the even tangential tensor T_{2g} . $\square$

The five components jointly exhaust the 12-dimensional nearest-neighbor space. Each propagates by the same wave equation but couples to the lattice geometry differently, with angular structure indexed by its O_h representation.

The physical interpretation. The scalar (A_{1g}) is the matter-density sector (the analog of Higgs-like scalar content). The polar vector (T_{1u}) is the polar gauge-field sector (the analog of the photon and the polar

parts of the gauge bosons). The deformation and tangential tensors ($E_g \oplus T_2g$) are the spacetime-geometry sector (the analog of the metric, with $E_g + T_2g$ jointly spanning the five independent components of a 3D traceless symmetric tensor). The second odd sector (T_2u) is the inversion-odd tensor partner of the even tangential tensor T_2g : it carries the odd-parity off-diagonal response that completes the twelve nearest-neighbor dimensions. The decomposition reproduces the structural skeleton of the field content of the Standard Model coupled to gravity as a scalar, a polar gauge vector, a two-piece symmetric-tensor geometry sector, and an additional odd-parity tensor.

The chiral structure of the weak interaction is supplied separately by the inversion-odd parent-record break d_z through the Jackiw–Rebbi domain-wall mechanism of § 8.4.4, which is where chirality is treated.

This is a structural correspondence to the field content of the Standard Model coupled to gravity. The TMS does not claim to derive that field content in full; that requires the explicit representation-theoretic mapping reserved as a structural correspondence in § VIII. The TMS claim is that the geometric structure of the FCC lattice naturally produces five irreducible field-type sectors covering all 12 dimensions of the nearest-neighbor space, with the structural roles of scalar matter, polar gauge, symmetric-tensor geometry, and an odd-parity tensor partner.

Bridge to § VIII. The gauge-symmetry inheritance and fermion-boson decomposition originally developed in this paper’s § 4.3 and § 4.4 are consolidated into § VIII below, where the bifurcation-derived cross-level commutator mechanism of § 8.2 provides the non-Abelian closure that completes the structural correspondence to the Standard Model gauge group.

V. Initial Conditions and the Big Bang Correspondence

5.1 The Initial State of $\Xi(R)$

By the construction of § 2.3, the substrate $\Xi(R)$ begins its operation with initial polarization landscape ψ' inherited from $\mathcal{B}(R)$ via the coarse-graining operator G . The initial state has five structural properties forced by the construction:

Maximum informational density. $\mathcal{B}(R)$ is the saturated biographical record of an exhausted region: $\Sigma_{\text{sampled}}(R, \tau*) \approx \Sigma(R)$. Every configuration in $\Sigma(R)$ has been sampled, meaning $\mathcal{B}(R)$ contains the maximum information density that R can hold. Translated to $\Xi(R)$ under $\hat{R}$, this becomes the maximum initial agent density and maximum initial polarization complexity. The new substrate begins at the upper bound of what its agent count can encode.

Near-uniform background. The exhaustion condition is regional, not point-wise. Different points in R have been sampled with comparable thoroughness — the random walk through $\Sigma(R)$ has visited the configuration space evenly. The polarization landscape across $\mathcal{B}(R)$, averaged over the imprint depth, is therefore close to uniform at large scales. Translated to $\Xi(R)$, this becomes a near-uniform initial U'_{sh} across the new substrate.

Small fluctuations from late-phase configurations. Σ_{sampled} approaches but does not exactly equal $\Sigma(R)$ at $\tau = \tau*$; some late-phase configurations are sampled less thoroughly than the bulk. These late-phase configurations contribute residual structure that deviates from the bulk uniformity. The size of the fluctuations is set by the ratio of late-phase configurations to bulk configurations, which is bounded by the per-flop noise injection rate divided by the imprinting rate (Paper 3 § 3.3). Translated to $\Xi(R)$, this becomes small residual fluctuations in the initial polarization landscape, on top of the near-uniform background.

Large-scale flatness. By the Principle of Generative Economy, the initial polarization landscape of $\Xi(R)$ is the minimum-curvature configuration consistent with the exhausted state. Curvature in the polarization landscape would correspond to an inhomogeneity favoring particular configurations over others, which would have driven Σ_{sampled} away from $\Sigma(R)$ at the parent level rather than toward saturation. Saturation is incompatible with large-scale curvature. Translated to $\Xi(R)$, the initial geometry (the tensor sector of § 4.2) is large-scale flat.

Causally connected past. $\mathcal{B}(R)$ is the biography of a single bounded region. Every confirmed bit in $\mathcal{B}(R)$ is part of a single causal record — they all trace back through the same parent-level causal stack. By the construction of $\hat{R}$, every new agent in $\Xi(R)$ inherits this property: they all derive from a single parent-level region whose own causal history is internally connected. Translated to $\Xi(R)$, the initial state is causally connected: every part of the new substrate has a past that was once in causal contact at the parent level, even if the corresponding regions are now spacelike-separated in $\Xi(R)$.

5.2 The Big Bang Correspondence Theorem

Theorem 0.0.10 (Big Bang Correspondence). *The initial state of $\Xi(R)$ under $\hat{R}$, viewed by an observer at hierarchical level $k \geq 1$ within $\Xi(R)$, exhibits the structural signatures of the observed cosmological initial state: maximum informational density (“hot dense early phase”), near-uniform background (CMB-like uniformity), small fluctuations on the uniformity (10^{-5} – scale CMB anisotropies), large-scale flatness, and a horizon-connected past. The correspondence is structural; numerical pinning of specific cosmological parameters is reserved for simulation.*

Proof. Each of the five structural properties of the initial state derived in § 5.1 maps to a structural feature of the observed Big Bang scenario:

- Maximum informational density (i) → hot dense early phase. The energy density of the early universe, in the cosmological standard model, is bounded above by the Planck scale; in the TMS reading, the initial $\Xi(R)$ has maximum agent density and maximum polarization complexity, which corresponds to maximum energy density in the field-theoretic continuum limit by the standard QCA-to-QFT identification of polarization activity with energy.
- Near-uniform background (ii) → CMB uniformity. The near-uniformity of the initial polarization landscape, averaged over imprint depth, corresponds in the continuum limit to a near-uniform initial field configuration with corresponding near-uniform temperature.
- Small fluctuations (iii) → CMB anisotropy. The residual structure from late-phase configurations of $\mathcal{B}(R)$ corresponds to small fluctuations on top of the uniform background, providing the seed inhomogeneities for cosmological structure formation. The size of these fluctuations is bounded by the per-flop noise-to-imprint ratio of Paper 3 § 3.3, which gives a structurally small but non-zero value — consistent with the observed 10^{-5} – scale CMB anisotropies though not pinned numerically at this level.
- Large-scale flatness (iv) → spatial flatness problem solved. The minimum-curvature property forced by saturation removes the need for inflationary fine-tuning of initial curvature. The TMS initial state is naturally flat at large scales because it is the PGE-selected minimum-specification continuation of a saturated parent record.
- Causally connected past (v) → horizon problem solved. Every part of $\Xi(R)$ traces back to a single parent-level region R that was internally causally connected. The horizon problem of standard cosmol-

ogy — how regions of the present-day CMB that are now spacelike-separated could have come into thermal equilibrium — is resolved structurally: they were in causal contact at the parent level, before reinterpretation, and their thermal equilibrium is inherited from that contact.

Each correspondence is structural rather than numerical. The numerical values of CMB temperature, scalar spectral index, baryon-to-photon ratio, and analogous quantitative observables depend on the specific $\mathcal{B}(R)$ being reinterpreted, the hierarchical level at which we are observing, and the specific combinatorics of the FCC lattice at successive levels. These numerical values are reserved for simulation work. The structural claim — that the TMS reinterpretation event reproduces the observed structural features of the Big Bang — is what is established here.

□

5.3 Matter–Antimatter Structural Correspondence

The sublattice bifurcation of § 6.1 produces, at every hierarchical level $k \geq 1$, two FCC sublattices A and B related by spatial inversion. This is the substrate-level analog of the matter / antimatter sector decomposition of QFT, with the sublattice exchange $A \leftrightarrow B$ playing the role of one CP-like discrete symmetry. We develop this correspondence in three claims.

Claim 0.0.11 (Structural — Derived). Every hierarchical level $k \geq 1$ has exactly two interpenetrating FCC sublattices, related by spatial inversion through the level-($k+1$) centroids. The sublattices share no vertices, no edges, and no faces at level k ; they share only centroids at level ($k+1$). Their $[[4,2,2]]$ stabilizer dynamics run independently at level k .

This is the geometric content of § 6.1 promoted to a structural law. The sublattices A and B at level k are forced by the bifurcation under PGE — each runs its own level- k FCC tetrahedral closure and its own $[[4,2,2]]$ code — and their only geometric overlap occurs at the next level’s actualization sites. Stage 5 verifies this through $L_{\text{box}} = 11$ ($n = 5$) and Stage 6 confirms identical structure at level 2 (Appendix E).

Claim 0.0.12 (Structural correspondence). The two-sublattice structure at every level $k \geq 1$ is the substrate-level analog of the matter / antimatter sector decomposition of QFT, with the sublattice exchange $A \leftrightarrow B$ playing the role of one CP-like discrete symmetry. Spatial inversion through any level-($k+1$) centroid is exact: every A-tetrahedron is mapped to a B-tetrahedron with the same level-($k+1$) centroid under coordinate inversion through that centroid.

The empirical content of the inversion-partnership is verified in Stage 5 (Appendix E) at 100% of shared centroids across $L_{\text{box}} = 5, 7, 9, 11$ — every A-tet centroid coincides with a B-tet centroid, and the B-tet is the exact coordinate inversion of the A-tet through their common centroid. In Stage 6, the same 100% inversion-partner pairing holds at level 2 across $L_{\text{box}} = 6, 8, 10, 12$.

The structural correspondence to QFT’s matter / antimatter sectors: in standard QFT, matter and antimatter fields are related by CP — a composite of charge conjugation C and spatial parity P. In the TMS picture, the substrate-level analog is the sublattice exchange $A \leftrightarrow B$, which is exact spatial inversion (the P component) combined with a sublattice-label flip (the C-analog component, with “charge” identified as the sum-mod-4 invariant $\in \{1, 3\}$ at the level- k site). The TMS sublattice symmetry is not literally CP — there is no charge conjugation in the usual QFT sense — but the structural role it plays in distinguishing the two sublattices and their independent dynamics is the same role CP plays in QFT. This is consistent with Harlow, Usatyuk, and Zhao’s (2025) holographic-gravity result that internal observer structure restores the complexity of a closed

universe: the substrate's two-sublattice structure provides exactly the observer/anti-observer doubling that their argument identifies as required for non-trivial closed-cosmos quantum gravity.

Claim 0.0.13 (Prediction / Conjecture). A reinterpretation event acting on a parent biographical record $\mathcal{B}(R)$ without exact sum-mod-4 symmetry produces level-1 substrates $\Xi(R)$ with unequal A and B populations. The matter–antimatter asymmetry of the observed universe is a generic consequence of reinterpretation from a non-perfectly-symmetric parent record.

The reinterpretation operator $\hat{R}$ maps the saturated parent record $\mathcal{B}(R)$ to a new level-1 substrate by the polarization-landscape transition of § 2.3. The sum-mod-4 content of $\mathcal{B}(R)$ determines, at the level-1 boundary, the relative weights of A versus B sublattice population in $\Xi(R)$. If $\mathcal{B}(R)$ carries equal sum-mod-4 weight, then $\Xi(R)$ has equal A and B populations and is matter-antimatter symmetric. If $\mathcal{B}(R)$ has a residual sum-mod-4 imbalance — as a generic consequence of the parent's late-phase polarization landscape — then $\Xi(R)$ inherits this imbalance as an unequal A versus B population, manifesting in the field-theoretic continuum limit as a matter-antimatter asymmetry.

This is the substrate-level structural account of baryogenesis. The TMS framework provides three conditions that jointly produce matter-antimatter asymmetry, structurally parallel to Sakharov's (1967) three conditions but adapted to TMS's symmetry structure:

(i) *Sum-mod-4-changing transitions exist at the reinterpretation boundary (analog of baryon-number violation).* $\hat{R}$ can redistribute the sum-mod-4 spectrum from $\mathcal{B}(R)$ into $\Xi(R)$'s new level-1 substrate. Within any single substrate level, $[[4,2,2]]$ dynamics preserves sum-mod-4 (A stays A, B stays B); only $\hat{R}$ can change it.

(ii) *The parent biographical record $\mathcal{B}(R)$ carries inversion-asymmetric structure (analog of CP violation).* The late-phase polarization landscape of an exhausted region generically has asymmetric content under the $A \leftrightarrow B$ sublattice exchange — a generic structural property of a saturated record produced by non-perfectly-symmetric parent dynamics.

(iii) *$\hat{R}$ is a phase transition, not an equilibrium process (analog of departure from thermal equilibrium).* By § 2.7, $\hat{R}$ has atomic, locality-preserving, generative-resetting structure; the asymmetry it produces in $\Xi(R)$ is not subject to detailed-balance relaxation because $\hat{R}$ is not reversible in the relevant sense.

The parallel is structural rather than literal. TMS does not have C and P as separately violable symmetries — the $A \leftrightarrow B$ sublattice exchange is a single discrete operation combining what QFT distinguishes as two — and TMS does not have thermal equilibrium in the QFT sense. But each TMS condition closes the same kind of structural leak its Sakharov counterpart closes, and all three are jointly necessary: take any one away and the asymmetry either does not generate or does not persist into $\Xi(R)$.

Remark on the sub-critical reading. Claim 3 frames the asymmetry as “unequal A versus B populations.” Analysis of the inversion-broken band structure on the A/B sublattice doublet sharpens this: a *uniform* (constant) population imbalance is structurally a featureless trivial insulator with no fermion content. Matter-bearing band structure requires the inversion-breaking term to be *structured* in k — to have nonzero spatial variation, not merely a nonzero mean. The “imbalance” of Claim 3 must therefore be read as the *sub-critical mean of a structured field*. Direct simulation of the triadic confirmation rule ($\bar{B} = 3\bar{\alpha}$ resolution under causal sweep, Paper 2 § 3.2, 4.1) shows that the substrate's own dynamics produce exactly this regime: the late-time record has a near-zero mean ($\sim 6 \times 10^{-2}$) embedded in spatial structure of amplitude order unity, giving a structure-to-mean ratio ~ 17 , well inside the sub-critical regime. The matter-bearing condition (sub-critical mean of a structured field) is therefore satisfied by the substrate's own confirmation dynamics, not as an external imposition. The condition for matter to exist at all is that this sub-critical regime is maintained —

reframing the observed smallness of the baryon asymmetry ($\sim 10^{-9}$) as a *necessary condition for matter to exist*, not a quantitative puzzle to explain away. The full geometric closure (FCC 3D, A/B doublet, $1 \rightarrow 4$ multiplier) and the resulting mass spectrum remain the subject of subsequent biographical-simulation work (cf. § 8.4.3).

Status of Results for § 5.3:

- *Derived*: The two-sublattice structure at every level $k \geq 1$ (Claim 1).
- *Structural correspondence*: The TMS sublattice exchange as substrate-level analog of QFT's CP symmetry (Claim 2).
- *Prediction / Conjecture*: Matter-antimatter asymmetry from non-symmetric reinterpretation (Claim 3).

5.4 The Horizon Problem Resolution in Detail

The horizon problem of standard cosmology is among the most striking observational puzzles: regions of the CMB separated by angles larger than the particle horizon at recombination exhibit thermal equilibrium that they could not have established through standard light-cone communication. Standard cosmology addresses this through inflation — a brief period of exponential expansion that places initially-nearby regions outside one another's causal horizons. The TMS provides a different and structurally more economical resolution.

Under $\hat{R}$, the entirety of $\Xi(R)$ traces back to a single bounded region R of the parent substrate. Before reinterpretation, R is a causally connected region under the parent substrate's connectivity — its internal points are within communication of one another through the parent-level FCC propagation. The biographical record $\mathcal{B}(R)$ is the integrated history of this internally connected region. When $\hat{R}$ acts, $\mathcal{B}(R)$ becomes the initial state of $\Xi(R)$, but the causal connectivity of the parent-level R is not preserved as causal connectivity in $\Xi(R)$ — it becomes prior-correlation between regions of $\Xi(R)$ that may now be spacelike-separated.

This is the structural origin of the horizon-problem resolution: regions of the present-day CMB that appear spacelike-separated in $\Xi(R)$'s spacetime were not spacelike-separated in the parent-level R that produced them. Their thermal equilibrium is not a coincidence to be explained by inflation; it is a structural inheritance from the parent-level causal connectivity.

This account makes a structural prediction that distinguishes it from inflationary cosmology: the CMB correlations should reflect the specific causal structure of the parent-level R , not the post-inflationary smoothing of an arbitrary initial state. In principle, statistical signatures of the parent-level causal topology should be detectable in high-precision CMB measurements. This is a falsifiable prediction whose explicit empirical content requires simulation work to compute the expected signatures.

5.5 The Flatness Problem Resolution

Standard cosmology requires a remarkable fine-tuning of the early universe's curvature: any deviation from spatial flatness at the Planck epoch grows rapidly under Friedmann evolution, so the observed near-flatness today requires Ω near 1 to one part in 10^{60} at early times. Inflation provides one resolution by driving curvature toward zero exponentially. The TMS provides a different structural resolution.

Under $\hat{R}$, the initial geometry of $\Xi(R)$ is the minimum-curvature configuration consistent with the exhausted parent state. This is forced by two arguments. First, by the Principle of Generative Economy, the minimum-specification continuation is selected; curvature in the initial polarization landscape would require additional specification beyond what the exhausted parent state determines. Second, by the exhaustion condition itself,

the parent-level Σ_{sampled} approached $\Sigma(R)$ evenly; an unevenly sampled exhausted state would correspond to a curvature in the polarization landscape that would have biased further sampling and prevented exhaustion. Saturation is geometrically incompatible with large-scale curvature.

The initial flatness of $\Xi(R)$ is therefore structural rather than fine-tuned. No inflationary mechanism is required to drive curvature toward zero; the initial state is flat because the structural conditions of saturation force it to be. Small residual curvature corresponds to the small fluctuations of § 5.1(iii), bounded by the same per-flop noise-to-imprint ratio.

5.6 The Initial Entropy Problem and Causal Anchoring

Penrose's Weyl-curvature hypothesis and the broader "initial entropy problem" of cosmology pose a structural question: why was the early universe in such a remarkably low-entropy state, given that random initial conditions should have produced very high entropy? The standard cosmological model has no internal mechanism to explain this; it must be posited as an initial condition.

The TMS supplies a structural mechanism. The initial state of $\Xi(R)$ is the saturated biography $\mathcal{B}(R)$, which has high informational density but also high structure: every program, every subsystem, every multi-level hierarchy that existed in R is encoded in $\mathcal{B}(R)$. The polarization landscape inherited as U'_{sh} of $\Xi(R)$ is therefore not random — it carries the algorithmic structure of all the programs that ran in R before reinterpretation. From the perspective of the new substrate, this is a low-entropy initial state: the polarization landscape is highly compressible because it encodes the program structure of the parent's biography.

The reinterpretation boundary transmits a two-part inheritance. The accessible record — the program structure of the parent's biography, scaling as the boundary area — is highly compressible and constitutes the daughter's low-entropy initial condition, fixing the matching condition and the arrow. The shed residue — the resolvable distinction discarded by the lossy fold at every confirmation, conserved under Axiom 3 but un-readable at the parent level — is, by the maximum-entropy character of the discarded micro-configurations, indistinguishable from fresh latent flux and constitutes the daughter's U_{sh} . The arrow $\hat{R}$ produces thus has its origin in this shed distinction: the irreversibility is not posited but is the accumulated per-fold loss. No information is created (Axiom 3) and none is lost; what changes is which part is accessible at which level.

This connects the resolution of the initial entropy problem directly to Axioms 3 and 4 of Paper 1. The Weyl-curvature hypothesis becomes, in TMS terms, the statement that the initial state inherits the structural compressibility of the parent-level programs. The compressibility is bounded above by the parent-level K_{ψ} at the moment of reinterpretation, which is itself bounded by the parent-level program count and complexity. The initial low entropy is therefore a derived property of the reinterpretation protocol, not a posited condition.

5.7 The Initial Singularity Resolution

Standard cosmology faces a singularity at $t = 0$: classical general relativity breaks down because curvature and density diverge. Quantum gravity programs typically replace the singularity with a non-singular initial state — a bounce, a quantum-pressure-supported minimum (loop quantum cosmology), or a primordial Bose-Einstein-condensate-like state (analog cosmology). The TMS provides a structural resolution: there is no singularity, only a reinterpretation event.

At the reinterpretation moment $\tau = \tau^*(R)$, the substrate is operating normally — flop-by-flop confirmation at the parent scale. The transition to $\Xi(R)$ is the change of read protocol, not a change in physical state. There is no infinite density, no infinite curvature, no divergent quantity. The parent substrate's $\mathcal{B}(R)$ is a

finite informationally-rich record; its reinterpretation as $\Xi(R)$ is a finite mapping that produces a finite new substrate.

What an observer in $\Xi(R)$ extrapolates backward through the daughter substrate’s history reaches, at $\tau' \rightarrow 0^+$, the boundary at which the daughter substrate begins to operate. Before $\tau' = 0$, the daughter substrate did not exist; the parent substrate was running. The “initial singularity” of standard cosmology corresponds in the TMS picture to the reinterpretation boundary, not to a physical singularity. The boundary is a phase transition (§ 2.7), and physical quantities are finite on both sides of it.

This is the structural origin of the standard “resolution” of the Big Bang singularity in quantum-gravity programs. The TMS provides a particularly clean version: the boundary is a structural identification rather than a dynamical event, and the apparent singularity is an artifact of extrapolating the daughter substrate’s dynamics into the region where the protocol changes.

5.8 Cosmological Natural Selection and Repeated Reinterpretation

Smolin (1992) proposed cosmological natural selection: universes that are good at producing black holes (and thus daughter universes through black-hole bounces) come to dominate the multiverse. Dowker and Zalel (2017) showed that causal-set classical sequential growth models exhibit a similar renormalization across creation events. The TMS provides a structural setting for this same idea: each reinterpretation event produces a daughter substrate whose initial conditions inherit the structural content of the parent’s biography.

If the daughter substrate $\Xi(R)$ is itself capable of forming subsystems, hierarchies, and bounded regions that eventually exhaust their own $\Sigma(R')$, then $\Xi(R)$ will itself undergo reinterpretations within its bounded regions. These produce grand-daughter substrates $\Xi(\Xi(R))$, and so on. The chain of reinterpretation events forms a generational hierarchy of substrates, each inheriting its initial conditions from the terminal biography of its parent.

Two structural predictions follow. First, daughter substrates inherit the algorithmic structure of their parents through the polarization-landscape boundary condition; substrates whose parents had richer program structure produce daughters with richer initial conditions, which in turn produce more programs, which exhaust faster, which produce more daughters. This is a structural form of cosmological natural selection. Second, the $c^{(k)}$ sequence at successive reinterpretation events forms a chain whose total dispersion is bounded by the product of per-level $(1/\sqrt{2})$ factors — providing a structural bound on how many generational reinterpretation steps separate any given substrate from the original parent.

A third structural prediction follows from the arrow-tower analysis of § 8.4.2 applied to the generational chain described above. Each reinterpretation event freezes one frozen-arrow direction into its daughter’s spatial geometry; the PGE-compressed tower saturates at the substrate’s spatial dimension. If each fermion generation corresponds to dominant alignment with one frozen arrow, then the observation of three generations implies the cosmological recursion has *depth at least three* ancestor substrates separating our current substrate from the original parent. This converts cosmological natural selection from a structural framework into a count with empirical traction: the observed generation count places a strict lower bound on the recursion depth.

All three predictions are structural correspondences whose specific empirical content requires simulation work to make quantitative. The TMS framework establishes that the generational structure exists and that its depth admits the lower bound supplied by § 8.4.2; the specific evolutionary trajectory it produces is a target for subsequent investigation.

VI. Numerical Closure: L_p , $c^{(k)}$, and the GUP Parameter

6.1 The Minimum Program Extent $L^{p(1)}$ (Bifurcated FCC Construction)

Paper 3 § 4.1 characterized L_p as the minimum spatial extent of a stabilizer-satisfying T-closed configuration with $K(C) < |C|$, estimated at 4–8 lattice spacings. Paper 4 § 7.5 reserved the exact value for combinatorial enumeration. We close this debt here by explicit bifurcated FCC construction, with the result empirically verified by the Stage 5 and Stage 6 simulation work documented in Appendix E.

The construction proceeds in three steps. First, we identify the geometric structure of the level-1 substrate (the sublattice bifurcation). Second, we specify the minimum configuration that satisfies all level-1 program conditions simultaneously (the stella-octangula-tile-plus-boundary, derived below). Third, we measure the spatial extent of this minimum configuration in substrate units and confirm $L^{p(1)} = 3\sqrt{2} \ell_p$.

6.1.1 The Sublattice Bifurcation of the Level-1 Substrate

By the Recursion-Preservation Theorem of § 2.5 and the coarse-graining operator G of Paper 3 § IV, the level-1 substrate produced by reinterpretation reproduces the substrate-level FCC lattice with the same triadic confirmation rule and the same $[[4,2,2]]$ stabilizer structure at the new scale. We adopt the *storage convention* throughout this section: at each hierarchical level, lattice positions are written in integer coordinates with the level- k FCC sublattice scaled by a factor of 2 relative to level $-(k-1)$. In this convention, the level-0 FCC consists of integer-coordinate sites with sum $i + j + k$ even; the level-1 sites lie on the all-odd-coord integer positions (the centroids of the level-0 tetrahedra after the $\times 2$ scaling).

The bifurcation. The all-odd-coord level-1 sites do not form a simple FCC. Under the squared-distance metric, the nearest-neighbor distance among all-odd-coord integers is 2 (e.g., from (1,1,1) to (1,1,3)) — a *simple cubic* coordination, not FCC. The simple-cubic NN structure supports no four-mutually-nearest tetrahedra, which would block the level-1 $[[4,2,2]]$ stabilizer dynamics altogether.

The structural resolution is that the all-odd-coord level-1 sites split into two interpenetrating FCC sublattices by the sum-mod-4 invariant:

- **Sublattice A** (sum $\equiv 1 \pmod{4}$): sites such as (1,1,3), (1,3,1), (3,1,1), (3,3,3) — all-odd-coord positions whose coordinate-sum is congruent to 1 modulo 4.
- **Sublattice B** (sum $\equiv 3 \pmod{4}$): the inversion partners (1,1,1), (1,3,3), (3,1,3), (3,3,1) — all-odd-coord positions whose coordinate-sum is congruent to 3 modulo 4.

Within each sublattice, the nearest-neighbor squared distance is 8 — the face-diagonal of the underlying cubic cell at edge 2. Each sublattice is therefore an FCC at NN distance $2\sqrt{2}$ in storage coordinates, with the standard FCC coordination number of 12 (Stage 5 verifies this empirically through $n = 5$; see Appendix E). The two sublattices are related by spatial inversion through any all-even-coord site (the level-2 sites), and they share no level-1 vertices, edges, or faces; they share only centroids at level 2.

This structure is analogous to the **diamond cubic** crystal familiar from solid-state physics, with the two FCC sublattices playing the roles of the two atom positions in the diamond unit cell. In TMS terms, the sublattice bifurcation is a structural consequence of the FCC + $[[4,2,2]]$ recursion under the storage scaling — not an additional axiom.

6.1.2 The Minimum Level-1 Program Configuration

By the bifurcated structure, a complete level-1 program configuration requires elements from *both* A and B sublattices, because each sublattice independently hosts an FCC tetrahedron and a $[[4,2,2]]$ stabilizer code, and the level-2 centroid where their dynamics jointly read out is the actualization site of the level-1 confirmed bit.

Definition 0.0.14 (Stella-Octangula Level-1 Tile). Let p be an all-even-coord (level-2) site. The *stella-octangula tile* at p consists of:

- The A-tetrahedron whose centroid is p : four sites in sublattice A at FCC nearest-neighbor positions to one another, surrounding p .
- The B-tetrahedron whose centroid is p : four sites in sublattice B at FCC nearest-neighbor positions to one another, surrounding p , related to the A-tetrahedron by inversion through p .

The union A-tet $\cup$ B-tet at p consists of eight level-1 sites arranged as a stella octangula (compound of two interpenetrating tetrahedra) centered on the level-2 site p .

Stage 5 verifies this geometry explicitly through $L_{\text{box}} = 11$ (see Appendix E): A-tet centroids and B-tet centroids coincide at 100% of shared sites, with each pair forming an inversion partnership through their common level-2 centroid.

The minimum program configuration consists of the stella-octangula tile at p together with the boundary neighbors required to evaluate the $[[4,2,2]]$ stabilizer parity at each of the eight level-1 sites. Each level-1 site within the tile has its own FCC neighborhood within its sublattice; only some of these neighbors are internal to the tile, and the remaining neighbors must be present in the configuration to evaluate stabilizer parity. The boundary neighbors lie on the same two sublattices and contribute to the program's spatial footprint.

6.1.3 The Spatial Extent

Theorem 0.0.15 (Minimum Program Extent at Level 1). *The minimum spatial extent of a stabilizer-satisfying T-closed configuration with $K(C) < |C|$ on the level-1 sublattice-bifurcated substrate is $L^{p(1)} = 3\sqrt{2} \ell_p$.*

Proof. We work in the storage convention of § 6.1.1, in which 1 storage unit equals $\ell_p/\sqrt{2}$ — fixed by Paper 1 § VI, where the FCC second-moment tensor over the 12 nearest neighbors of any agent evaluates to $4\ell_p^2 \delta_\mu \nu$ and forces ℓ_p to equal the level-0 nearest-neighbor distance (so $\ell_p = \sqrt{2}$ storage units).

The minimum configuration is the inversion-partner stella-octangula tile of § 6.1.2 — 8 level-1 vertices comprising 4 on sublattice A and 4 on sublattice B, all within the cube $[1, 3]^3$ in storage coords with centroid $(2, 2, 2)$ — together with the boundary sites required to evaluate the $[[4,2,2]]$ stabilizer parity at each of the 8 tile vertices.

For each tile vertex, 3 of its 12 in-sublattice nearest neighbors (at storage distance $2\sqrt{2}$ within the sublattice) lie inside the tile, namely the other three vertices of the same tetrahedron. The remaining 9 per vertex extend outside the tile. The cardinal range of the outward extension is bounded by the most negative and most positive coordinates these neighbors reach:

- From the A-tetrahedron vertex $(1, 1, 3)$, the negative-direction sublattice-A neighbors include $(-1, 1, 5)$, $(-1, 3, 3)$, $(-1, -1, 3)$, and $(-1, 1, 1)$, all at storage distance $2\sqrt{2}$ from $(1, 1, 3)$ and all with coordinate-sum $\equiv 1 \pmod{4}$ — these reach coord -1 .

- From the A-tetrahedron vertex (3, 3, 3), the positive-direction sublattice-A neighbors include (5, 5, 3), (5, 3, 5), and (3, 5, 5), all at storage distance $2\sqrt{2}$ and $\text{sum} \equiv 1 \pmod{4}$ — these reach coord +5.

The cardinal bounding-box extent of the full configuration (tile plus stabilizer-evaluation boundary) runs from coord -1 to coord $+5$, i.e., 6 storage units along any cardinal axis. Converting to ℓ_p :

$$L^{p(1)} = 6 \text{ storage units} \cdot (\ell_p / \sqrt{2} \text{ per storage unit}) = (6 / \sqrt{2}) \ell_p = 3\sqrt{2} \ell_p \approx 4.24 \ell_p.$$

This sits at the lower edge of the structural range of 4–8 lattice spacings posited in Paper 3 § 4.1, consistent with PGE forcing the minimum at the lowest geometrically permissible scale.

The three program conditions are satisfied as follows.

- *Stabilizer parity*: both A-tetrahedron and B-tetrahedron are in the +1 eigenspace of their respective level-1 $[[4,2,2]]$ stabilizers; the joint configuration is the ground state of the compound stabilizer code under inversion symmetry through (2, 2, 2). Stabilizer parity is evaluable for each of the 8 tile vertices because all 12 in-sublattice nearest neighbors of each vertex lie within the configuration by construction.
- *T-closure*: by the Recursion-Preservation Theorem of § 2.5, the level-1 T operator closes after a 2-flop cycle of substrate-scale confirmation steps. The first flop confirms a bit at the level-2 centroid (the inversion center (2, 2, 2)) — the unique level-2 site adjacent to at least three tile vertices, so the singular centroid is geometrically forced — and the second flop re-distributes confirmation across the eight level-1 sites of the tile. This two-step cycle is the isolated closure floor of the configuration; the operative level-1 flop period $T_p^{(1)}$ is set separately by the coupling-timescale constraint (§ 6.2).
- $K(C) < |C|$: the configuration is compressible because the inversion-partnership between A-tetrahedron and B-tetrahedron forces B from A via inversion through (2, 2, 2). The stella-octangula tile is O_h -invariant, so its orientation is a single orbit contributing $\log_2(1) = 0$ bits (App. B.4); specifying only the central $[[4,2,2]]$ logical state (one of 4, 2 bits) determines the entire tile, the boundary configuration being forced by the sublattice-bifurcation recursion. Total description length is $\log_2(4) = 2$ bits, much smaller than the $|C| \approx 17$ bits of the raw eight-site polarization count.

Smaller candidates fail at least one condition.

- A configuration restricted to one sublattice cannot host the level-1 $[[4,2,2]]$ dynamics in its full form, because the level-2 centroid at which the level-1 confirmed bit actualizes is reached only when both A-tet and B-tet are present. Stage 5 verifies this explicitly: shared A–B centroids occur at 100% of inversion-partner pairs at every L_{box} tested (Appendix E).
- A single tetrahedron without boundary is not closed — no surround to evaluate.
- A configuration with cardinal extent fewer than 6 storage units in any direction cannot enclose the stella-octangula compound together with its boundary.

The $3\sqrt{2} \ell_p$ extent is the geometric minimum at which all three program conditions are jointly satisfied.

□

Corollary 0.0.16 (Minimum Subsystem Extent at Level 1). $L^{\text{sub}(1)} = 3 L^{p(1)} + \delta = 9\sqrt{2} \ell_p + \delta$, with $\delta \leq 2 \ell_p$ from the meta-tetrahedral coupling overhead of Paper 3 § 5.3. Therefore $L^{\text{sub}(1)} \in [9\sqrt{2} \ell_p, 9\sqrt{2} \ell_p + 2 \ell_p] \approx [12.73 \ell_p, 14.73 \ell_p]$.

6.1.4 Empirical Verification (Forward to Appendix E)

The bifurcated structure underlying this construction is empirically verified through Stages 5 and 6 of the supplementary simulation work (Appendix E). The key results are:

- Each sublattice achieves the FCC coordination number 12 at interior sites, confirming the bifurcated geometry through $L_{\text{box}} = 11$ in storage units ($n = 5$).
- The stabilizer-satisfying configuration space on each sublattice has dimension $k = 3n$ over F_2 , where $n = (L_{\text{box}} - 1)/2$ is the level-local lattice depth. This is the same scale-invariant $k = 3n$ law that governs the substrate level (Paper 1 § 5.4, scale-invariant stabilizer null-space theorem).
- Tetrahedral count scales as n^3 , matching the bulk FCC count of unit cells in a volume of edge n .
- 100% of A-tet and B-tet centroids coincide at the same level-2 sites and form exact inversion partners through those centroids (verified at $L_{\text{box}} = 5, 7, 9, 11$).
- The bifurcation structure repeats identically at level 2 (Stage 6), confirming the constant bifurcation factor 2 per level rather than a compounding factor. This is the empirical content of the structural correspondence between hierarchical levels established by the Recursion-Preservation Theorem.

Status: *Derived* with empirical verification. The sublattice bifurcation and the resulting $L^{p(1)} = 3\sqrt{2} \ell_p$ are derived from FCC + $[[4,2,2]]$ + recursive coarse-graining under PGE, with the geometric closure structure (stella-octangula tile + boundary) explicitly constructed and the cardinal bounding-box extent computed in the storage frame. The empirical content is provided by Stages 5 and 6 of the simulation work, summarized in Appendix E.

6.2 The $c^{(k)}$ Sequence at Saturation

Paper 4 § 7.4 derived the upper bound $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$ from the coupling-timescale constraint of § 5.3. At saturation, this is realized with equality. We close the structural debt by explicit minimum-period construction.

Theorem 0.0.17 ($c^{(k)}$ Sequence at Saturation). *At the coupling-saturation bound, the hierarchical lightspeeds form the geometric sequence*

$$c^{(k)} = (1/\sqrt{2})^k \cdot c$$

with $c = c^{(0)}$ the substrate-level propagation speed. The corresponding $L_p^{(k)}$ and $T_p^{(k)}$ sequences are

$$L_p^{(k)} / L_p^{(k-1)} = R_L^{(k)}, T_p^{(k)} / T_p^{(k-1)} = \sqrt{2} R_L^{(k)}$$

where $R_L^{(k)}$ is the level- k minimum-program-extent ratio, with $R_L^{(1)} = 3\sqrt{2}$ from § 6.1.3. The recursive scale invariance forces $R_L^{(k)} = 3\sqrt{2}$ at every level $k \geq 1$ by the Recursion-Preservation Theorem of § 2.5; the absolute level- k minimum extent is therefore $L_p^{(k)} = (3\sqrt{2})^k \ell_p$ in substrate units.

Construction. Two temporal quantities must be distinguished. The isolated closure floor of the minimum program is 2 substrate-scale flops (the two-step cycle of § 6.1.2, verified in Appendix E). The operative level- k flop period $T_p^{(k)}$ is the larger of this floor and the coupling-timescale constraint of Paper 4 § 5.3;

the constraint dominates, giving $T_p^{(1)} = \sqrt{2} \cdot (3\sqrt{2}) \cdot t_p = 6 t_p$ at saturation. At each hierarchical level the same structure recurs: the level-k program closes after a 2-flop cycle of level $-(k-1)$ -scale steps, and the coupling constraint sets the operative period. The minimum $L_p^{(k)}$ is set by the level-k minimum tetrahedral configuration plus its boundary, which has the same structure as the substrate-level construction of § 6.1 scaled by $R_L^{(k)}$. At the coupling-saturation bound, $T_p^{(k+1)} = \sqrt{2} (L_p(k+1)/L_p(k)) T_p^{(k)}$, giving the saturation relation

$$c^{(k+1)} = (L_p(k+1)/L_p(k)) \cdot (T_p(k)/T_p(k+1)) \cdot c^{(k)} = (1/\sqrt{2}) c^{(k)}. \quad \square$$

The explicit $c^{(k)}$ sequence in substrate units, normalized to $c = c^{(0)} = c_{\text{TMS}}$, is given in the following table.

Level k	$L^{p(k)} / \ell_p$	$T^{p(k)} / t_p$	$c^{(k)} / c$ (exact)	$c^{(k)} / c$ (decimal)
0	1	1	1	1.000
1	$3\sqrt{2} \approx 4.24$	6	$1/\sqrt{2}$	0.707
2	18	36	1/2	0.500
3	$54\sqrt{2} \approx 76.37$	216	$1/(2\sqrt{2})$	0.354
4	324	1296	1/4	0.250
5	$972\sqrt{2} \approx 1374.6$	7776	$1/(4\sqrt{2})$	0.177

where $T^{p(k)} = (\sqrt{2} \cdot R_L)_k \cdot t_p = 6_k \cdot t_p$ (since $\sqrt{2} \cdot 3\sqrt{2} = 6$).

These are upper bounds at saturation. In non-saturated regimes, $c^{(k)}$ can be smaller than the saturation value. The sequence forms a discrete dispersion spectrum bounded above by the saturation cascade.

6.3 Identifying Our Universe's Level

6.3.0 The Level-Advance Relation

The hierarchical structure of TMS substrate requires a precise statement of what advances the level counter k. Without such a statement, the cross-level ratios of § 6.1.3 and § 6.2 and the level-identification framework of § 7.6 use k as an undefined parameter. We give the definition here and verify its consistency with the recursion structure of § 2.5.

Definition 0.0.18 (Level-Advance Relation). Let $O^{(k)}$ denote a primitive substrate operation at level k — a confirmation event in the sense of Paper 1 § III, taking three agents at level k as inputs and producing one resolved bit at level k. Level k+1 begins at the scale at which a minimum stabilizer-satisfying T-closed configuration with $K(C) < |C|$ forms among level-k confirmed bits acting as agents in a new primitive operation $O^{(k+1)}$ of the same structural form.

Three load-bearing features:

- (i) *Same structural form.* Each level's primitive operation must be a confirmation event satisfying $\bar{B} = 3\bar{\alpha}$. This excludes “more complex configurations of level-k events” from counting as level-(k+1) — only genuine role-shifts where outputs become new agents in a new triadic closure advance k.
- (i) *Single bit per operation.* Level advancement is a structural role-shift, not a numerical accumulation. The level counter does not advance because more bits get resolved at level k; it advances because the resolved bits begin functioning as agents in a structurally new operation.

- (i) *Quantization-by-closure criterion.* The “scale at which” a new level begins is operationalized through § 3.5’s quantization-by-closure: level $k+1$ is realized when the minimum $K(C) < |C|$ configuration successfully forms. This makes the criterion substrate-internal — it does not appeal to an external observer making the scale comparison.

Consistency with § 2.5 (Recursion-Preservation). The Recursion-Preservation Theorem states that confirmation-event structure is preserved under level transitions. The Level-Advance Relation is the formal statement of *how* it is preserved: by role-shift, where the bit-role at level k becomes the agent-role at level $k+1$.

Consistency with § 6.1.3 (length scaling). Under the Level-Advance Relation, the geometric extent of a level- $(k+1)$ operation is determined by the spacing of level- k confirmed bits (now functioning as level- $(k+1)$ agents). This spacing inherits the stella-octangula tile structure from level k , producing $L_p^{(k+1)}/L_p(k) = 3\sqrt{2}$ as derived in § 6.1.3.

Consistency with § 5.8 (reinterpretation cascades). The reinterpretation operator $\hat{R}$ at substrate exhaustion performs a level-advance at cosmological scale: level- k confirmed bits at exhaustion become the agents of the post- $\hat{R}$ substrate. $\hat{R}$ therefore advances k by exactly 1, not by the within-cycle level count and not by resetting.

Corollary 0.0.19. *The total level identity of an observer in TMS is the sum of (a) within-cycle role-shifts in the current cycle’s history, plus (b) the number of $\hat{R}$ –firings in the cosmic history leading to the current cycle.*

6.3.1 Inferring Our Level from Observation

If our universe is itself a level- k structure for some k determined by the chain of reinterpretation events that produced our substrate, then our measured speed of light c_{measured} is $c^{(k)}$ in substrate units. The substrate c_{TMS} is obtained by inverting the $c^{(k)}$ formula:

$$c_{\text{TMS}} = (\sqrt{2})^k \cdot c_{\text{measured}}$$

at the saturation bound. For $k = 1$ (one reinterpretation step removed from the substrate), $c_{\text{TMS}} = \sqrt{2} \cdot c_{\text{measured}} \approx 4.24 \times 10^8$ m/s. For $k = 2$, $c_{\text{TMS}} = 2 \cdot c_{\text{measured}} \approx 6 \times 10^8$ m/s. Higher k gives proportionally larger substrate-level c .

The actual value of k for our universe is a structural prediction whose empirical pinning requires comparison with multiple classes of observable. GUP scaling β depends on k (§ 6.4 and § VIII below); precision measurements of GUP corrections at varying preparation depths would in principle constrain k . Cross-level relativistic effects (§ VII) produce specific signatures whose magnitude depends on the local hierarchical structure. Vacuum-energy and dark-sector observables (§ 7.6.4) constitute a further empirical handle. All are accessible to next-generation precision experiments in principle.

Our universe’s k is therefore a derived structural quantity whose value is pinned by experimental signatures the framework predicts.

6.4 The Closed Form of β

Paper 4 § 8.1 specified the functional dependences of the refined GUP parameter:

$$\beta = \beta_0 \cdot f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}}, k)$$

with f monotonically increasing in all five arguments. We close the structural debt by deriving the closed form under additive Kolmogorov bounds.

Theorem 0.0.20 (Closed Form of f). *Under the additive Kolmogorov bound, where each algorithmic complexity contributes independently to the substrate footprint, the closed form is*

$$f(d, K_{\text{local}}, K_{\text{surround}}, K_{\text{subsys}}, k) = 1 + \alpha_d \cdot d + \alpha_{\text{local}} \cdot K_{\text{local}} + \alpha_{\text{surround}} \cdot K_{\text{surround}} + \alpha_{\text{subsys}} \cdot K_{\text{subsys}} + \alpha_k \cdot k$$

where each coefficient α_X is the per-unit weight of its corresponding argument's contribution to the bit's substrate footprint. The coefficients are dimensionless geometric constants determined by the FCC lattice connectivity and the $[[4,2,2]]$ code distance; specifically:

$$\alpha_d = 1, \alpha_{\text{local}} = 1, \alpha_{\text{surround}} = \sqrt{2}, \alpha_{\text{subsys}} = 3, \alpha_k = (1/\sqrt{2})^k$$

where the per-level structural overhead of meta-tetrahedral coupling, written δ_k in earlier drafts, is derived to be identically one by the form-invariance fixed point of Paper 3 § 4.7 (the meta-tetrahedral geometry is the same at every level, so its per-level ratio is one) and is therefore absorbed.

Proof sketch. The substrate footprint of a bit is the lattice volume over which its causal record extends. Each of the five arguments contributes additively to this volume in the limit of small contributions (independent contributions from each axis). The causal depth d contributes one lattice unit per flop of history. The local algorithmic complexity K_{local} contributes one lattice unit per bit of local algorithmic structure. The surround complexity K_{surround} contributes $\sqrt{2}$ per bit of imprinting history (weighted by the $\sqrt{2}$ imprint-depth factor of Paper 3 § 2.3). The subsystem complexity K_{subsys} contributes 3 per bit of program content (weighted by the 3-component (C_D, C_O, C_C) trichotomy). The hierarchical depth k contributes $(1/\sqrt{2})^k$ per level due to the $c^{(k)}$ dispersion (the meta-tetrahedral coupling overhead being unity by scale-invariance). The closed form follows by combining these contributions additively, with each α_X set by the corresponding geometric coupling. $\square$

The full derivation, including the multiplicative corrections beyond the additive bound, is given in Appendix C. The structural result here is sufficient: f is linear in its arguments under the additive Kolmogorov bound, with coefficients determined by geometric quantities. The empirical content of the prediction is that bits in deeper, more complex, more deeply hierarchically embedded configurations exhibit measurably larger GUP corrections; the linear functional form provides specific quantitative predictions accessible to precision GUP phenomenology (Hossenfelder, 2013).

Beyond the additive limit — when the contributions become large enough that cross-terms dominate — f acquires multiplicative structure. The additive form is the leading-order expression; the next-order correction is detailed in Appendix C and reserved for simulation work for full numerical pinning.

VII. Cross-Level Relativistic Dynamics: The Substrate-Level Origin of Gravity

7.1 Three Cross-Level Phenomena

Paper 4 § 8.2 flagged three structural effects of $c^{(k)}$ variation across levels when subsystems at different hierarchical levels share substrate at their interfaces: variable apparent propagation, frame-relative timing, and embedded-medium dilation. We derive each here as substrate-level mechanisms producing the kinematical and gravitational phenomena of general relativity in the continuum limit.

7.2 Variable Apparent Propagation: The Origin of Gravitational Redshift

Theorem 0.0.21 (Variable Apparent Propagation). *When a signal propagates from a region of level $-(k-1)$ hierarchical depth into a region of level- k hierarchical depth, its apparent propagation speed shifts from $c^{(k-1)}$ to $c^{(k)}$ at the interface, with $c^{(k)} = (1/\sqrt{2}) c^{(k-1)}$ at the saturation bound.*

Proof. The signal propagates at the local lattice speed of the level it occupies — by the lattice-relative invariance of the wave equation (Paper 3 § 4.5), this is $1/\sqrt{2}$ in level-local units. In substrate units, this corresponds to $c^{(k)}$ for level k and $c^{(k-1)}$ for level $(k-1)$. At the interface between levels, the signal transitions from one substrate to the other, and its apparent propagation speed in any global coordinate system reflects the level it currently occupies. The transition is geometric, not dynamical: no energy is added or removed; only the local propagation rate changes because the local substrate’s lattice structure differs. □

The physical interpretation: when a photon-like signal propagates outward from a deeply hierarchically embedded subsystem (a “mass” in level $-(k-1)$ substrate) into shallower substrate (level k with smaller hierarchical embedding), it appears to slow down. The slowing is what an external observer measures as gravitational redshift. In the TMS picture, redshift is not a frequency shift of the photon but a change in the local propagation speed of the substrate medium through which the photon propagates.

The factor $1/\sqrt{2}$ in the cross-level propagation ratio is not an empirical bound but a structural geometric law of the FCC lattice. By the FCC second-moment tensor $4\ell_p^2\delta$ over the 12 nearest neighbors (Paper 1 § VI), the lattice’s effective propagation rate is the face-diagonal-to-edge ratio of the cubic cell containing the FCC, which is exactly $1/\sqrt{2}$. By recursive scale invariance and the Recursion-Preservation Theorem of § 2.5, the same face-diagonal-to-edge ratio reappears at every hierarchical level. The $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$ bound is therefore a *structural law* of the FCC + recursive coarse-graining geometry, not an empirical fit. At saturation (the coupling-timescale bound of Paper 4 § 5.3), the bound is realized with equality at every level; non-saturated regimes have $c^{(k+1)} < (1/\sqrt{2}) c^{(k)}$ with the deviation controlled by the local hierarchical embedding.

This is structurally consistent with Volovik’s Fermi-point scenarios (Volovik, 2007), in which gravity emerges from defect structures in an underlying lattice substrate, and with the general entropic-gravity program (Verlinde, 2011; Bianconi, 2025), in which gravitational phenomena emerge from information-theoretic properties of the underlying substrate. The TMS provides a particularly clean instance: the gravitational redshift is the variable apparent propagation across hierarchical levels, with the level structure forced by the recursive coarse-graining of Paper 3.

7.3 Frame-Relative Timing: The Origin of Time Dilation

Theorem 0.0.22 (Frame-Relative Timing). *Events that are simultaneous in level $-(k-1)$ flop time $T_p^{(k-1)}$ may not be simultaneous in level- k flop time $T_p^{(k)}$, because the flop periods at different hierarchical levels are independent. Two events at level-mixed locations have ordering that depends on the level-frame of the observer.*

Proof. By the $c^{(k)}$ dispersion and the coupling-timescale constraint of Paper 4 § 5.3, $T_p^{(k)}$ is independent of $T_p^{(k-1)}$ modulo the saturation bound. An event at a level $-(k-1)$ location occurring at level $-(k-1)$ flop time $\tau_a^{(k-1)}$ and an event at a level- k location occurring at level- k flop time $\tau_b^{(k)}$ can both be assigned absolute substrate times. If the substrate times coincide, the events are simultaneous in absolute substrate time. But

their level-local timing depends on the level-frame: an observer at level $(k - 1)$ sees them as occurring at level $-(k - 1)$ units that differ from the level- k units by the $T_p(k)/T_p(k - 1)$ ratio. The ordering depends on the level-frame.

□

The physical interpretation: time dilation between observers in different gravitational potentials (different hierarchical-level configurations in TMS terms) follows from the difference in flop periods at their respective levels. An observer in a deeper hierarchically-embedded configuration experiences slower flop time relative to a shallower observer; this is the substrate-level mechanism of gravitational time dilation.

Standard general relativity describes this through the metric-tensor coupling of time to gravitational potential. The TMS provides the substrate-level origin: gravitational potential is a measure of local hierarchical embedding depth, and time dilation is the difference in flop periods at different hierarchical levels.

7.4 Embedded-Medium Dilation: The Mass-Geometry Coupling

Theorem 0.0.23 (Embedded-Medium Dilation). *A level $-(k - 1)$ subsystem deeply embedded in a level- k meta-subsystem experiences propagation through the meta-subsystem's substrate at the effective $c^{(k)}$ of the meta-medium, producing an apparent dilation of its internal dynamics relative to the bare $c^{(k-1)}$.*

Proof sketch. When a level $-(k - 1)$ subsystem is embedded inside a level- k meta-subsystem, the substrate at the subsystem's location is not pure level $-(k - 1)$ substrate — it is level- k substrate hosting a level $-(k - 1)$ configuration. Wave propagation within the level $-(k - 1)$ subsystem occurs through the level- k medium, with apparent speed $c^{(k)}$ rather than the bare $c^{(k-1)}$ the subsystem would experience in isolation. The dilation factor is the ratio $c^{(k-1)}/c^{(k)}$, which at saturation is $\sqrt{2}$. □

The physical interpretation: a massive object — a deeply embedded subsystem at lower hierarchical level — experiences slowed internal dynamics relative to its bare clock because its substrate is the level- k medium of the enclosing meta-subsystem. The slowing corresponds to the rest-mass energy in the field-theoretic continuum limit: the energy stored in the embedded configuration is the energy required to support its level $-(k - 1)$ operation within a level- k substrate.

This is the structural origin of mass-energy equivalence in the TMS picture: mass is energy that has been laid into the wake. A massive excitation is a standing, self-re-confirming program (§ 3.5) — it re-confirms itself in place with a rest period T_0 , giving it an intrinsic rest frequency $\omega_0 = 2\pi/T_0$ even at zero momentum. Its rest energy is the activity of that re-confirmation, $E_0 = \hbar\omega_0$, and its mass is that rest energy in inertial units, $m = \hbar\omega_0/c^2$. The conversion factor c^2 here is the within-level substrate wave speed of Paper 1 § 6.5 — the same c that propagates every excitation — not the cross-level $\sqrt{2}$ ratio. This is the crucial disentanglement: the within-level c^2 performs the mass-energy conversion ($E = mc^2$), while the cross-level $\sqrt{2}$ — per — level ratio performs gravity (the cross-level dispersion of § 7.5). One geometric primitive — the FCC face-diagonal — plays two distinct roles, within a level and across levels.

Corollary 0.0.24 (Relativistic Energy–Momentum Relation). *A standing excitation with rest frequency ω_0 propagating on the FCC substrate obeys, in the continuum limit, the dispersion $\omega^2(k) = \omega_0^2 + c^2|k|^2$, with c the universal, mass-independent, isotropic substrate speed. Identifying $E = \hbar\omega$, $p = \hbar k$, and $m = \hbar\omega_0/c^2$ yields $E^2 = (pc)^2 + (mc^2)^2$ — the relativistic energy–momentum relation.*

This relation is not an additional postulate; it is a corollary of four results already established in the volume. The second-order substrate dynamics (Paper 1 § 6.3) force the quadrature form $\omega^2 = \omega_0^2 + c^2k^2$

rather than a non-relativistic $\omega = \omega_0 + k^2/2m$. The isotropic second-moment tensor (Paper 1 § 6.5) forces c to be universal and direction-independent — the same c for every mass. Quantization-by-closure (§ 3.5) supplies the rest frequency ω_0 of a standing program. The PGE-forced wave-form theorem (Paper 2 § VII) forces every excitation to share the one substrate wave operator, so no mass modifies c . The relation holds exactly in the continuum limit; its leading correction is the $(E/E_{\text{Planck}})^2$ Lorentz violation of § 4.1 Lemma B, entering at the same quartic-in-momentum O_h invariant. Group velocity is zero at rest, rises with momentum, and remains below c , so propagation is causal. Status: the kinematic form is Derived; the rest-frequency spectrum ω_0 — the actual mass values — is the mass-spectrum problem deferred to § 8.4.3.

7.5 Gravity as Cross-Level Dispersion: The Unified Thesis

The three cross-level phenomena of § 7.2–§ 7.4 — variable apparent propagation, frame-relative timing, embedded-medium dilation — are not three separate phenomena. They are three manifestations of the same underlying substrate mechanism: the dispersion of effective propagation speed $c^{(k)}$ across hierarchical levels, with the dispersion rooted in the recursive FCC face-diagonal-to-edge ratio of § 7.2.

The unified thesis. Gravity is not a fundamental force in the TMS picture. It is the substrate-level expression of a single structural fact: subsystems at different hierarchical levels of embedding experience different effective propagation speeds, because each level's FCC + [[4,2,2]] substrate has its own local $c^{(k)}$. The standard phenomena of general relativity all reduce to manifestations of this cross-level dispersion:

- *Gravitational redshift* (§ 7.2) is the change in apparent propagation speed when a signal crosses between substrate levels. A photon-like signal moving outward from a deeply hierarchically embedded mass into shallower substrate appears to slow because the local $c^{(k)}$ changes by a factor $\sqrt{2}$ per level transition.
- *Gravitational time dilation* (§ 7.3) is the difference in flop periods $T^{p(k)}$ between substrate levels. An observer in deeper hierarchical embedding experiences slower local flop time relative to a shallower observer, because $T^{p(k)}$ increases by a factor of $\sqrt{2} \cdot R_L = 6$ per level (with $R_L = 3\sqrt{2}$ from § 6.1.3).
- *Mass-geometry coupling* (§ 7.4) is the embedded-medium dilation when a lower-level subsystem operates within a higher-level substrate. The energy required to support level $-(k-1)$ dynamics within a level- k medium is the rest-mass energy of the embedded configuration. Mass is hierarchical embedding depth, and gravitational binding is the substrate cost of maintaining the cross-level embedding.

Why this unifies as a thesis. Standard general relativity expresses the matter-geometry coupling through the Einstein equations, which couple stress-energy to the metric. In the TMS picture, both sides of this coupling are the same underlying object viewed at different scales: the matter side is the embedded subsystem at lower hierarchical level, and the geometry side is the host substrate at higher hierarchical level. The “coupling” is just the embedding relation, with the coupling strength set by the cross-level $c^{(k)}$ dispersion. This is why the Newton constant G has a structurally simple form in the TMS reading (§ 7.6): G is the ratio $(L^{p(1)}/L^{p(0)}) / (T^{p(1)}/T^{p(0)})^2 = (3\sqrt{2})/36$, in substrate-relative units, modulo the dimensional conversion factors.

The substrate-level origin of gravity. Gravity emerges in the TMS picture not as an additional force or field but as the structural consequence of having a hierarchical FCC substrate with recursive coarse-graining under PGE. The hierarchical structure exists because PGE selects the minimum sufficient closure at every scale (§ 2.3); the cross-level $c^{(k)}$ dispersion exists because the FCC face-diagonal-to-edge ratio is $1/\sqrt{2}$ at every level (§ 7.2); the matter-geometry coupling exists because lower-level subsystems can only operate by being embedded in higher-level substrate (§ 7.4). Take any one of these structural ingredients away and the

substrate either has no hierarchy or no gravitational phenomena. Gravity is what the TMS substrate exhibits when its hierarchical structure is viewed from inside one of its levels.

Status of § 7.5: *Derived* (the unified thesis). The structural identification of gravity as cross-level dispersion follows from § 7.2, § 7.3, § 7.4 jointly under recursive scale-invariance.

7.6 Cross-Level Gravitational Coupling and the Level-Identification Problem

The three cross-level phenomena of § 7.2–§ 7.5 — variable apparent propagation, frame-relative timing, and embedded-medium dilation — combine in the continuum limit to produce a coupling between matter (level – $(k - 1)$ subsystems embedded in level- k substrate) and the tensor sector of the polarization field. This coupling has the structural form of the Einstein equations. The structural ratio characterizing this coupling between successive levels is forced by substrate recursion. The SI numerical value of Newton’s constant in our universe depends on which level we sit at; this level identity is empirically determined.

7.6.1 The Structural Skeleton

The cross-level acceleration-like ratio characterizing gravitational coupling between successive levels of the substrate hierarchy is determined by three derived quantities: the length-ratio $L_p^{(k+1)}/L_p(k)$, the time-ratio $T_p^{(k+1)}/T_p(k)$, and the per-level lightspeed $c^{(k)}$. From § 6.1.3, the length-ratio is forced to $L_p^{(k+1)}/L_p(k) = 3\sqrt{2}$ by the stella-octangula tile geometry and the FCC second-moment storage convention (1 storage unit = $\ell_p/\sqrt{2}$).

From § 5.3, the coupling-timescale constraint requires that three level- k subsystems coherently meta-confirm at level $(k+1)$. Outputs propagate at $c^{(k)}$, and the meta-tetrahedral edge-to-interstice distance in level- k units is $\sqrt{2} \cdot L_p^{(k+1)}/L_p(k)$. For coherent simultaneous arrival of all three signals at the meta-interstice, the time-ratio must satisfy $T_p^{(k+1)}/T_p(k) \geq \sqrt{2} \cdot (L_p^{(k+1)}/L_p(k))$.

This is a lower bound; the substrate could in principle use a larger time-ratio (slower meta-confirmation). By the Principle of Generative Economy (Axiom 0), no unmotivated waiting between closures is admissible — the substrate selects the minimum-sufficient closure. The time-ratio therefore saturates: $T_p^{(k+1)}/T_p(k) = \sqrt{2} \cdot 3\sqrt{2} = 6$. Correspondingly, the lightspeed ratio saturates the geometric bound: $c^{(k+1)}/c(k) = (3\sqrt{2})/6 = 1/\sqrt{2}$.

The cross-level acceleration-like ratio that characterizes gravitational coupling at the substrate level is then:

$$(L_p^{(k+1)}/L_p(k)) / (T_p^{(k+1)}/T_p(k))^2 = 3\sqrt{2} / 36 = \sqrt{2}/12 \approx 0.118$$

This is Newton’s gravitational constant expressed as a cross-level structural ratio. The reduction is striking: the entire skeleton reduces to one geometric primitive (the FCC face-diagonal-to-edge ratio $\sqrt{2}$) plus the Principle of Generative Economy plus recursion. No dimensional analysis, no free constants, no tuning.

7.6.2 Scope of the Substrate Prediction

The framework predicts: (i) the cross-level ratios $L = 3\sqrt{2}$, $T = 6$, $c = 1/\sqrt{2}$ between any two successive levels; (ii) level-internal self-consistency: at each level k , the local Planck triple $(\ell_p^{(k)}, t_p^{(k)}, m_p^{(k)})$ satisfies the standard Planck relations using $c^{(k)}$ and $\hbar_{\text{TMS}} = 1$ atomic action quantum per substrate flop; (iii) mass-ratio $M_p^{(k+1)}/M_p(k) = 3\sqrt{2}/2$, forced by requiring local G-consistency at every level.

The absolute SI value of G in our universe depends on which level of the substrate hierarchy our universe occupies — the level-identification problem.

7.6.3 The Level-Identification Problem

If our universe occupies level k , then the substrate-level Planck length satisfies $\ell_{p,\text{substrate}} = \ell_{p,\text{observed}} / (3\sqrt{2})^k$. Sample magnitudes: at $k = 1$, substrate $\ell_p \approx 3.8 \times 10^{-36}$ m; at $k = 2$, substrate $\ell_p \approx 9.0 \times 10^{-37}$ m. The framework predicts the structural form of physical law; the empirical content lives in the discrete question of level identity, whose answer determines all level-dependent quantities simultaneously.

7.6.4 Empirical Routes Toward Level Identification

Several candidate empirical handles on k are flagged here for future work.

CMB power spectrum signature. The level-1 saturation cascade (§ 5.1–§ 5.2) produces a discrete-substrate fingerprint at small angular scales. The transition scale depends on $(3\sqrt{2})^k$, so the position of the fingerprint in the CMB power spectrum is a direct probe of k .

Hubble constant and dark-energy density. These quantities scale with cosmological time, which itself scales with $t_p^{(k)}$ through the saturation cascade. Their ratio to local atomic clock rates carries level information.

Gravitational-wave dispersion. If $c^{(k)}$ has cross-level structure, gravitational waves approaching substrate scale should show level-dependent dispersion. Current LIGO/Virgo frequencies are far below substrate scale, but future high-frequency detectors could probe it.

Neutrino mass hierarchy. The three generations occupy distinct positions in the recursive embedding structure (cf. § 8.4). If their mass eigenvalues encode the recursion depth at which each generation's fermionic content is anchored, the observed neutrino mass hierarchy ($\Delta m_{21}^2 \ll |\Delta m_{32}^2|$) constrains substrate parameters. The framework structurally accommodates a hierarchy but does not yet predict the specific ratios; making this prediction explicit is a candidate next-step calculation.

Black-hole interior structure. If black holes host recursive sublevels (§ 5.8), their evaporation spectrum and Hawking temperature would deviate from standard predictions in level-dependent ways.

Vacuum energy, dark matter, and dark energy. The framework produces a structural account of cosmological-scale vacuum phenomena via four nested results, presented here at honest epistemic statuses.

Result 1 (formally derived). Within-level S_3 cancellation. The 9 bilinear role-correlation operators on a triadic confirmation event decompose under the S_3 symmetry of agent permutations into 2 trivial-representation modes, 1 sign-representation mode, and 6 standard-representation modes. Only S_3 – trivial modes can carry nonzero vacuum expectation value in an S_3 – invariant substrate vacuum. Of the 9 bilinears, 7 therefore have vanishing vacuum stress-energy contribution by S_3 symmetry alone. The 2 surviving modes are the U(1) trace mode $T^{(k)}$ (“bit-channel residual”) and the symmetric-singlet mode $R^{(k)}$ (“relation-channel residual”). This 7-of-9 cancellation is forced by substrate primitives and does not require additional structural assumptions.

Result 2 (formally derived, as corollary of § 6.3.0). Recursive absorption. Under the Level-Advance Relation (§ 6.3.0), level- k confirmed bits become level- $(k+1)$ agents, and level- k confirmation relations become level- $(k+1)$ substrate manifold structure. Applied to the two surviving vacuum modes: $T^{(k)}$ is reabsorbed as matter content at level $k+1$ (since the bits carrying it become matter-carrier agents), and $R^{(k)}$ is reabsorbed as substrate structure at level $k+1$ (since the relations carrying it become the carrier lattice). The structural consequence: within-level vacuum residuals do not propagate to higher levels as vacuum. The “sub-Planckian vacuum integration” form of the cosmological constant problem does not arise in TMS, because sub-level vacuum modes have been reabsorbed into non-vacuum structure rather than summed.

Result 3 (candidate, assumption-dependent). Λ as steady-state equilibrium. If the substrate maintains a dynamical equilibrium between level- k bit-generation and bit-absorption into level- $(k+1)$ confirmation events, the unabsorbed-bit population reaches a steady state. The per-level density-suppression ratio is $\rho_p^{(k+1)}/\rho_p(k) = (M - \text{ratio})/(L - \text{ratio})^3 = (3\sqrt{2}/2)/(3\sqrt{2})^3 = 1/36$. Observed $\Lambda/\rho_p \approx 10^{-123}$ under this scaling gives a candidate level identity $k \approx 79$ for our universe. Three ingredients combine here and should not be conflated: the per-level suppression ratio $1/36$ is derived from the mass- and length-ratios; the steady-state of the unabsorbed-bit population is an assumption from rate-balancing, not a theorem; and the level identity $k \approx 79$ follows only when both are combined with the observed Λ/ρ_p . The derived content is the $1/36$ ratio alone. $k \approx 79$ is a candidate empirical bridge contingent on the steady-state assumption, and is falsified independently of $1/36$ if that assumption fails. The steady-state assumption is asserted from structural rate-balancing, not derived from a formal substrate-level theorem. A rigorous derivation would route through § 5.3 coupling-timescale machinery and § 5.1's saturation cascade dynamics.

Result 4 (candidate, structurally motivated). Bit and relation channels as dark sector candidates. The two surviving residual modes have distinct distribution properties. $T^{(k)}$ concentrates where confirmation events have occurred — i.e., where matter is — and behaves matter-like under gravitational coupling. $R^{(k)}$ is distributed across the substrate manifold structure itself, independent of event concentrations, and behaves field-like. These distribution properties match the observed phenomenologies of dark matter (clusters with matter, gravitates) and dark energy (smooth, drives expansion) respectively. Predicted ratio from substrate combinatorics: relation-channel : bit-channel = 3 : 1, matching the foundational substrate ratio $\bar{B} = 3\bar{\alpha}$. Observed ratio: dark energy : dark matter $\approx 2.65 : 1$. The leading-order prediction of 3:1 differs from observation by $\sim 12\%$; higher-order corrections in mode-weighting could move the prediction in either direction and have not been calculated. The structural identification is clean; the residual numerical gap is flagged as open.

Underlying structural identification (proposed but not yet ratified): substrate agents correspond to the energy-side of physical content, and substrate bits correspond to the mass-side. Under this identification, agents : bits = 3 : 1 from $\bar{B} = 3\bar{\alpha}$ is the substrate-side origin of the energy-mass split in observable physics. This identification has not been formally introduced in Papers 1-4 and requires consistency audit before ratification.

7.6.5 The Structural Reduction

Standard physics has four independent fundamental quantities in the gravitational sector: $\hbar$, c , G , and the mass scale. TMS reduces these to one geometric primitive (FCC face-diagonal-to-edge ratio = $\sqrt{2}$), one axiom (Principle of Generative Economy, Axiom 0), and one empirical question (level identity k). The Einstein field equations are the continuum limit of a substrate recursion whose coupling is structurally forced and whose absolute scale is a property of our universe's location in the hierarchy.

7.7 Connection to Independent Programs

The TMS account of gravitational phenomena connects with several independent programs that derive gravity from underlying substrate dynamics. Verlinde's entropic-gravity proposal (Verlinde, 2011) identifies gravity as an entropic force arising from holographic screens; in the TMS reading, the holographic screens correspond to the boundaries between hierarchical levels, with the entropic content being the algorithmic complexity of the level-coupling. Bianconi's gravity-from-entropy framework (Bianconi, 2025) derives gravity from a relative-entropy action between the spacetime metric and the matter-induced metric; in the TMS reading, the two metrics correspond to the level- k tensor sector and the level- $(k-1)$ tensor sector that meet at the embedding

boundary. Padmanabhan (2014) derives the cosmological constant from holographic mode-counting under emergent-gravity assumptions; the TMS framework parallels this structurally through its level-counting of substrate modes.

Two cellular-automaton programs warrant explicit contrast. The QCA-to-QFT programs of D’Ariano and Perinotti (2014) and Mlodinow and Brun (2025) develop continuum-limit field theory from discrete substrate dynamics; the TMS framework derives the equivalent lift substrate-internally (Section 4.1, Lemmas A and B) using only TMS primitives, with these external programs serving as comparison references rather than as load-bearing imports. The QICT (Quantum Information Copy Time) framework (Quantum Rep., 2026) derives emergent gravity from gauge-coded QCAs; structurally close to TMS in spirit, with the level-identification problem in TMS finding a natural analog in QICT’s copy-time geometry.

A direct contrast with empirical implications: Wetterich’s cellular-automaton spinor-gravity construction (Wetterich, 2022) predicts exact local Lorentz symmetry on the discrete level for a non-FCC lattice designed specifically to preserve it. TMS on the FCC lattice predicts residual Lorentz violation suppressed by $(E/E_{\text{Planck}})^2$ in the inversion-symmetric (scalar and gauge-vector) sectors, with a stronger $(E/E_{\text{Planck}})^1$ signal only in the inversion-broken matter sector, per Section 4.1, Lemma B. These are competing falsifiable predictions; future high-energy cosmic-ray observations will distinguish the two substrate hypotheses.

None of these independent programs starts from the same geometric primitives as TMS — circles, triadic confirmation, and the FCC lattice. The convergence at the level of structural results — gravity as an emergent, geometric, holographic, information-theoretic phenomenon — supports the broader picture that classical gravitational dynamics is not a fundamental force but a structural consequence of the underlying substrate’s organization across scales.

VIII. Structural Correspondence to Standard Model Gauge Structure

The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$, its representation content for matter fields, and the fermion-boson decomposition are among the most striking empirical inputs to modern physics. We address their structural origin here at the level of structural correspondence. The TMS does not derive the Standard Model in full; it provides the substrate-level carrier structure for an SM-form gauge theory, with the explicit representation-theoretic mapping reserved for subsequent work.

8.1 The Substrate-Level Stabilizer Group

The $[[4,2,2]]$ stabilizer code at the substrate level has stabilizer group $Z_2 \times Z_2$, generated by $S_x = X_1X_2X_3X_4$ and $S_z = Z_1Z_2Z_3Z_4$ (Paper 1 § V). By the Recursion-Preservation Theorem (§ 2.5), each hierarchical level reproduces the $[[4,2,2]]$ structure at its own scale, giving a $Z_2 \times Z_2$ stabilizer group at every level $k \geq 0$.

The continuous gauge group is not obtained from any general theorem that finite Abelian stabilizers must extend to continuous Lie groups — there is no such theorem, and many discrete gauge structures remain discrete. It is obtained by explicit construction: the push-forward transport of § 8.3 builds the generators from substrate primitives and closes them, yielding $\mathfrak{su}(3)$ with the trace mode carrying a separate $U(1)$ (the commutator computation is given in Appendix H). The role of the $Z_2 \times Z_2$ stabilizer hierarchy here is to supply, through its cross-level structure, the non-commutativity that the transport requires; the continuous group itself is the transport’s closure, not an assumed Abelian lift.

8.2 Cross-Level Commutator Mechanism (Bifurcation-Derived)

The non-Abelian closure that converts $U(1) \times U(1)$ into $SU(2) \times U(1)$ and ultimately $SU(3) \times SU(2) \times U(1)$ is supplied, in the TMS picture, by the sublattice bifurcation of § 6.1 and § 5.3. At each level $k \geq 1$, the substrate splits into two FCC sublattices A and B related by spatial inversion. The stabilizer dynamics on A and B are independent at level k but couple at level $k + 1$ through the shared centroids that host the level- $(k+1)$ confirmed bits.

The cross-level coupling between A-sublattice and B-sublattice stabilizers at level k provides the commutator structure required for non-Abelian gauge closure at level $k + 1$. Specifically, the A-stabilizer S_x^A and the B-stabilizer S_x^B at level k do not generally commute when their actions are evaluated at the same level- $(k+1)$ centroid, because the A and B dynamics run independently at level k and their joint readout at the shared centroid is sensitive to the order in which the readout combines them. This non-commutativity is the substrate-level origin of the non-Abelian closure of the gauge group at the meta-level.

This is the structural mechanism that the original gauge appendix (now relabeled Appendix F) hand-waved. The bifurcation supplies the cross-level commutator naturally, without requiring any additional structure to be posited.

8.3 Hierarchical Carrier Structure for $SU(3) \times SU(2) \times U(1)$

The Standard Model gauge group emerges from the substrate primitives at distinct levels of the hierarchy. This section gives the revised structural derivation; earlier drafts placed $SU(3)$ at level 2 via sublattice bifurcation acting on level-1 doubling, but that account gave $2 \times 2 = 4$ sublattices rather than 3 colors. The corrected derivation places $SU(3)$ at the level-0 substrate primitive itself.

SU(3) at level 0. Every confirmation event involves three agents closing around one bit ($\bar{B} = 3\bar{\alpha}$, Paper 1 § III). The event is invariant under S_3 permutations of agent labels. In the QCA-to-QFT continuum lift (§ 4.1), this discrete S_3 symmetry on label space lifts to continuous local gauge symmetry on the 3-dimensional internal space of agent labels. S_3 is the Weyl group of $SU(3)$; standard gauging of “3 indistinguishable label slots” produces $SU(3)$ as the continuum gauge group. The 8 gluons of QCD correspond to the 8 traceless bilinear role-correlation operators on the 3-agent space (the 9th, trace, mode being $U(1)$). Color confinement is the substrate-side requirement that only triadic-closed configurations are observable substrate events: single agents (single quarks) do not appear in isolation because triadic closure is structurally required for actualization.

SU(3), rigorously: the push-forward derivation and its global form. The label-permutation account above identifies the three-dimensional internal space but does not by itself fix the gauge group, and a literal reading of the confirmation readout does not yield $SU(3)$: the confirmation rule $P = (1 + \Pi)/2$ reads a real bias and so preserves only real-orthogonal symmetry, giving $so(3)$ at the readout. The gauge group does not live in the readout. It lives in the connection — the transport that pushes a confirmed configuration forward to the next meta-flop. Because the substrate wave equation is second order (Paper 1 § 6.3; Paper 2 § VII), the transported object is not the value ψ alone but the complex amplitude $z = \psi + i\Pi$ carrying the momentum Π (the rate of change $\dot{\psi}$); the factor i is required by propagation, not imported. This is the cross-level temporal $\rightarrow$ spatial flip of § 2.3 acting as the carry.

On the three-agent label space the transport decomposes into two substrate pieces. Value transport — the real rotations of the agent labels — generates the real antisymmetric matrices ($so(3)$, three generators). The momentum carry, to which the forced i attaches symmetric reorientations, generates i times the real symmetric traceless matrices (five generators). These eight close under commutation to exactly the eight-dimensional

algebra of anti-Hermitian traceless 3×3 matrices, with no overshoot — neither to $sl(3, \mathbb{C})$ (dimension 16) nor to $sp(6)$ (dimension 21). The generators are anti-Hermitian and traceless: anti-Hermiticity is the norm-preservation of the transport (the stochastic resolution term is zero-mean, $E[\chi|\Pi] = \Pi$ exactly, and the mean-field re-confirmation spectral radius never exceeds 1; verified in Appendix G), and tracelessness is the absence of the $U(1)$ trace mode.

These three properties fix the continuum algebra uniquely. Anti-Hermiticity selects the compact real form; a non-degenerate, negative-definite Killing form with an irreducible adjoint action (commutant of dimension one) makes the algebra simple and compact; and the rank is two with six non-zero roots — the A_2 root system. Because the unique compact simple Lie algebra of dimension eight is $su(3)$, the continuum lift of the transport symmetry is forced to be $su(3)$: the eight gluons are the eight transport generators, with the trace mode removed by tracelessness carrying the separate $U(1)$. The check is non-circular, and the absence of overshoot is not an empirical observation about the closure but a structural fact: the eight generators are anti-Hermitian and traceless by construction, and the commutator of any two anti-Hermitian traceless matrices is again anti-Hermitian ($[X, Y]^\dagger = -[X, Y]$) and traceless ($\text{tr}[X, Y] = 0$ identically). The space of anti-Hermitian traceless 3×3 matrices is exactly eight-dimensional and is $su(3)$; the bracket cannot leave it, so there is nowhere to overshoot. A control algebra of the same dimension, compactness, and anti-Hermitian character ($su(2) \oplus su(2) \oplus u(1)^2$) fails the simplicity test, distinguishing $su(3)$ from same-dimension imposters. The explicit structure constants, the negative-definite Killing form, the trivial center, and the rank-two A_2 root system are computed in Appendix H.

The global form is fixed by the matter content. The three agents carry the fundamental representation 3, which is irreducible (the only operators commuting with all eight generators are the scalars). The centre of the group is therefore the scalars $\zeta \mathbf{1}$ with $\zeta^3 = 1$, namely $\{1, \omega, \omega^2\} \cong \mathbb{Z}_3$, acting with triality ω on a single agent, ω^2 on an agent pair (the $\bar{3}$), and $\omega^3 = 1$ on a triadically closed event. Single agents are substrate-real primitives on which the centre acts non-trivially, so faithful action requires the cover: the global gauge group is $SU(3)$, not $SU(3)/\mathbb{Z}_3$. Confinement is then exactly the statement that only $\mathbb{Z}_3$ – neutral configurations — the triadically closed events — actualize, the substrate-level form of the unbroken-centre confinement criterion of QCD. The same forced i that lifts $so(3)$ to $su(3)$ is what makes the centre $\mathbb{Z}_3$: the value-only transport generates the real group $SO(3)$, whose centre is trivial.

Status. Given the QCA-to-QFT continuum lift of § 4.1 (itself a structural correspondence), the identification of the colour gauge group is *Derived*: the algebra is $su(3)$ (no overshoot, simple, compact), the global group is $SU(3)$ (fundamental triality of the substrate-real agents), and confinement is the unbroken – $\mathbb{Z}_3$ – centre statement. What remains a structural correspondence is the lift itself — that the discrete transport symmetry promotes to a gauged continuous symmetry — not the group identity, which the lift now determines. The earlier label-permutation reading (S_3 as the Weyl group of $SU(3)$) is recovered as the readout-side shadow of this transport symmetry and is consistent with it: $so(3)$ at the readout, $su(3)$ in the connection.

SU(2) at level 1. The level-1 substrate carries two interpenetrating FCC sublattices (the A and B sublattices of § 6.1.1). Under inversion through the level-2 centroid of each stella-octangula tile, the A and B sublattices exchange. The polarization field ψ at level 1 decomposes as a two-component doublet $\psi = (\psi_A, \psi_B)$. Gauging this doublet structure produces $SU(2)_L$ electroweak. (This $SU(2)$ is an *internal gauge* symmetry on the sublattice-membership label; the fermion spinor phase is a separate structure, deriving from the triadic oddness of § 4.1. Each carries its own role: gauge $SU(2)$ here, spinor from the threeness there.) The W and Z bosons correspond to the 3 traceless bilinear operators on the 2-component doublet (the 4th, trace, mode contributing to $U(1)$).

The group identity here is fixed by the same push-forward argument that fixes the colour group, now applied to the two-component doublet rather than the three-agent space. On the A/B doublet the transport again splits into value transport — the real rotation of the $\{A, B\}$ labels, generating the single real antisymmetric 2×2 matrix ($\mathfrak{so}(2)$) — and the momentum carry, the forced i attaching to the two real symmetric traceless 2×2 reorientations. These three generators close to exactly the anti-Hermitian traceless 2×2 algebra, $\mathfrak{su}(2)$, with no overshoot to $\mathfrak{sl}(2, \mathbb{C})$ (real dimension 6); a control adding the forbidden directions reaches dimension 6, so the closure test again detects overshoot. The algebra is compact (anti-Hermitian), simple (non-degenerate Killing form, irreducible adjoint), and rank one with two roots — the A_1 system — so the continuum lift is $\mathfrak{su}(2)$ uniquely. The norm-preservation of Appendix G applies verbatim, the doublet propagating under the same second-order wave operator.

The global form is again fixed by the matter content. The doublet (ψ_A, ψ_B) is the irreducible fundamental 2; the centre of the group is the scalars $\zeta \mathbf{1}$ with $\zeta^2 = 1$, namely $\{+1, -1\} \cong \mathbb{Z}_2$. The non-trivial element -1 acts as -1 on a single sublattice component (the 2) but trivially on the W-boson adjoint (the 3). A single sublattice site is a substrate-real primitive on which the centre acts non-trivially, so faithful action requires the cover: the level-1 gauge group is $SU(2)$, not $SO(3) = SU(2)/\mathbb{Z}_2$. As with colour, the value-only transport would generate the real rotation group $SO(2)$, whose centre is trivial; it is the momentum-carried i that produces the $\mathbb{Z}_2$ centre.

Status and the two boundaries. Under the § 4.1 lift, the level-1 gauge factor is *Derived* as $\mathfrak{su}(2)$ with global form $SU(2)$, by the same closure and matter-content arguments as colour. Two boundaries are stated honestly. First, *chirality* — that the gauge group acts on left-handed components only, the $SU(2)_L$ of the Standard Model — is not part of the gauge closure. The closure fixes the group; the left-handedness is supplied by the inversion-odd parent-record break d_z (§ 5.3, § 8.4.4), the same asymmetry that carves the fermion sectors, and is treated there rather than here. Second, the precise electroweak *global quotient* — whether the combined electroweak group is $SU(2) \times U(1)$ or the quotient $(SU(2) \times U(1))/\mathbb{Z}_2 \cong U(2)$, which in the Standard Model is fixed by the hypercharge assignments of the matter multiplets — depends on the matter representation content and remains a structural correspondence co-located with the fermion-roster work. What is Derived is the $SU(2)$ factor itself and its $U(1)$ partner as the trace mode; the way they are glued is the reserved piece.

U(1) at level 0. The overall phase of the bilinear role-correlation operators (the trace mode on the 3-agent space) carries $U(1)$. This is the single S_3 – trivial component of the bilinear decomposition that is not part of the $SU(3)$ generators.

This is a structural derivation rather than an external import. The mapping from discrete substrate symmetries to continuous Lie groups follows the substrate-internal QCA-to-QFT lift of § 4.1 (Lemmas A and B), which routes through Paper 1 § III, Paper 1 § VI, Paper 2 § IV, and Paper 3 § IV without inheriting external machinery. The connection to independent QCA-to-QFT programs (Mlodinow & Brun, 2025; D’Ariano & Perinotti, 2014) is now a comparison citation, not a load-bearing import — see § 7.7.

8.4 Matter Content: Fermionic-Bosonic Sectors and Generations

Fermionic content originates in the half-integer spin forced by the triadic oddness of $\bar{B} = 3\bar{\alpha}$ (§ 4.1, Lemma A): a single confirmation node recouples three $\text{spin-}\frac{1}{2}$ agents to a half-integer and is a fermion, while a bound pair recouples to an integer and is a boson (Corollary A.1). The A/B sublattice doublet plays a separate role, supplying the internal gauge structure — the $SU(2)_L$ label space of § 8.3 on which the electroweak break d_z acts. Bosonic content emerges as the coarse-grained polarization decomposition of § 4.2: scalar (Higgs-like), polar vector (gauge), symmetric-tensor geometry, and the odd-parity tensor partner.

8.4.1 The Three-Generation Count from 2D Minimum Closure

The three generations of Standard Model fermions correspond to the three distinguishable agent-slots in the primitive triadic confirmation event. The argument proceeds in three steps.

Step 1: Confirmation events are intrinsically two-dimensional. The primitive substrate operation is $\bar{B} = 3\bar{\alpha}$ (Paper 1 § III): three agents close around one bit in a triangular configuration on the manifold M_α . The triangle is a 2D object. The minimum closure at 2D requires exactly 3 participating agents; the minimum closure at 3D would be tetrahedral ($\bar{B} = 4\bar{\alpha}^3$, requiring 4 participating agents).

Step 2: The substrate hierarchy embeds 2D events in 3D packing, not the other way. The FCC lattice (Paper 1 § VI) is the 3D packing structure that supports level-1 confirmation events. The 3D structure governs spatial packing of agent-spheres but does not change the dimensionality of the confirmation event itself. Each level-1 site carries triadic structure inherited from the 2D primitive beneath.

Step 3: Matter-carrier count inherits the 2D minimum, not the 3D packing minimum. Under recursive embedding through the hierarchy (§ 2.5), distinguishable agent-slots are preserved by the recursion-preservation structure. Each agent-slot in the primitive 2D event becomes a distinguishable matter-carrier under the recursive lift. Three agent-slots produce three matter-carrier generations.

The framework predicts three generations, not four, because confirmation events live in 2D, not 3D. The 3D structure governs spatial packing rather than the count of distinguishable confirmation roles.

The nature of this result should be stated precisely, because it is easy to overread. The “three” of the generation count is the *same* three as the 3 of $\bar{B} = 3\bar{\alpha}$ — the founding valence, re-expressed through the recursion as a count of distinguishable matter-carrier slots. This is not an independent parameter that happened to come out equal to three; it is the founding number reappearing in a new role. Whether that recurrence is a deep structural echo (the same threeness manifesting at two levels of description) or a tautology (three put in, three read out) is not something the framework can adjudicate from the inside, and it is not claimed to be an independent prediction of the number three. What *is* substantive is the mechanism: that the generation count is tied to the confirmation valence at all — that “why three generations” and “why triadic confirmation” are the same question rather than two — and that the count is forced to follow the 2D confirmation minimum rather than the 3D packing minimum. The result is therefore labeled a structural correspondence tying generation number to confirmation valence, not a first-principles derivation of the numeral three from independent structure.

The informal invocation of § 2.5 in Step 3 above can be promoted to a formal theorem, completing the derivation of the count “three” within the framework.

Theorem 0.0.25 (Recursion-Preservation for Matter Content). *Under $\hat{R}$, the structural form of matter content — the count of generation slots forced by $\bar{B} = 3\bar{\alpha}$, the O_h irrep decomposition of the polarization field on the FCC nearest-neighbor space, and the inversion-symmetry character that distinguishes fermion from boson sectors — is preserved at every recursion level. The matter content at level $k+1$ has the same structural form as at level k .*

Proof. Three components.

(i) $\bar{B} = 3\bar{\alpha}$ at every level. By the recursion-preservation theorem of § 2.5, $\Xi(R)$ at level $k+1$ reproduces the level- k substrate’s triadic confirmation rule under the same five axioms. The triadic minimum $\bar{B} = 3\bar{\alpha}$ in 2D, derived from the angular-closure argument of Paper 1 § II.1, applies to the substrate primitive at any scale. Therefore the count three of the triadic minimum is preserved at every level.

(ii) *O_h irrep decomposition at every level.* Section § 4.2 gives the complete five-irrep decomposition of the polarization field under O_h on the FCC nearest-neighbor space, exhausting all twelve dimensions. The FCC connectivity and O_h symmetry are preserved by § 2.5 at every level. Therefore the irrep decomposition is preserved at every level.

(iii) *Fermion-versus-boson character at every level.* The substrate-level distinction between fermion sectors and boson sectors in the polarization field is determined by two structures: the O_h irrep decomposition (which classifies modes by their transformation under the substrate's full point group, § 4.2), and the action of inversion-breaking from the parent-record on those modes (which carves the multifold sectors into chiral fermion content, § 5.3, while leaving inversion-symmetric sectors as bosons, § 9.6). The irrep decomposition is preserved at every level by component (ii) above. The parent-record break is structurally guaranteed at every $\hat{R}$ – produced level by the recursion structure itself: by the definition of reinterpretation (§ 2.3), every level has a parent whose biographical record provides the inversion-breaking environment for the child substrate. The existence of a break is therefore not a feature specific to our level but an automatic structural consequence of being an $\hat{R}$ – produced substrate. Both the classifier (the O_h irrep decomposition) and the actor (the parent-record break) are preserved at every recursion level; therefore the fermion-versus-boson distinction is preserved at every level.

The theorem preserves *structural form*, not specific instantiation: the matter content at level $k+1$ has the same count of generation slots, the same irrep decomposition, and the same fermion-boson character as at level k . The specific configurations realizing matter content at each level differ — this is what makes hierarchical recursion non-trivial — but the structural form they realize is identical.

□

Corollary 0.0.26 (Structural Count Preservation). *The structural count “three matter-carrier slots” is preserved at every recursion level. The observed three fermion generations in our substrate result from the combination of (a) this structural preservation at every level and (b) the depth- ≥ -3 arrow-tower realization that selects three frozen-arrow carriers from the accumulated recursion (§ 8.4.2). Together, (a) and (b) upgrade the count “three” from substrate-forced to formally derived.*

Remark 0.0.27. The theorem makes the inheritance explicit but does not yet identify *which* three generations are realized — that question concerns specific configurations and mass values, both of which are governed by the biographical record $\mathcal{B}(R)$ and remain the subject of § 8.4.3.

Status. With this theorem the three-generation count of § 8.4.1 is promoted from substrate-forced to Derived; the identification of the three structural carriers with the three observed fermion generations remains a structural conjecture (§ 8.4.2), and the mass values remain open (§ 8.4.3).

8.4.2 The Same 3 Thrice

The “3 of three colors” (§ 8.3, $SU(3)$ gauging) and the “3 of three generations” (§ 8.4.1) are the same 3, distinguished by which symmetry-axis is projected. Color is *which-agent-now*: the S_3 action within a single confirmation event. Generation is *which-agent-slot-as-matter-carrier*: the S_3 action across the recursive embedding that produces observable matter content. Two structurally distinct symmetry-axes acting on the same underlying triadic primitive.

Independent corroboration: Jegerlehner (2019) argues that three fermion generations are required to make CP violation and baryogenesis emerge naturally in low-energy effective Standard Model scenarios. This is a different reason for the count 3, derived from low-energy phenomenology rather than substrate combinatorics, arriving at the same number.

A third independent route comes from the arrow tower under recursive reinterpretation. Each reinterpretation event freezes its parent's causal-succession axis (§ 2.3, applied at the meta level via the recursion-preservation theorem of § 2.5) into the child's spatial geometry as a frozen directional structure. Successive reinterpretation events accumulate a tower of frozen arrows in the descendant substrate, one per ancestor level. The Principle of Generative Economy compresses linearly-dependent directions as not-minimum-sufficient, so the tower saturates at the linear-independence ceiling of the substrate's spatial dimension. With substrate dimension 3 (Paper 1 § II.4), **the tower saturates at exactly three frozen arrows for any recursion depth ≥ 3** . If each fermion generation corresponds to dominant alignment with one frozen arrow, the count of generations equals the spatial dimension, locked.

Three structurally distinct routes — substrate combinatorics, low-energy phenomenology, and recursion-tower saturation — converging on the same integer indicate that the count is overdetermined rather than coincidental.

Two side-predictions fall out of the arrow-tower route. First, a recursion-depth lower bound: the observation of three generations implies the cosmological recursion has depth ≥ 3 ancestor substrates, sharpening § 5.8's cosmological natural selection from structural prediction to a count with empirical traction. Second, a dimensionality–generations lock: the generation count is not a free number; it is identical to $\dim(\text{space})$. A substrate framework deriving a different spatial dimension would predict a correspondingly different generation count.

Status of Results for § 8.4.2:

- *Derived:* Tower saturation at $\min(\text{depth}, \text{dim})$ under PGE compression; for our substrate ($\text{dim} = 3$, $\text{depth} \geq 3$), the count is 3.
- *Structural correspondence:* The convergence of three independent geometric routes on the count 3.
- *Prediction / Conjecture:* The mapping from frozen arrows to specific fermion generations via dominant alignment; the mass values themselves reside in $\mathcal{B}(R)$ and are not derivable from the tower alone (cf. § 8.4.3 and keystone idealization 2).

8.4.3 Mass Spectrum

The mass spectrum's dimensionful values are set by the band-crossing dispersion together with the biographical record, and are calibrated against experiment in the manner of the Standard Model's Yukawa couplings (developed below and in § 8.4.3's status block).

Mass spectrum values follow from (i) the detailed band-crossing structure on the electroweak A/B gauge doublet under the FCC dynamics (§ 8.3), (ii) the substrate-side coupling structure $J(k) = J_0 \cdot |F(k)|^2$ where $F(k)$ is the A-tetrahedron structure factor (§ 6.1.2), and (iii) the interaction of band-crossing dispersion with the cross-level coupling structure of § 7.5.

This calculation is developed as future work. The structural claim is that mass differences between generations correspond to differences in how each generation's agent-slot is anchored in the recursive embedding, with the mass spectrum following from band-structure analysis under the substrate dynamics.

A structural decomposition of the spectrum is nonetheless available, separating what the framework forces from what the biography supplies. The fermion rest frequencies ω_0 are the protected standing-mode frequencies of the re-confirmation dynamics: a standing program re-confirms in place with rest frequency $\omega_0 = \Omega(0)$, giving mass $m = \hbar\omega_0/c^2$ (§ 7.4). The inversion-odd structured break d_z (§ 8.4.5, § 5.3) acts on the vector sector of the O_h field decomposition (§ 4.2), and each frozen arrow of the recursion tower (§ 8.4.2)

contributes an inversion-odd break aligned with its axis. The break-induced mass-squared operator on the three-dimensional vector sector is therefore $M^2 = \sum_g w_g \hat{n}_g \hat{n}_{gT}$, summed over the tower's frozen arrows $\hat{n}_g$ with biographical amplitudes $w_g \geq 0$; the rest frequencies are the square roots of its eigenvalues.

Two consequences are structural. First, the count: because the PGE-compressed tower saturates at exactly $\dim(\text{space}) = 3$ linearly independent arrows (§ 8.4.2), M^2 is a 3×3 matrix and the spectrum has exactly three rest frequencies — locked to the spatial dimension, so a substrate of different dimension would predict a correspondingly different count. Second, the existence of a hierarchy: for any linearly independent arrow triad the three eigenvalues of M^2 are generically distinct, so a non-degenerate mass splitting is forced rather than assumed — degeneracy occurs on a measure-zero set that includes both the orthonormal equal-amplitude case and symmetric triads — such as a tetrahedron-vertex triad along the $\langle 110 \rangle$ directions — at equal amplitudes; a non-degenerate three-level hierarchy therefore requires amplitude asymmetry or triad asymmetry, both supplied by the biography $\mathcal{B}(R)$. Direct evaluation on a generic saturated triad gives rest-frequency ratios of order a few even at equal amplitude, widening as the biographical amplitudes spread.

One premise underlies both consequences and deserves explicit statement: they assume the tower's frozen arrows are linearly independent. The arrow-tower argument (§ 8.4.2) establishes independence for generic arrows under PGE compression, but whether the arrows a given biography actually freezes are independent — rather than collapsing onto a single repeated succession axis — is a property of the reinterpretation operator $\hat{R}$ that has not yet been demonstrated from the substrate dynamics. The count and the hierarchy are accordingly conditional on arrow independence, which a faithful construction of $\hat{R}$ must establish.

Localized matter is massive; matter has a minimum mass. Two structural facts sharpen the mass picture below the level of specific values. First, a localized triadic confirmation must re-confirm in place to persist — an isolated confirmation that does not re-confirm decays back into the U_{sh} background — so any standing, localized program has a non-zero rest re-confirmation frequency $\omega_0 > 0$ and hence, through $m = \hbar\omega_0/c^2$ (§ 7.4), a non-zero rest mass. Localization forces mass. Second, the minimum stable standing program under the operational flop dynamics has size four — a bent four-cluster with alternating chirality — so there is a smallest standing re-confirmation frequency and matter carries a minimum mass.

The re-confirmation amplitude spectrum is discrete and rational. The self-re-confirmation amplitude $|\Pi|$ of a standing program — the net agreement of a node with its own prior confirmation, divided by its coordination — is a ratio of integers and takes only a small set of robust values under the flop dynamics: $|\Pi| \in \{1, \frac{1}{2}, \frac{1}{3}, 0\}$. The value $|\Pi| = 1$ is maximal freeze (the most tightly bound, heaviest standing mode); $|\Pi| = \frac{1}{3}$ is the triadic signature of $\bar{B} = 3\bar{\alpha}$ directly; $|\Pi| = \frac{1}{2}$ is the paired (bosonic) amplitude; and $|\Pi| = 0$ is the massless limit. These are the harmonic values $1/n$, so the re-confirmation amplitudes are mode numbers rather than a continuous parameter. This is the structural content of the mass spectrum: a mode structure in Π -amplitude space, discrete and rational, forced by the counting.

The map from $|\Pi|$ to particle mass values is open. The discrete Π -structure fixes the mode architecture; the dimensionful mass values in MeV are set by the full band-crossing dispersion together with empirical calibration, exactly as the Standard Model takes its Yukawa couplings from experiment. The framework derives the mode structure and leaves the value map open, at the honest boundary between the two.

Status of Results for § 8.4.3:

Derived: The three-rest-frequency count and the mass-squared operator $M^2 = \sum_g w_g \hat{n}_g \hat{n}_{gT}$ on the vector sector, with the count locked to $\dim(\text{space}) = 3$ through the arrow tower (§ 8.4.2); that localized matter is massive ($\omega_0 > 0$ from the persistence-requires-re-confirmation argument); that matter has a minimum mass (smallest stable standing program, size four); and that the re-confirmation amplitude spectrum is discrete and rational, $|\Pi| \in \{1, \frac{1}{2}, \frac{1}{3}, 0\}$, a mode structure in Π -amplitude space.

Structural correspondence: The existence of a non-degenerate mass hierarchy as a generic (non-fine-tuned) property of M^2 .

Prediction / Conjecture: The specific rest-frequency ratios and the overall mass scale, set by the arrow geometry and amplitudes of $\mathcal{B}(R)$; co-located with the biographical simulation (keystone idealization 2).

Open: The map from the discrete Π -structure to particle mass *values* in MeV, set by the full band-crossing dispersion plus empirical calibration, as the Standard Model takes its Yukawa couplings from experiment. The framework derives the mass *architecture* (the counting, the minimum, the mode structure); the dimensionful values live at the calibration boundary.

8.4.4 Open Items

Chirality and parity violation. The mechanism by which the parent-record break breaks inversion is now structurally identifiable: under § 2.5’s recursion-preservation theorem, a reinterpretation event has the structure of a meta-confirmation, and its meta-bit inherits the inversion-odd cyclic chirality that every confirmed bit carries (cf. § 4.1, Lemma A). *The break breaks inversion because it is a bit, and bits are chiral.* The local mechanism by which this break produces chirality is now demonstrated. The break $d_z = \langle \mathcal{B}_A - \mathcal{B}_B \rangle$ is odd under the $A \leftrightarrow B$ inversion, and the $\hat{R}$ boundary relates its two sides by exactly that inversion, so d_z changes sign across the boundary without further assumption. A sign-changing mass across a codimension-1 surface is the Jackiw–Rebbi / Kaplan domain-wall construction: it binds exactly one zero-energy mode, localized on the wall, of definite chirality. Direct computation confirms this on the d_z profile — a single mid-gap mode at zero energy bound to the wall, carrying a definite chirality eigenvalue, with its handedness tracking $\text{sign}(d_z)$: reversing the break reverses the bound mode’s group velocity. The substrate’s inversion-odd break is therefore what selects left over right, converting the $SU(2)$ of § 8.3 into the chiral $SU(2)_L$ of the Standard Model at the level of the local wall mode. The companion script ($d_{z\text{chirality}}$) records the verification together with the Nielsen–Ninomiya control (a naive lattice doubles, netting zero) that shows the codimension-1 mechanism is necessary. Global weak parity violation (net $L \neq R$ coupling) remains open: Nielsen–Ninomiya doubling pins net chirality to zero on any local Hermitian translation-invariant lattice, so the $L \neq R$ asymmetry must come through a codimension-1 mechanism — a domain-wall or overlap-fermion construction on the $\hat{R}$ boundary, with the hierarchy direction as the extra dimension. The local wall mode is now established; what remains open is specifically the *net global* asymmetry: on a closed hierarchy a wall and an anti-wall bind modes of opposite handedness that cancel (Nielsen–Ninomiya reasserting globally), so a net chiral count requires the hierarchy direction to be effectively one-sided at $\hat{R}$. The existence and sign of a net asymmetry are established from results already in hand; only the explicit integer magnitude is deferred. Two inherited facts are decisive. First, $\hat{R}$ is not a generic mass domain wall but a role-promotion map: by its definition (§ 2.3) each confirmed bit becomes an agent one level up, a center-to-vertex, outcome-to-confirmer promotion whose build direction is fixed by irreversibility (parent to child, never the reverse; Axiom 3, the frozen arrow). Second, and decisively for the geometric obstruction, Corollary A.0 established that the substrate carries *exactly one orientation invariant* — a single global handedness — and this is a fact about the lattice geometry (the sign of the triad triple product), derived from the recoupling geometry with no reference to the fermion spectrum. The Nielsen–Ninomiya cancellation requires a handedness-symmetric lattice, on which every mode of one handedness has a mirror partner of the other; that symmetry is the theorem’s premise. The substrate’s geometry denies that premise at the root: there is no opposite global orientation for a mirror partner to carry. This identifies a genuine geometric obstruction to the global cancellation — the wall and anti-wall that would cancel bind modes referred to a lattice orientation that is not symmetric. What this does *not* do, and the distinction is essential, is establish the net *spectral* index: the step from “the lattice orientation is single-valued” (a geometric fact, Corollary A.0) to “the fermion spectrum carries a nonzero

net chiral charge, of definite sign” (a spectral fact) is an inference that requires the explicit overlap-operator index on the $\hat{R}$ boundary lattice, and that computation is not carried out here. The framework therefore establishes the *geometric precondition* for a net global chirality — $\hat{R}$ ’s irreversibility gives the one-sidedness, and Corollary A.0 removes the mirror symmetry that would force cancellation — while the existence, sign, and magnitude of the resulting net spectral index remain to be demonstrated by that index calculation. This is a narrower and more honest claim than a derived net chirality, and it is worth stating precisely what the framework’s results do and do not require. The corpus’s substantive claim about the matter sector is that the framework produces *chiral* matter rather than vector-like matter — that left and right couple differently, as in the Standard Model — and that claim rests on the *local* chiral wall mechanism (§ 8.4.4), which is established, together with the identification of the geometric obstruction just given (Corollary A.0 removes the mirror symmetry that a no-go theorem would otherwise use to force the matter back to vector-like). Neither of those depends on the value of the net global integer. The exact net chiral count — whether it is one unit per generation or another value — is a well-posed further question, answerable in principle by the overlap-operator index on the explicit $\hat{R}$ boundary lattice, but it is not load-bearing for the framework’s claim that the matter it produces is chiral. It is therefore recorded as a genuine but non-essential open computation: the local chirality is derived, the obstruction to global cancellation is identified and is not circular, and the precise net integer is neither claimed nor required here.

Neutrino masses and their hierarchy. As flagged in § 7.6.4, the neutrino mass hierarchy is potentially a probe of level identification through how each generation’s agent-slot anchors into the recursive embedding. Specific predictions deferred to future work.

8.4.5 Higgs Mechanism Structural Correspondence

The Higgs mechanism of standard electroweak theory — a scalar field acquires a non-zero vacuum expectation value, giving mass to gauge bosons and (through Yukawa couplings) to fermions — has a structural correspondence in the TMS framework. We identify the substrate-level Higgs analog using the gauge structure of § 8.3, in which $SU(3)$ is the continuum lift of the within-event S_3 on agent labels (level 0) and $SU(2)$ is the *internal gauge* symmetry acting on the A/B label space of the level-1 doublet. This internal gauge- $SU(2)$ is the object on which the Higgs mechanism depends. It is a distinct structure from the fermion spinor phase, which derives from the triadic oddness (§ 4.1, Lemma A): the electroweak $SU(2)$ here acts on the sublattice-membership label, and it is this gauge symmetry that the Higgs analog uses below.

Theorem 0.0.28 (Existence of a Substrate-Inherited Scalar Background). *Every $\hat{R}$ — produced substrate inherits, in addition to the inversion-odd structured break of § 5.3, a generically non-zero inversion-even scalar background — the A1g (trivial irrep) component of the parent biography’s projection onto the polarization field decomposition (§ 4.2).*

Proof. By § 4.2, the polarization field decomposes under O_h on the FCC nearest-neighbor space into five irreps including the trivial A1g. The biographical record $\mathcal{B}(R)$, projected onto this decomposition, contains generically non-zero components in all five irreps. The zero-A1g case is measure-zero and not preserved under generic substrate dynamics; direct simulation of the TMS triadic confirmation rule (companion materials) confirms a non-zero A1g component in the late-time record.

□

The Higgs is d_z , not the A1g background. The natural first guess — that the inversion-even A1g background plays the role of the Higgs VEV — fails by symmetry. A1g is the trivial irrep of the full O_h point group, and a fortiori a singlet under every subgroup of O_h , including both the within-event S_3 that lifts to

SU(3) color and the A/B doublet structure on which SU(2) acts (§ 8.3). A scalar in the trivial representation of any gauge group does not contribute to gauge-boson mass terms by the standard $\langle H^\dagger T^a H \rangle = 0$ mechanism. The A1g background therefore gives mass to no gauge boson.

The Higgs of the TMS framework is the inversion-odd structured break d_z of § 5.3. The break $d_z = \langle \mathcal{B}_A - \mathcal{B}_B \rangle$ is by construction non-trivial under $A \leftrightarrow B$ exchange, placing it in the SU(2)-doublet direction of the level-1 sublattice doublet structure. It is therefore a non-trivial SU(2) representation, and its non-zero spatial-mean component plays the role of an SU(2)-doublet Higgs vacuum expectation value, measured directly in the FCC + A/B biographical simulation (companion materials) as a 6σ non-zero, time-stable, spatially structured mean: it breaks $SU(2) \times U(1)$ to a U(1) subgroup determined by the specific direction of d_z in the doublet space, giving mass to the SU(2) gauge bosons that do not preserve the unbroken U(1) subgroup.

Color stays massless. The within-event S_3 that lifts to SU(3) color acts on the three agent labels within a single confirmation event (§ 8.3), independently of the level-1 A/B sublattice structure on which d_z lives. d_z transforms trivially under this within-event S_3 : it is a property of the sublattice membership at level 1, not of the agent labels within an event at level 0. By the same $\langle H^\dagger T^a H \rangle = 0$ mechanism, an SU(3)-trivial scalar gives no mass to the SU(3) gauge bosons. The color gauge bosons therefore remain massless, consistent with their observed character.

Two roles, one object. The structural identification $d_z = \text{Higgs}$ assigns a single substrate object two distinct field-theoretic roles: (a) the band-structure carver that produces fermion sector content (§ 5.3, § 8.4.4), and (b) the SU(2)-doublet VEV that breaks electroweak symmetry. Both roles emerge from the same band-structure analysis of d_z acting on the A/B sublattice doublet. This is a single substrate object playing both roles, verified by the existence and stability of d_z in the substrate's own dynamics, not a correspondence between two separate substrate quantities and two separate Standard Model operators.

Status of the A1g background. The non-zero A1g scalar background established by the theorem above exists but does not play the role of the electroweak Higgs. Its substrate-level role is open: candidate identifications include a $U(1)_Y$ hypercharge background, a substrate-level vacuum energy contribution, or a quantity without an immediate Standard Model analog at this level of the framework.

Status of Results for § 8.4.5:

- *Derived (under § 8.3's SU(2)-from-A/B-doublet structural correspondence):* d_z exists, is time-stable, and is spatially structured in the substrate's own dynamics. The identification $d_z = \text{Higgs VEV}$ holds modulo the § 8.3 continuum lift.
- *Derived:* The existence of a non-zero inversion-even scalar background (A1g component) inherited via $\hat{R}$ from the parent biography.
- *Structural correspondence:* The dual role of d_z as both fermion-content carver (§ 5.3) and electroweak symmetry breaker (this section), as one substrate object in two field-theoretic roles. The fermion-carver role is established in § 5.3; the EW-breaker role is established here.
- *Prediction / Conjecture:* The specific direction of d_z in the SU(2)-doublet that determines the unbroken $U(1)_{\text{EM}}$ subgroup; the W and Z mass values; the substrate-level role of the A1g scalar background. All co-located with the keystone idealization 2 biographical simulation and the § 8.3 representation-theoretic mapping.

8.4.6 The Fermion Roster from Anomaly Freedom

With the gauge algebra Derived (§ 8.3) and the chirality mechanism in place (§ 8.4.4), the matter content can be assembled. The substrate fixes the representation structure of one generation: quarks carry colour, participating directly in the three-agent triadic confirmation, and so transform as SU(3) triplets (confined, § 8.3); leptons are triadically neutral and transform as colour singlets; the left-handed components are the chiral A/B doublet bound to the d_z wall (§ 8.4.4) and so transform as SU(2) doublets, while the right-handed components do not couple to the wall and are SU(2) singlets. In left-handed Weyl form the multiplets are $Q = (3, 2, Y_Q)$, $u_c = (\bar{3}, 1, Y_u)$, $d_c = (\bar{3}, 1, Y_d)$, $L = (1, 2, Y_L)$, and $e_c = (1, 1, Y_e)$, with the hypercharges Y left open at this stage. This representation assignment is a structural correspondence: the substrate motivates each rep, but the identification of a substrate excitation with an SM multiplet is not a theorem.

The hypercharges, however, are not free. A chiral gauge theory is quantum-mechanically consistent only if all gauge and gravitational anomalies cancel; an uncanceled anomaly breaks gauge invariance at one loop and renders the theory ill-defined. Because § 8.4.4 establishes that the substrate produces a genuinely chiral gauge theory, anomaly-freedom is not an optional feature to be imposed but a consistency requirement the substrate must already satisfy. Imposing it on the rep structure above fixes the hypercharges. The mixed $SU(2)^2 - U(1)$ condition gives $3Y_Q + Y_L = 0$; the gravitational U(1) condition then gives $Y_e = 6Y_Q$; the mixed $SU(3)^2 - U(1)$ condition gives $Y_u + Y_d = -2Y_Q$; and the cubic $U(1)^3$ condition fixes the remaining split, its only solutions being $Y_u = -4Y_Q$, $Y_d = 2Y_Q$ or the $up \leftrightarrow down$ reversal. Setting the single overall normalization $Y_Q = 1/6$ reproduces the Standard-Model hypercharges exactly: $Y = (1/6, -2/3, 1/3, -1/2, 1)$ for (Q, u_c, d_c, L, e_c) . The cubic condition has isolated roots, not a continuum, so the split is genuinely pinned. Every hypercharge is therefore fixed up to one overall normalization and the convention of which weak partner is named “up” — no free numerical parameter, consistent with the volume’s no-tuning stance.

One dependency is stated honestly rather than hidden. The anomaly-cancellation conditions used here are the one-loop triangle-anomaly polynomials of continuum quantum field theory; the substrate does not yet derive them internally, but inherits them as the consistency requirement any chiral gauge theory it produces in the continuum limit must satisfy. This is therefore a *structural correspondence with a continuum-QFT dependency*: the hypercharges are pinned *given* that the continuum theory obeys the standard anomaly conditions, and deriving those conditions from substrate primitives — rather than importing them — is an open item. What is internal is the chirality (§ 8.4.4) that makes anomaly-freedom binding in the first place; what is imported is the anomaly polynomial whose vanishing then fixes the charges.

The content that results is exactly one Standard-Model generation: fifteen Weyl fermions (six in Q , three each in u_c and d_c , two in L , one in e_c), replicated three times by the generation count of § 8.4.2. Two consequences are worth stating. First, the right-handed neutrino $\nu_c = (1, 1, 0)$ is a total gauge singlet that contributes to no anomaly; it is therefore neither required nor forbidden by consistency — the framework predicts it as an optional sterile state, matching its anomalous absence-or-presence in the Standard Model. Second, the content is the minimal anomaly-free *chiral* completion of the substrate rep structure: dropping the right-handed electron, recolouring the leptons, altering a hypercharge, or rendering the content vector-like each breaks anomaly-freedom (and the vector-like case additionally destroys the parity violation the substrate produces). The Standard-Model generation is thus singled out, not assumed.

Status of Results for § 8.4.6. *Derived* (given the rep structure and the chirality of § 8.4.4): anomaly-freedom fixes all hypercharges to the SM values up to overall normalization and the $up \leftrightarrow down$ label, with no free parameter; the optional ν_c as a consistency-invisible singlet. *Structural correspondence*: the assignment of substrate excitations to the colour-triplet / colour-singlet and weak-doublet / weak-singlet representations. *Open*: the overall hypercharge normalization (the analog of the level index k elsewhere in the volume), and

the net global parity violation flagged in § 8.4.4, on which the chirality — and therefore the requirement of anomaly-freedom itself — ultimately rests.

8.4.7 The Periodic-Table Skeleton: Subshell Capacities, Madelung Ordering, and the Noble-Gas Atomic Numbers

The same nearest-neighbor O_h decomposition that organizes the field sectors (§ 4.2), combined with the exclusion principle derived in § 4.1 (Corollary A.2) and the single flop-clock of the substrate dynamics, fixes the combinatorial skeleton of atomic structure — the subshell capacities, the filling order, and the noble-gas atomic numbers — as pure counting, with no free parameter. What follows is the architecture of the periodic table; the dimensionful energies that set the actual term values are not claimed here, exactly as the mass values are deferred in § 8.4.3.

Subshell capacities $2(2\ell + 1)$. The angular content of an atomic shell is the set of standing polarization modes of definite angular momentum on the sphere of confirmation directions. The number of independent modes of angular momentum ℓ is the dimension of the ℓ -th irreducible representation of the rotation group, $2\ell + 1$ — and Paper 5 § 4.1 derives that the triadic recoupling reconstructs the full rotation group $SO(3)$, so this count is available at every ℓ , not merely the small values a single shell resolves. The count $2\ell + 1$ is the number of independent degree- ℓ harmonic polynomials on $\mathbb{R}^3$ (the angular standing modes at that degree), which is $1, 3, 5, 7, 9, \dots$ for $\ell = 0, 1, 2, 3, 4, \dots$. Multiplying by the two-fold spin polarity (the derived spin factor of § 4.1, Corollary A.2) gives the subshell capacities $2(2\ell + 1)$: $s : 2, p : 6, d : 10, f : 14, g : 18$. The twelve nearest-neighbor directions of a single FCC shell resolve only the low-angular-momentum part of this tower: under the subduction $SO(3) \downarrow O_h$, $L = 0$ subduces to A_{1g} , $L = 1$ to T_{1u} , and $L = 2$ to the pair $E_g \oplus T_{2g}$ (the standard five-fold d -shell splitting), reproducing $s : 2, p : 6, d : 10$ and matching the harmonic count $1, 3, 5$ exactly. The f -shell capacity 14 is the seven angular modes of $\ell = 3$ — present in the reconstructed group though beyond the single shell's resolution — times the spin factor two. No value is fitted.

Madelung $(n + \ell)$ ordering. The substrate advances one lattice step per flop tick; radial closure of a shell costs on the order of n ticks and angular closure on the order of ℓ ticks, counted in the *same* unit because there is one clock. The filling cost is therefore additive in n and ℓ , which is the Madelung rule (Madelung 1936; Klechkowski 1962): subshells fill in order of increasing $n + \ell$. This dissolves the standard puzzle of why the radial and angular quantum numbers enter the filling rule with equal weight — they are ticks of one clock. The tie-break among equal $n + \ell$ (fill lower n first) follows from loop compactness: the more radially compact loop is more tightly bound. That tie-break is a sound physical tendency rather than a rigorous consequence, and is labeled Derived-conditional accordingly.

A point of honesty about what this result is and is not. In ordinary atomic physics the $n + \ell$ ordering is the empirical signature of electron *screening*: in a pure Coulomb $1/r$ potential the energy depends on n alone (the accidental degeneracy), which would fill strictly by n and give the wrong closures — the third noble gas would fall at $Z = 28$ rather than 18. It is the screened many-electron potential that lifts the degeneracy into the $n + \ell$ order actually observed. The substrate reproduces that same $n + \ell$ order from a different and more primitive source: the additivity of radial and angular closure costs under a single clock. That the clock-counting yields precisely the screened ordering, rather than the bare-Coulomb ordering, is a genuine and non-trivial structural match — pure $1/r$ does *not* give the periodic table, and the substrate does. Whether the single-clock additivity is, at a deeper level, the substrate's expression of screening physics, or an independent route that happens to coincide with it, is not settled here. The honest status is therefore a derived counting rule that matches the empirically screened order exactly, with the physical identification of the two mechanisms marked as open.

Noble-gas atomic numbers and period lengths. Filling the derived capacities in the derived order gives cumulative electron counts closing each period at

$$2, 10, 18, 36, 54, 86, 118$$

— exactly the atomic numbers of helium, neon, argon, krypton, xenon, radon, and oganesson, the heaviest element yet synthesized — with period lengths 2, 8, 8, 18, 18, 32, 32. The two length-32 periods are exactly the two that contain an f -block: the $4f$ subshell fills fourteen electrons across the lanthanides (ending at $Z = 70$) and the $5f$ subshell fourteen across the actinides (ending at $Z = 102$). This is pure counting of the two preceding results, with no adjustable input: the real atomic numbers of the entire periodic table — heavy elements, lanthanides, and actinides included — emerging from $\bar{B} = 3\bar{\alpha}$. It is the strongest concrete arithmetic anchor in the matter sector.

Status of Results for § 8.4.7. *Derived:* the subshell capacities $2(2\ell + 1)$ — $s : 2, p : 6, d : 10, f : 14$ — from the $SO(3)$ angular-mode count (Paper 5 § 4.1) times the spin factor; the Madelung $n + \ell$ additivity from the single flop-clock; and the noble-gas atomic numbers (2, 10, 18, 36, 54, 86, 118) and period lengths (2, 8, 8, 18, 18, 32, 32), including the $4f$ lanthanide and $5f$ actinide blocks, by counting. *Derived-conditional:* the lower- n Madelung tie-break, a sound compactness tendency. *Open:* the dimensionful orbital energies (the analog of the mass values of § 8.4.3). The framework derives the periodic table’s full skeleton — capacities, ordering, the noble gases, and the f -block placement across all seven periods — not its spectroscopic energies.

8.5 Status Summary

The substrate-level $Z_2 \times Z_2$ at each level and the cross-level commutator mechanism from the sublattice bifurcation are *Derived* — they follow from the $[[4,2,2]]$ stabilizer structure of Paper 1 § V and the bifurcation of § 6.1 respectively. Given the QCA-to-QFT continuum lift of § 4.1, the colour gauge group is now *Derived* (§ 8.3): the algebra is $\mathfrak{su}(3)$ by a non-circular closure of substrate-built generators with no overshoot, simple and compact; the global group is $SU(3)$ — fixed by the fundamental triality of the substrate-real agents — rather than $SU(3)/\mathbb{Z}_3$; and confinement is the unbroken $-\mathbb{Z}_3$ – centre statement. The level-1 $SU(2)$ factor and its $U(1)$ trace-mode partner are now *Derived* on the same footing (§ 8.3): $\mathfrak{su}(2)$ closes from the A/B doublet transport with no overshoot, with global form $SU(2)$ rather than $SO(3)$ by the faithful fundamental doublet. What remains reserved is the electroweak gluing — the quotient $SU(2) \times U(1)$ versus $(SU(2) \times U(1))/\mathbb{Z}_2 \cong U(2)$, fixed by hypercharge — the chirality $SU(2)_L$ supplied by the d_z break (§ 8.4.4), and the three-generation fermion content; these are *Structural correspondences* in the sense developed throughout the volume: the TMS framework produces the structural skeleton of the SM, with the explicit representation-theoretic mapping reserved for those factors. The quantitative consequences (mass spectrum, mixing angles, etc.) are *Predictions / Conjectures* whose empirical bridges require subsequent simulation work. One quantitative layer below the gauge group is, however, now fixed without tuning: given the substrate rep structure, the requirement that the chiral theory (§ 8.4.4) be anomaly-free Derives the entire one-generation fermion roster and every hypercharge to the Standard-Model values, up to one overall normalization and the $up \leftrightarrow down$ label, with the right-handed neutrino predicted as an optional gauge singlet (§ 8.4.6). What stays open at the numerical level is the overall hypercharge normalization and, beneath it, the net global parity violation on which the chirality rests.

IX. Substrate-Forced Predictions and Empirical Bridges

The structural content of Papers 1-5 produces a number of predictions about the observable universe that follow from substrate primitives without free parameters. We organize them by epistemic status: *formally*

derived (proof chain complete within stated framework), *substrate-forced* (follows from primitives but with a stated structural assumption not yet a theorem), and *falsifiable* (specific quantitative form that observation could decisively test). Each prediction is annotated with its substrate provenance.

9.1 Gauge Group Structure (substrate-forced)

The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges from the triadic confirmation primitive at distinct levels of the substrate hierarchy.

SU(3) at level 0. Every confirmation event involves three agents closing around one bit (B-bar = 3 alpha-bar, Paper 1 Section III). The S_3 permutation symmetry of agent labels lifts in the QCA-to-QFT continuum limit (Section 4.1) to continuous local gauge symmetry on the 3-dimensional internal space. S_3 is the Weyl group of $SU(3)$; the standard gauging produces $SU(3)$ as the continuum symmetry. The group identity is fixed not by this label-permutation reading alone but by the push-forward derivation of § 8.3: the second-order transport carries the complex amplitude $z = \psi + i\Pi$, whose substrate-built generators close to $\mathfrak{su}(3)$ (anti-Hermitian, traceless, simple, compact, dimension 8, no overshoot) and whose fundamental-triality matter content fixes the global group as $SU(3)$ with an unbroken $\mathbb{Z}_3$ centre (confinement). With that derivation the colour factor is Derived under the § 4.1 lift, not merely substrate-forced.

SU(2) at level 1. The level-1 substrate carries two interpenetrating FCC sublattices (the A and B sublattices of Sections 6.1.1, 6.1.2). The doublet structure under inversion through the level-2 centroid produces $SU(2)_L$ electroweak as the gauging of the sublattice exchange.

U(1) at level 0. The overall phase of the bilinear role-correlation operators on the 3-agent space (the trace mode of Section 9.2) carries $U(1)$.

Provenance: Paper 1 Section III, Paper 5 Sections 4.1, 6.1.1, 6.1.2, 8.3. **Status:** Substrate-forced under the QCA-to-QFT lift (lift itself substrate-internal per Section 4.1 with Lemmas A and B).

9.2 Gluon Count (formally derived)

For a triadic event with 3 agents, the bilinear role-correlation operators number $3 \times 3 = 9$. One is the trace ($U(1)_{\text{baryon}}$). The remaining 8 are traceless and form the $SU(3)$ adjoint generators. The 8 gluons of QCD correspond to these 8 traceless bilinears. The count $3^2 - 1 = 8$ is forced once the substrate primitive is triadic.

Provenance: Paper 1 Section III, Paper 5 Section 8.3. **Status:** Formally derived.

9.3 Fractional Electric Charge (formally derived)

Three agents share one bit's $U(1)$ charge in a triadic confirmation event. By S_3 symmetry no agent is privileged, so each carries $1/3$ of the bit's charge. Anomaly cancellation ($3 Y_q + Y_L = 0$, with the 3 being the number of colors — i.e., the same 3 as triadic confirmation) combined with $Y_L = -1$ forces $Y_q = +1/3$, then $Q = T_3 + Y/2$ gives $Q = +/-1/2 + 1/6$, producing $+2/3$ (up) or $-1/3$ (down). The “ $1/3$ ” is literally “one of three” — the agent count itself.

With the full hypercharge assignment now fixed by anomaly-freedom (§ 8.4.6), the electric charge $Q = T_3 + Y$ of every fermion follows with no further input: u, c, t at $+2/3$; d, s, b at $-1/3$; neutrinos at 0; charged leptons at -1 . Two consequences are worth drawing out. First, charge is quantized in units of $1/3$, and the denominator is the colour count three — the same three agents of $\tilde{B} = 3\tilde{\alpha}$; leptons, being colour singlets, carry integer charge. Second, the proton (uud) has charge exactly $+1$ and the electron exactly -1 , so a hydrogen atom is exactly neutral. This exactness is not numerical luck: the proton–electron cancellation is

the same per-generation charge sum that the gravitational–U(1) anomaly condition sets to zero, so charge quantization and the neutrality of atoms are corollaries of the very consistency requirement that fixed the roster. The measured bound $|Q_p + Q_e|/e < 10^{-21}$ is, within the framework, an exact algebraic identity rather than a fine-tuned coincidence.

Provenance: Paper 1 Section III, Paper 5 Section 8.3. **Status:** Formally derived modulo the import of anomaly cancellation from standard QFT.

9.4 Three-Generation Fermion Count (formally derived)

Confirmation events are intrinsically 2-dimensional: three agents close around one bit in a triangular configuration on the manifold M_{α} (Paper 1 Section III, $B\text{-bar} = 3\alpha\text{-bar}$). The 3D FCC structure (Paper 1 Section VI) is the embedding lattice for the events, not the events themselves. The minimum-closure count for distinguishable agent-slots is therefore the 2D minimum: 3.

Three generations correspond to three distinguishable agent-slots in the primitive triadic event, instantiated as matter carriers under recursive embedding through the hierarchy. The “3 of three colors” and the “3 of three generations” are the same 3, distinguished by which symmetry-axis is projected. The framework predicts not 4 generations (which would be the 3D tetrahedral minimum $B\text{-bar} = 4\alpha\text{-bar}^3$) because confirmation events live in 2D, not 3D.

Independent corroboration: Jegerlehner (2019) argues that three fermion generations are required to make CP violation and baryogenesis emerge naturally in low-energy effective Standard Model scenarios — a different reason for the count 3, arriving at the same number.

Provenance: Paper 1 Section III, Paper 1 Section VI, Paper 5 Sections 2.5, 8.4. **Status:** Formally derived (count), via the § 8.4.1 recursion-preservation theorem and the § 8.4.2 arrow-tower saturation; the identification of the three structural carriers with the three observed fermion generations remains a structural conjecture (§ 8.4.2, § 8.4.3).

9.5 Cross-Level Gravitational Coupling (formally derived)

The cross-level acceleration-like ratio characterizing gravitational coupling at the substrate level is $G_{\text{skeleton}} = (L_p^{(k+1)}/L_p(k)) / (T_p^{(k+1)}/T_p(k))^2 = (3 \sqrt{2})/36$, with T-ratio = 6 forced by PGE saturating the c-bound inequality of Sections 5.3 + 6.2 to equality. The framework predicts the structural ratio between successive levels and the internal self-consistency of Planck units at each level. The SI value of G depends on which level we sit at — see Section 9.10.

Provenance: Paper 1 Section VI, Paper 5 Sections 5.3, 6.1.3, 6.2, 7.5, 7.6. **Status:** Formally derived as a cross-level structural ratio.

9.6 Lorentz-Violation Residual Scaling (falsifiable)

The substrate-internal continuum-symmetry argument (Section 4.1, Lemma B) predicts residual Lorentz violation suppressed by $(E/E_{\text{Planck}})^2$ for all inversion-symmetric sectors — scalar and gauge-vector alike — because lattice inversion symmetry forbids odd-momentum terms and the leading anisotropy enters at the quartic-in-momentum O_h invariant. A stronger $(E/E_{\text{Planck}})^1$ contribution is predicted only in the inversion-broken matter sector (§ 5.3), with magnitude set by the parent-record asymmetry. Future cosmic-ray observations measuring a fractional Lorentz-violation exponent inconsistent with these predictions — quadratic in the boson sectors, with any linear signal confined to and scaling with the matter-sector asymmetry — would falsify the framework’s QCA-to-QFT lift.

Provenance: Paper 1 Section VI, Paper 5 Section 4.1 (Lemma B). **Status:** Falsifiable. Contrast with Wetterich (2022), whose cellular-automaton spinor-gravity construction predicts exact Lorentz symmetry on the discrete level for a non-FCC lattice; observation will distinguish the two substrate hypotheses.

9.7 GUP Scaling with Hierarchical Depth (falsifiable)

The Generalized Uncertainty Principle deformation parameter β scales with causal depth d rather than being universal: $\beta(d) = \beta_0 * f(d, K_{\text{psi,local}}, K_{\text{psi,surround}}, K_{\text{psi,subsystem}}, k)$, with closed form in Appendix C. Distinguishes TMS from string-theory and loop-quantum-gravity GUP predictions; specific experimental targets include quantum decoherence measurements at varying gravitational potentials (matter-wave interferometry at varying altitudes, atomic clock decoherence comparisons between ground-level and ISS-level platforms, gravitational-redshift-corrected decoherence times).

Provenance: Paper 1 Section VII, Paper 5 Section 6.4, Appendix C. **Status:** Falsifiable.

9.8 Dark Sector Identification (candidate, structurally motivated)

Within-level vacuum stress-energy decomposes under S_3 into 7 vanishing modes and 2 surviving residual modes: $T^{(k)}$ (bit-channel) and $R^{(k)}$ (relation-channel). These have distinct distribution properties: $T^{(k)}$ concentrates with matter (clusters, gravitates like dark matter); $R^{(k)}$ is distributed across substrate structure (smooth, field-like, behaves like dark energy). Predicted ratio from substrate combinatorics: relation-channel : bit-channel = 3 : 1, matching the foundational substrate ratio $\bar{B} = 3 \alpha\text{-bar}$. Observed ratio: dark energy : dark matter $\sim 2.65 : 1$. Leading-order prediction differs from observation by $\sim 12\%$; higher-order corrections in mode-weighting could move the prediction in either direction and have not been calculated.

Provenance: Paper 5 Sections 7.6.4, 8.3. **Status:** Candidate, structurally motivated. The 12% gap is a real residual flagged as open.

9.9 Recursive Vacuum Absorption (formally derived)

Under the Level-Advance Relation (Section 6.3.0), within-level vacuum residuals are reabsorbed as non-vacuum structure at higher levels: $T^{(k)}$ becomes matter at level $k+1$, $R^{(k)}$ becomes substrate structure at level $k+1$. The “sub-Planckian vacuum integration” form of the cosmological constant problem does not arise in TMS.

Provenance: Paper 5 Sections 6.3.0, 7.6.4. **Status:** Formally derived as corollary of the Level-Advance Relation.

9.10 Level-Identification (empirical bridge)

The framework predicts cross-level structural ratios and level-internal Planck consistency, but not the SI values of fundamental constants directly. Asking “what is G in SI?” reduces to “what level k are we at?” Empirical handles on k include: CMB power spectrum signature (Section 7.6.4); Hubble constant and dark-energy density ratio to atomic clock rates; gravitational-wave dispersion at future high-frequency detectors; neutrino mass hierarchy as probe of recursive embedding depth; black-hole interior evaporation spectrum deviations from standard predictions. Under steady-state vacuum dynamics (candidate, Section 7.6.4 Result 3), observed Λ corresponds to $k \sim 79$.

Provenance: Paper 5 Sections 6.3, 7.6. **Status:** Empirical bridge — primary point of contact with cosmological data.

9.11 CMB Anisotropy Structural Signature (substrate-forced)

The level-1 saturation cascade produces a discrete-substrate fingerprint in the CMB power spectrum at small angular scales, distinguishable from inflationary alternatives. Two structural features predicted: structural maximum at characteristic late-phase configuration scale (empirically consistent with first acoustic peak at $l \sim 220$, structural identification requires simulation), and super-horizon correlations from parent-level causal structure (empirically consistent with observed super-horizon correlations that motivate inflationary cosmology).

Provenance: Paper 5 Sections 5.1-5.2. **Status:** Substrate-forced; specific signature form requires numerical simulation.

9.12 Open Items

Items where the framework has structural commitments but the formal derivation is incomplete:

- Anomaly cancellation as consequence of QCA-to-QFT consistency (used in Section 9.3)
- Formal $SU(2)$ -vs- $SO(3)$ representation verification for Section 4.1 Lemma A
- Steady-state vacuum dynamics derivation from substrate primitives (closes Section 9.8 Result 3 conditionality)
- Higher-order mode-weighting calculation for dark sector 3:1 ratio (resolves Section 9.8 gap)
- Consistency audit of “agents = energy-side, bits = mass-side” foundational identification against Papers 1-4
- Chirality / parity violation in $SU(2)_L$
- Higgs mechanism: the structural correspondence (d_z as the $SU(2)$ -doublet vacuum expectation value) is established in Section 8.4.5; open are the W and Z mass values, the precise $U(1)_{EM}$ embedding, and the substrate-level role of the A1g scalar background
- Asymptotic freedom and running couplings from triadic-confirmation behavior at different substrate scales
- Specific numerical CMB signature form (Section 9.11)

9.13 Summary Table

Prediction	Status	Substrate Origin
$SU(3)$ color from triadic S_3	Substrate-forced	3 agents per confirmation
8 gluons	Formally derived	$3^2 - 1$ traceless bilinears
Quark charges $\pm 1/3, \pm 2/3$	Formally derived*	1 bit charge / 3 agents + anomaly
3 generations	Derived (count)	2D minimum + recursion-preservation (§ 8.4.1)
$SU(2)_L$ from sublattice doublet	Substrate-forced	A/B inversion structure
3 weak bosons (W $\pm$, Z)	Formally derived	$2^2 - 1$ traceless bilinears

Prediction	Status	Substrate Origin
G structural skeleton (3 sqrt 2)/36	Formally derived	Sections 5.3 + PGE + 6.1.3
Lorentz-violation residual (E/E_{Planck}) ² (boson sectors); (E/E_{Planck}) ¹ only if P-broken (matter)	Falsifiable	Inversion symmetry + quartic O_h invariant
GUP beta scaling with depth	Falsifiable	Paper 1 Section VII
Dark energy / dark matter ~ 3:1	Candidate	Substrate combinatorics
Recursive vacuum absorption	Formally derived	Level-Advance Relation
Level-identification	Empirical bridge	Sections 6.3, 7.6

*modulo imported anomaly cancellation

X. Conclusion: From Circles to a Universe

Volume 1 of the Triadic Measurement Substrate has, across five papers, derived a generative chain from geometric primitives to cosmological initial conditions. The chain has no free *dimensionless* numerical parameters and no extraneous axioms: every dimensionless quantity is forced by the five axioms and the Principle of Generative Economy, and a single dimensionful input (one action scale) sets the physical scale.

Paper 1 established the geometric and ontological architecture: the sphere primitive of Axiom 2, the hexagonal manifold $M\alpha$, the triadic confirmation rule, the $1 \rightarrow 4$ multiplier, FCC necessity, the $[[4,2,2]]$ stabilizer structural correspondence, the identification $\ell_0 \equiv \ell_p$ and $\tau_{\text{top}} = t_p$, the conjugate field structure, and the wave equation at propagation speed $c_{\text{TMS}} = 1/\sqrt{2}$ forced by FCC face-diagonal geometry.

Paper 2 developed the computational dynamics: the polarization field ψ with its wave equation, the conjugate pair (ψ, Π_ψ) , the polarization-biased resolution rule $P(\bar{B} = 1 \mid \Pi_\psi) = (1 + \Pi_\psi)/2$ as the Born-rule structural correspondence, the 3-input majority gate at every confirmation event, universal computation from $\{\text{MAJ, constant}\}$, and the program formalism with stabilizer-based selection.

Paper 3 established the emergence of computation: surround-driven repair as the $[[4,2,2]]$ stabilizer projection at boundary sites, the Imprinting Theorem, the Hebbian Corollary, the coherence bistability threshold, the recursive coarse-graining operator G producing meta-level primitives, and the first computing subsystem as the (C_D, C_O, C_C) trichotomy at meta-tetrahedral extent.

Paper 4 developed the macroscopic consequences: the recursive coarse-graining $G^{(k)}$ at arbitrary hierarchical depth, three growth modes (annexation, imprinting, incorporation), self-modeling as a structural property of every subsystem, three stressors (mortality, predation, starvation), the Borel-Cantelli inevitability of subsystem formation, region biography, novelty exhaustion as the monotonic decay of $v(R, \tau)$, and the hierarchical lightspeed dispersion bounded above by $c^{(k+1)} \leq (1/\sqrt{2}) c^{(k)}$.

Paper 5 has closed the structural loop. The reinterpretation operator $\hat{R}$ acting on the exhausted biography $\mathcal{B}(R)$ is the unique PGE-selection forced by Axioms 3 and 4 jointly closing every other continuation. The new substrate $\Xi(R)$ is structurally identical to the parent substrate under the same axioms applied at the new scale, with confirmed bits playing the role of agents and the polarization landscape playing the role of U'_{sh} . The two-wave structure that appeared to threaten consistency is resolved as a single wave equation at successive recursion levels. The fundamental fields of the new substrate are the irreducible components of the polarization field on the FCC lattice — scalar, vector, tensor — with the substrate-internal QCA-to-QFT

lift of § 4.1 (Lemmas A and B) providing the structural bridge to standard QFT. The initial conditions of the new substrate exhibit the observed structural signatures of the Big Bang: maximum informational density, near-uniform background, small fluctuations, large-scale flatness, and a horizon-connected past — the Big Bang Correspondence. The numerical closures of Papers 1–4 are supplied: $L_p^{(1)} = 3\sqrt{2} \ell_p$, the $c^{(k)}$ saturation cascade $c^{(k)} = (1/\sqrt{2})^k c$, and the closed form of f under additive Kolmogorov bounds. The cross-level relativistic phenomena are derived as the substrate-level origins of variable apparent propagation (gravitational redshift), frame-relative timing (gravitational time dilation), and embedded-medium dilation (mass-energy equivalence). The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges from substrate primitives at distinct levels: $SU(3)$ from the triadic S_3 symmetry of confirmation events at level 0, $SU(2)_L$ from the A/B sublattice doublet at level 1, and $U(1)$ from the trace mode of the bilinear role-correlation operators. Three generations follow from the 2D-minimum-closure count $\bar{B} = 3\bar{\alpha}$. The dark sector is structurally identified with the two surviving S_3 – trivial vacuum residual modes, with predicted dark-energy-to-dark-matter ratio 3:1 (observed $\sim 2.65:1$, 12% gap flagged as open). The full predictions consolidation is given in § IX.

The full arc — from the sphere primitive of Paper 1 § 1.2 to the cosmological initial conditions of Paper 5 § V — is one unbroken geometric chain. Axiom 0 supplies the maximum-entropy ground state. Axiom 2 supplies the sphere. The Principle of Generative Economy selects the unique minimum-sufficient generative geometry at every step. The five axioms apply at every recursion level by axiom scale-independence. The reinterpretation operator $\hat{R}$ produces new substrates that inherit the same structural content at coarser scales. The chain produces universes inside the substrate because the geometry of confirmation, applied recursively at the structural problem of exhaustion, has no other available move.

The promise of the volume — that a discrete, informationally closed confirmation substrate built from geometric primitives is both necessary and sufficient to generate wave dynamics, computational substrate, hierarchical organization, and cosmological emergence — is brought to its conclusion here. The substrate is closed under its own dynamics. The model has no numerical degrees of freedom left to tune. The remaining open questions — explicit numerical pinning of $L_p^{(k)}$ and $c^{(k)}$ at higher levels, full representation-theoretic mapping to the Standard Model gauge group, detailed CMB-spectrum computations, and the agent-side framing reserved for Volume 2 — are continuations within the closed framework, not fundamental gaps.

The TMS proposes that physics is what happens when the simplest possible structure that can measure itself is forced to keep measuring. Volume 1 has shown that this proposition, fully unpacked, generates a universe.

Appendix A: Detailed Derivation of the Reinterpretation Operator $\hat{R}$

A.1 The Constraint Set K at the Exhaustion Boundary

We give the formal constraint set K invoked in § 2.2 in explicit form. At $\tau \geq \tau^*(R)$, the substrate's continuation is constrained by:

- K_{A0} (Axiom 0): The new substrate's ground state must be informationally fundamental — a maximum-entropy field with no preferred direction, scale, or target.
- K_{A1} (Axiom 1): The new substrate must support measurement.
- K_{A2} (Axiom 2): The new substrate's primitives must be isotropic immutable spheres (or their structural analogs).

- K_{A3} (Axiom 3): The transition from parent to daughter substrate must preserve the signals that produced the parent's confirmed bits.
- K_{A4} (Axiom 4): The transition must preserve the parent's confirmed bits as immutable causal record.
- $K_{\text{Continuity}}$: The substrate must continue operating; there is no mechanism by which any region can halt.
- $K_{\text{PGE},i}$: The continuation must be minimum-specification — no structural commitment beyond what is forced by K_{A0} through $K_{\text{Continuity}}$.
- $K_{\text{PGE},ii}$: The continuation must be generatively non-trivial — it must admit downstream structure under the new substrate's dynamics.

A.2 Verification That $\hat{R}$ Satisfies K

We verify each constraint:

- K_{A0} : $\Xi(R)$ has ground state U'_{sh} = the polarization landscape across $\mathcal{B}(R)$. This is a maximum-entropy field by construction — it has no preferred direction (the polarization values are scalar), no preferred scale (the lattice structure inherited from R is regular), and no preferred target ($\Sigma(\Xi(R))$ is the full configuration space at the new scale). ✓
- K_{A1} : The new substrate supports measurement by inheriting the triadic confirmation rule at the new scale. ✓
- K_{A2} : The new agents (confirmed bits of $\mathcal{B}(R)$) are isotropic (no orientation parameters) and immutable (Axiom 4 of the parent). The structural analogy to spheres is via the T-void centroid identification of § 2.4. ✓
- K_{A3} : Signals that produced $\mathcal{B}(R)$'s confirmed bits are preserved — they remain consumed and locked in the parent record, which is now the new substrate's pre-geometric substrate. No signal is lost. ✓
- K_{A4} : $\mathcal{B}(R)$'s confirmed bits are preserved as immutable substrate of $\Xi(R)$. They cannot be overwritten by the daughter substrate's dynamics. ✓
- $K_{\text{Continuity}}$: The transition is atomic at the regional scale, and the daughter substrate begins operating immediately upon reinterpretation. ✓
- $K_{\text{PGE},i}$: $\hat{R}$ introduces no new mechanism beyond the identification of confirmed bits as new agents. This is the minimum specification — no other identification adds fewer structural commitments while satisfying K_{A2} . ✓
- $K_{\text{PGE},ii}$: $\Xi(R)$ is generatively non-trivial — it has its own $\Sigma(\Xi(R))$ which is initially unsampled, so the new substrate's direct-read protocol can extract novelty for at least the time until $\Xi(R)$ itself exhausts. ✓

All eight constraints are satisfied by $\hat{R}$. No other candidate continuation satisfies all eight simultaneously, as established by elimination in § 2.2. $\hat{R}$ is therefore the unique element of $\mathcal{C}_K$ satisfying both PGE conditions, and the unique PGE-selection.

A.3 Uniqueness Up to Equivalence

Strictly speaking, $\hat{R}$ as defined in § 2.3 is unique up to permutation of agent labels and translation of the new lattice's origin — both trivial equivalences that do not affect the structural content of $\Xi(R)$. Up to these equivalences, $\hat{R}$ is the unique mapping satisfying K and PGE.

This uniqueness statement is the formal content of the structural claim that the Big Bang Correspondence (§ V) is not a contingent fact but a forced consequence of the framework. The daughter substrate is what it is because $\hat{R}$ is what it is; the initial conditions of the daughter are forced by the parent's terminal state under the unique PGE-selection.

Appendix B: Combinatorial Enumeration of L_p at the Substrate Level

B.1 The Minimum Stabilizer-Satisfying T-Closed Configuration

The load-bearing derivation of $L_p^{(1)} = 3\sqrt{2} \ell_p$ is given in § 6.1.3, which uses the storage convention (1 storage unit = $\ell_p/\sqrt{2}$, forced by Paper 1 § VI's FCC second-moment tensor) and the stella-octangula tile construction to produce the minimum configuration. This appendix provides supporting combinatorial enumeration: the compressibility check $K(C) < |C|$, elimination of smaller candidates, and counting of distinct minimum configurations. The geometric construction itself is given in § 6.1.3 and is not duplicated here.

Earlier drafts of this appendix began the construction at positions $\{(0,0,0), (\ell_p, 0, 0), (0, \ell_p, 0), (0, 0, \ell_p)\}$ and described this as an FCC tetrahedron. This is geometrically incorrect — in the FCC lattice, the nearest-neighbor distance is $\sqrt{2}$ (not 1) in unit-edge cubic coordinates, so those positions are cubic-lattice positions rather than FCC nearest-neighbor positions. The “ $L_p^{(1)} = 4\ell_p''$ ” result obtained from that construction was a downstream artifact of the wrong starting geometry. The current appendix has been revised to defer to § 6.1.3's correct construction.

The configuration described by § 6.1.3 has the structural properties relevant to this appendix: agent count 16 (one stella-octangula tile worth of agent-sphere vertices, including the boundary needed for stabilizer parity evaluation), one confirmed bit at the centroid, and spatial extent $L_p^{(1)} = 3\sqrt{2} \ell_p \approx 4.24 \ell_p$.

B.2 Compressibility ($K(C) < |C|$)

The configuration is described completely by specifying: (a) the FCC orientation of the central tile, which contributes $\log_2(1) = 0$ bits, because the stella-octangula tile is O_h -invariant — its orientation orbit has size 1 (see § B.4); (b) the logical state of the $[[4,2,2]]$ code at the central tetrahedron (one of 4 states, 2 bits); (c) the boundary configuration, which is forced by the higher-order tetrahedral closure of § 2.5 once the central logical state is fixed. The boundary is not determined by stabilizer parity alone — the Z-stabilizer constraint at the boundary box admits $k = 3 \cdot L_{\text{box}}$ free parameters (see § B.5) — but by the T-closure dynamics, which restrict the 2^{12} stabilizer-satisfying configurations to a much smaller set characterized by (a) and (b). The total description length is therefore $\log_2(4) = 2$ bits.

The information content $|C|$ of the configuration is the polarization values at 16 agents plus the confirmed bit, giving $|C| \approx 17$ bits if each polarization is one bit. Since $K(C) = 2 < 17 = |C|$, the configuration is compressible: $K(C) < |C|$. The corrected $K(C) = 2$ bits coincides exactly with the per-confirmation richness of 2 bits established independently; the minimum program's description length equals one confirmation's worth of form. ✓

B.3 Smaller Candidates Eliminated

We verify that no smaller configuration satisfies all three conditions. Consider candidates by their spatial extent:

- $L = 1\ell_p$: a single agent. No tetrahedral structure; stabilizer parity not applicable. Eliminated.
- $L = 2\ell_p$: a pair of agents. Cannot form a tetrahedron in 3D. Eliminated.
- $L = 3\ell_p$: a triangular triple of agents. Forms a 2-simplex but not a 3-simplex (tetrahedron); the $[[4,2,2]]$ code requires 4 vertices. Eliminated.
- $L = 3\sqrt{2}\ell_p$: the minimum stella-octangula tile configuration of § 6.1.3. All three conditions satisfied. ✓

Therefore $L_p^{(1)} = 3\sqrt{2}\ell_p$ is the geometric minimum. The construction is forced; the answer is unique up to symmetry.

B.4 The Combinatorial Count of Distinct Minimum Configurations

The minimum configuration admits four distinct logical states under the $[[4,2,2]]$ code. Its orientation multiplicity is a single orbit: the stella-octangula tile's eight vertices *are* the eight corners of the cube, so the tile is invariant under the full O_h point group and its orbit under O_h has size 1. The A/B sublattice exchange (the red/blue polarization swap) is the internal polarization degree of freedom already counted among the four logical states.

The count is therefore $4 \times 1 = 4$ distinct minimum configurations up to translation and O_h symmetry: the four $[[4,2,2]]$ logical states of the single O_h -invariant stella-octangula tile. This is the substrate's first "program zoo" — the set of distinct minimum programs that can exist at the substrate level. (The explicit two-flop closure is verified in Appendix E: period exactly 2 in confirmed support in every tested realization of the corpus cycle, and period 2 in the signed pattern in the coherent-surround limit of the Paper 2 § 6.2 definition.)

B.5 The $k = 3 \cdot L_{\text{box}}$ Null-Space Regularity

The dimension of the F_2 null space of the Z-stabilizer constraint on an FCC patch of integer extent L_{box} satisfies $k = 3 \cdot L_{\text{box}}$ exactly for $L_{\text{box}} \in \{1, 2, 3, 4\}$, with $k = 3, 6, 9, 12$ respectively. This linear relation expresses a structural fact about the FCC + $[[4,2,2]]$ combinatorics: each unit of patch extent admits exactly three additional independent stabilizer-satisfying degrees of freedom, after accounting for the rank- $(N - 3 \cdot L_{\text{box}})$ coupling among the tetrahedral parity constraints.

The regularity is the small-L manifestation of the inclusion-exclusion among shared-vertex tetrahedra: as the patch grows, additional tetrahedra are added but their parity constraints become increasingly redundant with existing ones. At $L_{\text{box}} \in \{1, 2, 3, 4\}$, the redundancy structure yields exactly the rank $N - 3 \cdot L_{\text{box}}$. The site and tetrahedron counts at these extents are $(N, T) = (4, 1), (14, 8), (32, 27), (63, 64)$, with constraint ranks 1, 8, 23, 51 respectively.

The regularity persists beyond $L_{\text{box}} = 4$: direct computation of the F_2 null-space rank continues to give $k = 3 \cdot L_{\text{box}}$ through $L_{\text{box}} = 7$, with ranks 93, 154, 235 yielding $k = 15, 18, 21$ respectively (Appendix E, Stage 12). This establishes $k = 3 \cdot L_{\text{box}}$ as a stable geometric property of FCC + $[[4,2,2]]$ across the verified range rather than a small-L coincidence, and it belongs in Paper 1's combinatorial inventory as such. Whether the linear relation holds for all L_{box} remains a conjecture; were it ever to break, the breakdown point would itself be

structurally informative, indicating the extent at which higher-order constraint redundancies (plausibly tied to closed geodesic loops on the FCC lattice) enter.

Appendix C: Closed Form of $f(d, K_{\psi, \text{local}}, K_{\psi, \text{surround}}, K_{\psi, \text{subsystem}}, k)$

C.1 The Additive Kolmogorov Bound

The GUP deformation parameter β was introduced in Paper 1 § 7.2 as scaling with the substrate footprint of a confirmed bit. The footprint accumulates contributions from five independent sources, each identified in Papers 1–4. Under the additive Kolmogorov bound, the total footprint is the sum of the individual contributions.

Each source contributes to the substrate footprint a number of lattice units proportional to its information content:

- Causal depth d contributes d lattice units (one per flop of history).
- Local algorithmic complexity K_{local} contributes K_{local} lattice units (one per bit of local algorithmic structure).
- Surround complexity K_{surround} contributes $K_{\text{surround}} \times \sqrt{2}$ lattice units (one per bit of imprinting history, weighted by the $\sqrt{2}$ imprint-depth factor of Paper 3 § 2.3).
- Subsystem complexity K_{subsys} contributes $K_{\text{subsys}} \times 3$ lattice units (one per bit of program content, weighted by the 3-component (C_D, C_O, C_C) trichotomy of Paper 3 § V).
- Hierarchical level k contributes $(1/\sqrt{2})^k$ lattice units, the per-level meta-tetrahedral structural overhead being unity by the form-invariance fixed point (Paper 3 § 4.7).

C.2 The Closed Form

Combining the five contributions:

$$f(d, K_{\text{local}}, K_{\text{surround}}, K_{\text{subsys}}, k) = 1 + \alpha_d \cdot d + \alpha_{\text{local}} \cdot K_{\text{local}} + \alpha_{\text{surround}} \cdot K_{\text{surround}} + \alpha_{\text{subsys}} \cdot K_{\text{subsys}} + \alpha_k \cdot k$$

with the coefficients identified by the geometric factors above:

$$\alpha_d = 1, \alpha_{\text{local}} = 1, \alpha_{\text{surround}} = \sqrt{2}, \alpha_{\text{subsys}} = 3, \alpha_k = (1/\sqrt{2})^k$$

These are dimensionless geometric constants forced by the FCC lattice connectivity and the $[[4,2,2]]$ code distance. The leading constant (1) ensures $f \geq 1$ for all values of the arguments, recovering the bare GUP minimum length scale in the limit of vanishing arguments.

C.3 The β Formula

The GUP deformation parameter β is related to f by

$$\beta = \beta_0 \cdot f(d, K_{\text{local}}, K_{\text{surround}}, K_{\text{subsys}}, k)$$

where β_0 is the bare deformation parameter at $d = K_{\text{local}} = K_{\text{surround}} = K_{\text{subsys}} = 0$, $k = 0$. In the literature on Generalized Uncertainty Principles (Hossenfelder, 2013), β_0 is typically a dimensionless number of order unity. In the TMS framework, β_0 is forced by the FCC lattice structure at the substrate level and is computable in principle from the per-flop noise statistics; we estimate $\beta_0 \approx 1$ from the dimensionality of the FCC lattice.

C.4 Beyond the Additive Bound

The additive closed form is the leading-order expression valid when the individual contributions are small relative to one another. When two or more arguments become large enough that their cross-correlations dominate, the form acquires multiplicative structure. For example, when both d and K_{local} are large, the joint contribution to f is bounded above by

$$d \cdot K_{\text{local}} / \ln 2$$

via the Kolmogorov bound $K(\text{string of length } n \text{ with structure}) \leq n \cdot \log_2(\text{alphabet}) / \ln 2$. The full f beyond the additive limit is a piecewise-linear function in each argument with additional multiplicative cross-terms at large argument values.

The leading additive form is sufficient for empirical predictions at moderate GUP corrections (β corrections of order 10^{-3} or smaller). The full multiplicative form is reserved for simulation work.

Appendix D: Numerical Estimates of $c^{(k)}$ and Identification of Our Level

D.1 The Saturation Cascade

From § 6.2, the hierarchical lightspeed sequence at saturation is $c^{(k)} = (1/\sqrt{2})^k \cdot c$. In SI units, taking $c_{\text{measured}} = 299,792,458$ m/s as our observed lightspeed, the substrate-level c_{TMS} for various candidate hierarchical depths of our universe is:

- If our universe is at $k = 0$ (we are the substrate level): $c_{\text{TMS}} = c_{\text{measured}} \approx 3 \times 10^8$ m/s. This is the minimum possible value.
- If our universe is at $k = 1$: $c_{\text{TMS}} = \sqrt{2} \cdot c_{\text{measured}} \approx 4.24 \times 10^8$ m/s.
- If our universe is at $k = 2$: $c_{\text{TMS}} = 2 \cdot c_{\text{measured}} \approx 6 \times 10^8$ m/s.
- If our universe is at $k = 3$: $c_{\text{TMS}} = 2\sqrt{2} \cdot c_{\text{measured}} \approx 8.48 \times 10^8$ m/s.
- If our universe is at $k = 4$: $c_{\text{TMS}} = 4 \cdot c_{\text{measured}} \approx 12 \times 10^8$ m/s.

The $k = 3$ case is particularly intriguing if the $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ gauge structure of § 8.3 corresponds to three hierarchical levels above substrate. Under this identification, our universe is at hierarchical depth $k = 3$, and the substrate-level c_{TMS} is $2\sqrt{2} \approx 2.83$ times our observed lightspeed.

D.2 Consistency with Cosmological Observations

The TMS framework predicts specific structural features of cosmological observations under different hierarchical-depth assignments. Each prediction provides a partial constraint:

- CMB anisotropy spectrum: the position of the first acoustic peak at $\ell \approx 220$ corresponds to a characteristic angular scale of $\sim 1^\circ$, which under standard cosmology is interpreted as the sound horizon at recombination.

In the TMS reading, this scale corresponds to the typical late-phase configuration scale of $\mathcal{B}(R)$, set by the parent-level program structure.

- Large-scale flatness: the observed near-flatness ($\Omega \approx 1$ to one part in 100) is consistent with the structural flatness derived in § 5.4 for any hierarchical depth, since flatness is a structural property of the reinterpretation event rather than a numerical fine-tuning.
- Horizon problem: the structural horizon-problem resolution of § 5.3 applies at any hierarchical depth.

None of these constraints uniquely pins our universe's k . The structural framework is compatible with multiple hierarchical-depth assignments; pinning requires the combinatorial enumeration of $L_p^{(k)}$ and the comparison to cosmological observables at sufficient precision.

D.3 The Newton Constant and Planck Scale

Under the hierarchical-depth assignment, the relationship between substrate-level and observed Planck length is:

$$\ell_p^{(\text{observed})} = (\sqrt{2})^k \cdot \ell_p^{(\text{substrate})}$$

at saturation. For $k = 3$, the observed Planck length is $2\sqrt{2} \approx 2.83$ times the substrate-level Planck length. This implies that the substrate-level Planck length is $\ell_p^{(\text{substrate})} \approx 5.7 \times 10^{-36}$ m, which is the same order as the observed Planck length and structurally compatible with the identification $\ell_0 \equiv \ell_p$ of Paper 1 § 3.2.

The Newton constant G satisfies a similar level-scaling relation under the cross-level relativistic phenomena of § VII: G corresponds in the TMS picture to the coupling between hierarchical levels, with the observed value reflecting the level-coupling at our hierarchical depth. The detailed identification requires the full continuum-limit Einstein-equation derivation flagged in § 7.5.

Appendix E: Empirical Verification of the Sublattice Bifurcation Structure

The sublattice bifurcation of § 6.1 and the bifurcation-derived maximum novelty output theorem of § 2.6, the matter-antimatter structural correspondence of § 5.3, and the cross-level commutator mechanism of § 8.2 all depend on the level-1 substrate having a specific geometric structure: two interpenetrating FCC sublattices A and B on the all-odd-coord integer lattice (in the storage convention with $\times 2$ scaling per hierarchical level), with each sublattice independently supporting $[[4,2,2]]$ stabilizer dynamics, and the two sublattices related by spatial inversion through the level-2 centroids.

This appendix documents the empirical verification of this structure through two simulation programs (Stages 5 and 6) that test the bifurcation hypothesis at level 1 and at level 2, respectively. The simulations are documented in the supplementary files *lp1_stage5.py* and *lp1_stage6.py* (available as ancillary files with this preprint).

E.1 Lattice Convention

The simulations work in the *storage convention* used throughout § 6.1. The level-0 agent FCC consists of integer-coordinate sites with $\text{sum } i + j + k$ even. Each hierarchical level scales by a factor of 2 relative to the previous level's lattice spacing, so the level-1 sites lie at all-odd-coord integer positions and the level-2 sites lie at all-even-coord integer positions. In substrate units, one storage unit equals $\ell_p/\sqrt{2}$, since the level-0

nearest-neighbor distance is $\sqrt{2}$ storage units and ℓ_p is fixed to that nearest-neighbor distance by Paper 1's wave-equation derivation (second-moment tensor $4\ell_p^2 \delta_{\mu\nu}$ over 12 nearest neighbors).

E.2 Stage 5: Per-Sublattice FCC Verification at Level 1

Hypothesis. The all-odd-coord level-1 sites split by sum-mod-4 into two interpenetrating FCC sublattices A (sum $\equiv 1 \pmod{4}$) and B (sum $\equiv 3 \pmod{4}$), each with FCC nearest-neighbor distance $2\sqrt{2}$ storage units (face-diagonal of the underlying cubic cell of side 2 storage units). Each sublattice independently supports tetrahedral closure (four-site clusters at mutual NN distance squared = 8) and an associated $[[4,2,2]]$ stabilizer structure with null-space scaling $k = 3n$ in the level-local depth n .

Method. For each $L_{\text{box}} \in \{3, 5, 7, 9, 11\}$, enumerate the sublattice-A sites (all-odd-coord integers in $[1, L_{\text{box}}]^3$ with sum $\equiv 1 \pmod{4}$), count nearest neighbors at squared distance 8, identify all four-site clusters with mutual NN at this distance (the tetrahedra), and compute the F_2 null-space dimension of the parity matrix relating sites to tetrahedra. Repeat for sublattice B (sum $\equiv 3 \pmod{4}$). For shared centroids, verify that each A-tetrahedron's centroid coincides with a B-tetrahedron's centroid and that the two tetrahedra are exact inversion partners through the shared centroid.

Results. Stage 5 verifies the bifurcation hypothesis across all tested L_{box} . The per-sublattice structural numbers:

L_{box}	A: sites	A: tets	A: k	A: max NN	B: sites	B: tets	B: k	B: max NN
3	4	1	3	3	4	1	3	3
5	13	8	6	12	14	8	6	8
7	32	27	9	12	32	27	9	12
9	62	64	12	12	63	64	12	12
11	108	125	15	12	108	125	15	12

Three structural features are confirmed:

- **FCC coordination 12 at interior sites.** Each sublattice achieves the standard FCC coordination number of 12 at interior sites for $L_{\text{box}} \geq 7$, confirming that each sublattice is a genuine FCC lattice at NN distance $2\sqrt{2}$ storage units. The low max-NN values at small L_{box} reflect boundary effects (no sites are fully interior in the small box).
- **n^3 tetrahedral count.** The tetrahedron count scales as n^3 where $n = (L_{\text{box}} - 1)/2$ is the level-local depth: $n = 1 \rightarrow 1$, $n = 2 \rightarrow 8$, $n = 3 \rightarrow 27$, $n = 4 \rightarrow 64$, $n = 5 \rightarrow 125$. This is the standard FCC cubic-volume scaling of regular tetrahedral cells.
- **$k = 3n$ null-space scaling.** The stabilizer-satisfying configuration space dimension over F_2 scales linearly with depth: $k = 3n$ through $n = 5$. This is the same scale-invariant $k = 3n$ law that governs the substrate level (Paper 1 § 5.4, scale-invariant stabilizer null-space theorem), now empirically confirmed at level 1 on each sublattice independently.

Inversion-partnership verification. For each $L_{\text{box}} \in \{5, 7, 9\}$, every A-tetrahedron centroid coincides with a B-tetrahedron centroid, and (sampled at 3 centroids per L_{box}) the A-tetrahedron and B-tetrahedron are exact inversion partners through their shared centroid:

L_{box}	A centroids	B centroids	Shared	A-only	B-only	Sample inversion verified
5	8	8	8	0	0	3/3 ✓
7	27	27	27	0	0	3/3 ✓
9	64	64	64	0	0	3/3 ✓

100% of A-tetrahedron centroids coincide with B-tetrahedron centroids; 100% of sampled pairs are exact inversion partners. The structural claim of § 5.3 Claim 2 — that every level-1 spatial site hosts an inversion-paired (A-tet, B-tet) pair through the level-2 centroid — is verified empirically.

E.3 Stage 6: Level-1 vs Level-2 Structural Identity

Hypothesis. The bifurcation pattern repeats at level 2 with identical structural numbers. Specifically, the level-2 sites (all-even-coord integers in the storage frame) split by sum mod 4 $\in \{0, 2\}$ into two interpenetrating FCC sublattices A' (sum $\equiv 0 \pmod{4}$) and B' (sum $\equiv 2 \pmod{4}$), each independently supporting tetrahedral closure with the same n^3 count and the same $k = 3n$ null-space scaling as level 1.

Method. Repeat Stage 5's per-sublattice and inversion-partnership analyses, this time at level 2 (all-even-coord integers in $[0, L_{\text{box}}]^3$ with the appropriate sum-mod-4 filter for sublattices A' and B'). For comparison, rerun the level-1 analyses at matched L_{box} values and compare the structural numbers side-by-side.

Results. At every matched level-local depth n , the level-1 and level-2 structural numbers are identical:

n	Level-1: tets	Level-2: tets	Match	Level-1: k	Level-2: k	Match
2	8	8	✓	6	6	✓
3	27	27	✓	9	9	✓
4	64	64	✓	12	12	✓
5	125	125	✓	15	15	✓

The bifurcation pattern is **structurally identical** at level 1 and level 2. This confirms the constant bifurcation factor of 2 per level (not a compounding factor 2^k), and provides empirical content to the Recursion-Preservation Theorem of § 2.5: the level-1 substrate and the level-2 substrate have the same geometric structure at the same level-local depth.

Inversion-partnership at level 2. At every L_{box} tested at level 2, 100% of A'-tetrahedron centroids coincide with B'-tetrahedron centroids, and sampled pairs are exact inversion partners:

L_{box}	A' centroids	B' centroids	Shared	Sample inversion verified
6	27	27	27	5/5 ✓
8	64	64	64	5/5 ✓
10	125	125	125	5/5 ✓
12	216	216	216	5/5 ✓

The level-2 bifurcation is empirically as exact as the level-1 bifurcation.

E.4 Summary of Empirical Content

The simulation evidence supports the following structural claims of the main text:

- **§ 6.1.3** ($L^{p(1)} = 3\sqrt{2} \ell_p$): the bifurcated structure on which the derivation rests is geometrically realized at level 1 (Stage 5).
- **§ 2.6** (Maximum Novelty Output Theorem): the bifurcation doubles the agent count per level (one A-flavor + one B-flavor agent per shared centroid), empirically verified.
- **§ 5.3 Claim 1** (two sublattices at every level $k \geq 1$): verified at level 1 (Stage 5) and level 2 (Stage 6).
- **§ 5.3 Claim 2** (inversion-partner pairing): 100% verified at both levels.
- **Scale-invariant $k = 3n$ null-space theorem** (Paper 1 § 5.4): empirically verified through $n = 5$ at level 1 and through $n = 5$ at level 2 on matched depth.
- **§ 8.2** (cross-level commutator mechanism): the bifurcation structure on which § 8.2 builds is empirically the actual structure of the level-1 substrate.

E.5 Code Availability

The Python source for the simulations is available as ancillary files with this preprint: *lp1_stage5.py* (Stage 5, per-sublattice FCC verification at level 1) and *lp1_stage6.py* (Stage 6, level-1 vs level-2 structural identity). The raw output logs are similarly available as *lp1_stage5_results.txt* and *lp1_stage6_results.txt*. The simulations are deterministic and reproducible; reviewers and readers can verify the structural numbers tabulated above by direct execution. The full stage suite backing the remaining structural results of this volume (§ E.6) is provided on the same basis — each script standalone under Python 3 with NumPy and SciPy, each accompanied by its captured output log.

E.6 The Full Verification Suite (Reproducibility)

The structural results introduced in this volume are each backed by a standalone, deterministic script with a captured output log, provided as ancillary files. Each recomputes a structural quantity from the substrate rules and checks it against the value quoted in the text; none fits a parameter to a target.

Geometry and code structure. *lp1_stage5–lp1_stage12* verify the L_p bifurcation numbers, the $[[4, 2, 2]]$ classical and quantum distance, the stella-octangula tile with $L_p = 3\sqrt{2} \ell_p$, and the $k = 3 \cdot L_{\text{box}}$ null-space regularity, which Stage 12 extends through $L_{\text{box}} = 7$ (ranks 93, 154, 235 giving $k = 15, 18, 21$; § B.5). *lp1_stage13* computes the minimum-configuration count and returns 4: the stella tile is O_h -invariant (orbit size 1), so the only multiplicity is the four $[[4, 2, 2]]$ logical states (§ B.4). *tms_dynamics.py* defines the operational flop operator T imported by the later stages; *spec_verify_T.py* checks the T specification against the corpus’s own arithmetic (the four Π_ψ values and both resolution branches; the $1 \rightarrow 4$ transport signal accounting). The two-flop T -closure of the minimum program is verified directly: the corpus-described cycle (tile $\rightarrow$ level-2 centroid $\rightarrow$ tile, tetrahedral adjacency) closes with period exactly 2 in confirmed support in every tested variant and branch, with the singular centroid geometrically forced, and closes with period 2 in the full signed polarization pattern in the coherent-surround limit — the regime named by the Paper 2 § 6.2 program definition — with finite-coherence runs deviating and recovering exactly as the probability-approaching-1 clause states (*claim5_corrected.py*; *lp1_stage11* retains the within-sublattice realization as a negative control, which correctly fails to cycle because its domain omits the centroid).

Matter sector. $lp1_{stage18}$ verifies the spinor result: the resolution rule $P(+) = (1 + \Pi)/2$ equals $\cos^2(\theta/2)$ to machine precision across the sampled angles (agents are $SU(2)$ spin- $\frac{1}{2}$), and the product of three 2π rotations is $-I$ (the fermion sign is the oddness of three; § 4.1).

Closed constants. $lp1_{stage19}$ verifies the area-coupling factor of two by the exact up/down triangular-face bipartition of the horizon manifold, and $\varepsilon^{(k)} = 0$ by the integer-coordinate fit of the regular tetrahedral closure at every scale (§ 5.5).

Mass and chemistry. $lp1_{stage20}$ verifies the minimum standing program (size four, a bent cluster) and the discrete rational amplitude set $|\Pi| \in \{1, \frac{1}{2}, \frac{1}{3}, 0\}$ (§ 8.4.3). $lp1_{stage17}$ verifies the subshell capacities (2, 6, 10) from the $SO(3) \downarrow O_h$ subduction, the Madelung $n + \ell$ order, and the noble-gas atomic numbers (2, 10, 18, 36, 54, 86) by direct counting.

Spin geometry, dynamics, form-invariance, heavy elements. $lp1_{stage21}$ verifies the spin-geometry reconstruction (§ 4.1): angles defined from recoupling overlaps embed in exactly three dimensions (positive-semidefinite, rank three, to $N = 200$, with the spin-1 and non-inner-product controls failing as required); three is the minimal valence carrying a fermionic top representation and a single orientation; and the trivalent node is a faithful $SU(2)$ representation with the $2\pi \rightarrow -I$ double cover. $lp1_{stage22}$ verifies that the diagonal whole-event action is forced by the flop — the S_3 -symmetric resolution rule admits only the diagonal action, and the real probabilistic flop is covariant with no preferred axis (the covariance holds to machine precision while the mismatched-axis control differs by order one). $lp1_{stage23}$ verifies the coarse-graining fixed point (§ 4.7): the meta-lattice is FCC with coordination 12 and 24 triangles and 8 tetrahedra per site, identical across levels 0 through 3. $lp1_{stage24}$ verifies the full periodic table (§ 8.4.7): the capacity $2(2\ell + 1)$ from the harmonic angular-mode count (giving $f = 14$), and the noble-gas atomic numbers (2, 10, 18, 36, 54, 86, 118) with period lengths (2, 8, 8, 18, 18, 32, 32) and the $4f/5f$ blocks by counting.

Dispersion, gauge closure, and the selection functional. $dispersion_{25}$ computes the FCC dispersion relation of § 4.1 in closed form: the exact small-momentum expansion, the isotropic quadratic term, the anisotropic quartic remainder with coefficient $1/48$, the $[100] < [110] < [111]$ angular signature, and the $(E/E_{\text{Planck}})^2$ magnitude against experimental Lorentz-violation bounds. $su3_{closure}$ verifies the $su(3)$ closure of Appendix H: eight independent generators, all commutators closing to machine precision, trivial center, negative-definite Killing form (eigenvalues all -12), rank two, irreducible fundamental. pge_{moduli} verifies the Generative-Economy selection functional: the moduli-dimension (specification cost) of the sphere and the alternative primitives, the integer triadic-enclosure count, and the derivation of the FCC-selecting constraint set from Axiom 4 on the contested HCP case.

Every script is deterministic and standalone; the quoted structural numbers can be reproduced by direct execution, and the captured logs are provided alongside the source.

Appendix F: Cross-Level Commutator Structure (Slim Sketch)

This appendix sketches the two-sublattice cross-level commutator structure of § 8.2 in slightly more detail, as a target for subsequent representation-theoretic work. Specifically: the A-stabilizer S_x^A and the B-stabilizer S_x^B at level k can be defined as operators on the level-1 polarization landscape that act independently on their respective sublattices. Their product $S_x^A \cdot S_x^B$ acts at the level- $(k+1)$ shared centroid as a combined operator whose value depends on the order of action when A and B are not in a symmetric configuration. The non-vanishing commutator $[S_x^A, S_x^B]$ at the shared centroid is the substrate-level origin of the non-Abelian gauge closure of $SU(2)$ at level $k + 1$.

The full continuum-limit derivation of this commutator structure as the $SU(2)$ algebra of electroweak theory requires the QCA-to-QFT apparatus, with the specific identification depending on detailed analysis of the polarization-field representation theory under the bifurcated FCC + $[[4,2,2]]$ symmetry. This is a target for subsequent work in Volume 1 (or Volume 2) and is sketched here as the structural direction in which the representation-theoretic completion should proceed.

Appendix G: Norm-Preservation of the Transport (the Anti-Hermiticity Plank of § 8.3)

The $su(3)$ closure of § 8.3 rests on the transport generators being anti-Hermitian and traceless: anti-Hermiticity selects the compact real form and bounds the closure to $u(3)/su(3)$ rather than the non-compact $sl(3, \mathbb{C})$, and tracelessness removes the $U(1)$ trace mode. Anti-Hermiticity is equivalent to the statement that the transport preserves the norm of the propagated amplitude — that the re-confirmation dynamics neither amplify nor damp the internal carry. This appendix establishes that norm-preservation directly from the substrate primitives, in three independent checks. The mean-field map M whose spectral radius is computed here is the FCC nearest-neighbour re-confirmation averaging of Paper 3 § IV; the resolution step is the confirmation primitive $P = (1 + \Pi)/2$ of Paper 2 § IV.

G.1 The resolution term is zero-mean. The confirmation rule resolves a bias $\Pi \in [-1, 1]$ into a bit $\chi \in \{-1, +1\}$ with $P(\chi = +1) = (1 + \Pi)/2$. The conditional expectation is then $E[\chi | \Pi] = (+1)(1 + \Pi)/2 + (-1)(1 - \Pi)/2 = \Pi$, exactly and for every Π . The resolution therefore injects no mean bias: the stochastic part of re-confirmation is a martingale increment, contributing variance but not drift. A Monte-Carlo check across twenty-one biases spanning $[-1, 1]$ at four million samples each gives a worst-case deviation $|\langle \chi \rangle - \Pi|$ of 8×10^{-4} , consistent with the analytic identity. This closes the term previously flagged in the volume as “asserted, not proved”: it is provable, and the proof is one line.

G.2 The mean-field re-confirmation has spectral radius exactly one, saturated only by the gauge mode. The re-confirmation map averages a site’s value over its twelve FCC nearest neighbours (the face-diagonal directions of Paper 1 § VI). Because the FCC lattice is inversion-symmetric, the Fourier symbol of this averaging operator is real: $\lambda(k) = (1/12) \sum_d \cos(k \cdot d)$, summed over the twelve neighbour vectors d . At the zone centre $\lambda(0) = 1$ exactly; a scan over the Brillouin zone gives a maximum modulus of 1 attained only at the Γ point, with $|\lambda(k)| \leq 0.996$ at every grid point away from it and a representative finite- k value $\lambda(0.4, 0, 0) \approx 0.97$. The spectral radius is therefore exactly 1, never greater. The unit eigenvalue is the uniform ($k = 0$) internal mode — the gauge carry that the transport is meant to preserve — while all finite- k spatial modes are strictly contracted: this is ordinary lattice dispersion, isotropic at leading order by the isotropic FCC second-moment tensor (Paper 1 § 6.5), and is not the gauge direction.

G.3 The full stochastic dynamics never amplify the signal. Combining the two steps — resolve each site, then re-confirm by neighbour averaging with reinforcement weight r — and sweeping r from 0 to 0.95, the growth rate of the mean field is 1 at the Γ mode (measured 1.000 ± 0.002 across the sweep) and strictly below 1 at every finite wavevector tested. The effective spectral radius does not exceed 1. One methodological caution is recorded because it was a live trap: measuring the growth of the total field norm from a near-zero initial condition gives a spurious rate of about 1.21, because the zero-mean resolution noise (G.1) inflates the field variance without carrying any net signal. That figure is noise power, not a spectral radius; the correct object is the growth rate of the mean field with the noise ensemble-averaged out, which is what the Γ – mode measurement reports. With that correction the three checks agree.

The three checks together establish that the re-confirmation transport is norm-preserving, with the single conserved mode being the uniform internal carry. A norm-preserving transport has unitary propagation and hence anti-Hermitian generators; together with tracelessness this confines the generated algebra to $su(3)$,

with no overshoot to the non-compact $sl(3, \mathbb{C})$ or the larger $sp(6)$ — the plank on which § 8.3 stands. **Status: Derived.** The reproducibility script (*normpres_G*) accompanies the companion materials.

Appendix H: Explicit $su(3)$ Closure of the Transport Generators

Section 8.3 derives the colour algebra as the closure of the eight transport generators built from substrate primitives: the three real antisymmetric matrices from value transport (the $so(3)$ rotations of the agent labels) and the five i -times-symmetric-traceless matrices from the momentum carry. This appendix records the explicit computation confirming that these eight generators close to exactly $su(3)$, so that the claim rests on a displayed calculation rather than an assertion. All quantities are reproducible by direct evaluation (*su3_{closure}*) in the companion materials.

H.1 The generators. Write the three antisymmetric generators A_1, A_2, A_3 (the $so(3)$ basis) and the five $S_1, \dots, S_5 = i$ times the real symmetric traceless 3×3 basis (two diagonal, three off-diagonal). Each of the eight is anti-Hermitian ($G^\dagger = -G$) and traceless by construction.

H.2 Closure, by construction and by computation. The commutator of any two anti-Hermitian matrices is anti-Hermitian, and every commutator is traceless identically. The real vector space of anti-Hermitian traceless 3×3 matrices has dimension eight (three imaginary-diagonal parameters constrained to sum to zero, giving two, plus three complex off-diagonal entries, giving six) and is precisely $su(3)$. The bracket of two generators therefore cannot leave this eight-dimensional space: closure is automatic, with no overshoot. Computing all 8×8 commutators and projecting onto the generator span confirms this directly, with a maximum residual outside the span of 9×10^{-16} — zero to machine precision. The eight generators are linearly independent (the rank of the flattened set is eight), so none is redundant and none is missing.

H.3 The algebra is $su(3)$, not a same-dimension imposter. Three invariants fix the algebra uniquely among eight-dimensional real Lie algebras. (i) The Killing form $K_{ab} = \text{tr}(\text{ad}_a \text{ad}_b)$ is negative-definite — all eight eigenvalues equal -12 — so the algebra is compact, and nondegenerate, so by Cartan’s criterion it is semisimple. (ii) The center of the algebra is trivial: the common kernel of all eight adjoint maps has dimension zero, so there is no hidden central extension and the algebra is simple. (iii) The Cartan subalgebra has dimension two (the two diagonal generators commute), so the rank is two, with the six remaining generators supplying the six roots of the A_2 root system. The unique compact simple real Lie algebra of dimension eight with rank two is $su(3)$. A control algebra of the same dimension and compactness, $su(2) \oplus su(2) \oplus u(1)^2$, has a two-dimensional center and fails the simplicity test, confirming that the invariants discriminate.

H.4 The fundamental is faithful and irreducible. The three agents carry the defining three-dimensional representation. By Schur’s lemma the only matrices commuting with all eight generators are scalars (the commutant has dimension one), so the representation is irreducible; the center of the group is then the cube roots of unity $\mathbb{Z}_3$ acting by triality, and faithful action on single substrate-real agents requires the cover, fixing the global group as $SU(3)$ rather than $SU(3)/\mathbb{Z}_3$ (§ 8.3). **Status: Derived.**

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All theoretical commitments, the choice of axioms, the Principle of Generative Economy, the sublattice bifurcation insight, and the framework’s overall architecture are the author’s; the AI systems’ role was to teach, formalize, critique, verify, and document.

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Volume 2

The Confirmation Relation: From the Now-Surface to the Reader in the Frame